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**DEVELOPMENT OF A CFD-COMPATIBLE TRANSITION MODEL BASED
ON LINEAR STABILITY THEORY**

A Dissertation in
Aerospace Engineering
by
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ABSTRACT

A new laminar-turbulent transition model for low-turbulence external aerodynamic applications is presented that incorporates linear stability theory in a manner compatible with modern computational fluid dynamics solvers. The model uses a new transport equation that describes the growth of the maximum Tollmien-Schlichting instability amplitude in the presence of a boundary layer. To avoid the need for integration paths and non-local operations, a locally defined non-dimensional pressure-gradient parameter is used that serves as an estimator of the integral boundary-layer properties. The model has been implemented into the OVERFLOW 2.2f solver and interacts with the Spalart-Allmaras and Menter SST eddy-viscosity turbulence models. Comparisons of predictions using the new transition model with high-quality wind-tunnel measurements of airfoil section characteristics validate the predictive qualities of the model. Predictions for three-dimensional aircraft and wing geometries show the correct qualitative behavior even though limited experimental data are available. These cases also demonstrate that the model is well-behaved about general aeronautical configurations. These cases confirm that the new transition model is an improvement over the current state of the art in computational fluid dynamics transition modeling by providing more accurate solutions at approximately half the added computational expense.

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LIST OF NOMENCLATURE

A	=	amplification ratio
AR	=	wing aspect ratio
C_D	=	drag coefficient
C_L	=	lift coefficient
C_M	=	pitching-moment coefficient
C_F	=	skin-friction coefficient
C_p	=	pressure coefficient
c	=	chord length
c_d	=	section profile-drag coefficient
c_l	=	section lift coefficient
c_m	=	section pitching-moment coefficient about the quarter chord
d	=	distance to nearest wall
H_{12}	=	integral boundary-layer shape factor
H_L	=	local boundary-layer shape factor
k	=	turbulent kinetic energy
N	=	amplification factor
N_{crit}	=	critical amplification factor
$\tilde{n}$	=	instability envelope amplification factor
P	=	pressure
R	=	Reynolds number based on chord length and free-stream conditions
R_v	=	vorticity Reynolds number
R_x	=	running Reynolds number based on streamwise distance
R_y	=	Reynolds number based on distance to and velocity relative to nearest wall

R_δ	=	Reynolds number based on boundary-layer thickness and edge velocity
R_{δ_2}	=	Reynolds number based on boundary-layer momentum thickness and edge velocity
Re_{θ_t}	=	transitional momentum thickness Reynolds number
S	=	strain-rate magnitude
s	=	streamwise direction
Tu	=	turbulence intensity
U	=	velocity magnitude
u_j	=	Cartesian velocity component
x_j	=	Cartesian coordinate
x	=	airfoil abscissa
y	=	local wall-normal coordinate
α	=	angle of attack relative to x -axis
α	=	flow instability wave number
δ_1	=	boundary-layer displacement thickness
δ_2	=	boundary-layer momentum thickness
γ	=	ratio of specific heats
γ	=	turbulent intermittency
$\mathcal{A}$	=	Pohlhausen pressure-gradient parameter
$\mathcal{A}_2$	=	Thwaites pressure-gradient parameter
μ	=	molecular viscosity
μ_t	=	turbulent eddy viscosity
ν	=	kinematic viscosity, μ/ρ
$\tilde{\nu}$	=	modified eddy viscosity
ρ	=	fluid density
θ	=	boundary-layer momentum thickness

Ω = vorticity magnitude

ω = turbulent specific dissipation rate

ω = flow instability frequency

Subscripts

e = estimated boundary-layer edge conditions based on pressure

t = transition

0 = conditions at which instabilities are first possible

∞ = free-stream conditions

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DEDICATION

For my loving wife Julie and our wonderful son William James...

Chapter 1

Introduction

Accurate prediction of aircraft drag has long been a goal of computational fluid dynamics (CFD) solutions of the Reynolds-averaged Navier-Stokes (RANS) equations. When solving the RANS equations, a closure method for the Reynolds stress term must be specified. Typically this is done with the Boussinesq approximation and a one- or two-equation model that is active everywhere in the flow field (an approach termed “fully turbulent”). Much work in the development and calibration of these models has been focused on high-Reynolds number applications, such as swept-wing commercial transport configurations. In such cases, the fully turbulent assumption is actually quite reasonable, as the flow typically transitions very close to the nose of the fuselage and the leading edge of the wing (provided there is no attachment-line contamination making the flow immediately turbulent). As a result, the large number of aircraft characterized by lower Reynolds numbers, where laminar-turbulent transition is a dominant feature in the flow field, have been historically underserved by RANS CFD capabilities. Additionally, recent research indicates that the benefits of laminar flow can be realized at commercial transport Reynolds numbers [1]. Having the ability to accurately treat transition in RANS calculations is now also important for this class of aircraft. Considering the usage of such aircraft, even a small reduction in drag can yield significant fuel savings per year for the operator [2]. Some newer commercial design concepts approach this goal through configuration trade-offs and/or innovative configurations in order to delay the onset of transition until further aft on the wing. The MIT D8.5 double-bubble [3] and the Boeing SUGAR High/Volt aircraft configurations [4], which are both design concepts for an N+3 timeframe (i.e. entering service around 2030-

2035), are designed with the intent of achieving natural laminar flow on the wings. The D8.5, pictured in Fig. 1-1, is to have a laminar run on the lower surface of the wing to the 60%-chord location. The wing has less leading-edge sweep than the 737-800 baseline aircraft (the mission requirements of which were used to design the D8.5) to facilitate the laminar flow. The design trade-off for reducing the sweep is that the cruise Mach number must be reduced. The Boeing SUGAR High, pictured in Fig. 1-2, and SUGAR Volt aircraft utilize unswept wings to support the achievement of natural laminar flow on the wing. More recently, the DLR LamAiR design study [5] looked at the possibility of using forward-swept wings to delay cross-flow transition without sacrificing cruise Mach number relative to the baseline A320-200 aircraft. This comes from reducing the leading-edge sweep without affecting the shock angle, as shown in Fig. 1-3. This aircraft, which is pictured in Fig. 1-4, takes advantage of composite structural tailoring as an enabling technology and is designed to maintain laminar flow up to the shockwaves on the both the upper and lower surfaces. While the MIT and Boeing designs assumed the extent of laminar flow in the respective trade studies, the DLR LamAiR design team actually verified the design capabilities [5] with boundary-layer stability calculations [6,7]. Other aircraft studies, such as that in Ref. [8], look to lessen the design trade-offs by using laminar flow control in addition to natural laminar flow methodologies to give what is referred to as hybrid laminar flow control. Beyond conventional fixed-wing aircraft, the attainment of natural laminar flow is even being considered in rotorcraft design studies such as for a high-efficiency tilt-rotor aircraft [9].

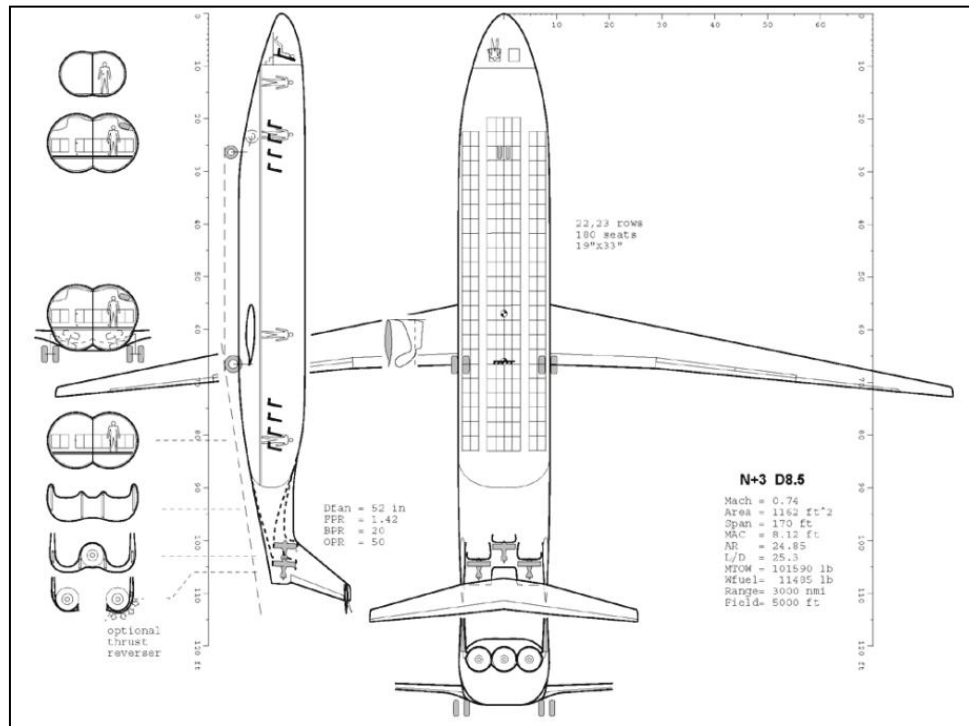


Figure 1-1. MIT D8.5 "Double-Bubble" low-wing-sweep design concept (from Ref. [3]).

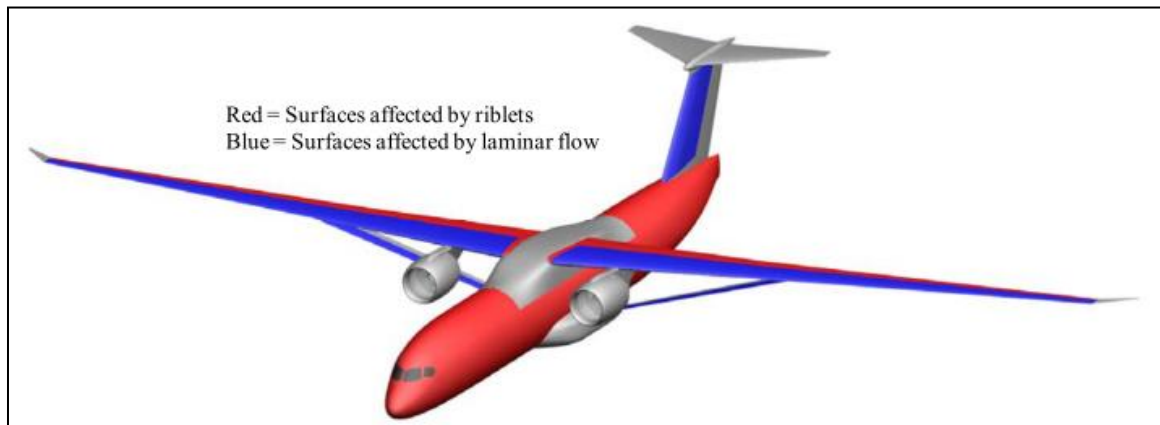


Figure 1-2. Boeing SUGAR High (high, unswept wing) design concept with the anticipated distribution of drag-reduction technologies highlighted (from Ref. [4]).

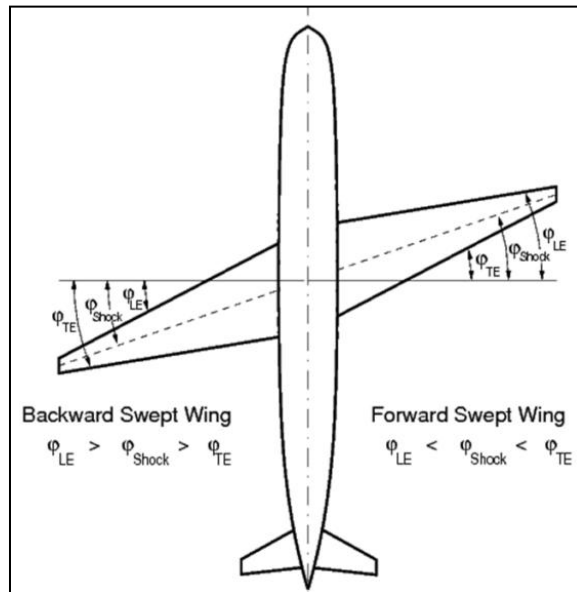


Figure 1-3. Diagram of a forward-swept wing compared to a backward-swept wing with the same shock angle (from Ref. [5]).

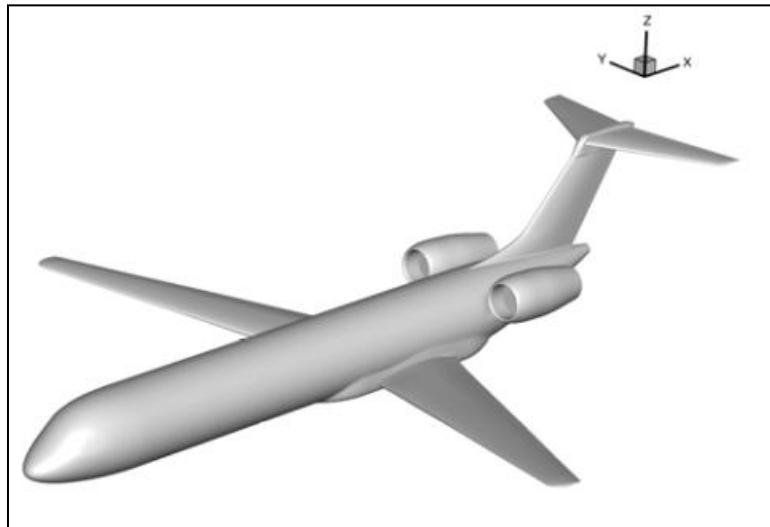


Figure 1-4. DLR LamAiR forward-swept-wing design concept (from Ref. [5]).

For these design concepts, the traditional fully turbulent assumption is not reasonable, and CFD calculations lead to erroneous performance predictions. When such simulations are run for geometries that have relatively long laminar runs, the drag predictions can show significant

error [10]. Thus, it is necessary for RANS solvers to accurately and consistently predict laminar-turbulent transition if those codes are to be reliable for aerodynamic performance prediction. Some CFD codes and their implemented turbulence models allow the specification of “trip lines” to suppress turbulent production upstream of the line. This does not capture the behavior of transition and requires *a priori* knowledge of where transition occurs. Instead, it is desirable to have the transition location predicted and treated automatically by the solver. One method is to use a “post-processing” step during the solution procedure by making use of an external integral-boundary-layer/transition solver [6,7,11], as was done to verify the LamAiR design, but this requires pre-defined integration paths and non-local operations where the entire flow solution needs to be available. The method is, by all means, state-of-the-art and can reliably predict different modes of transition, including Tollmien-Schlichting, cross-flow, and bypass transition; however, the non-local qualities make this approach undesirable to some users since it is not well suited to massively parallel computational methods [12].

An alternative approach is to solve coupled, locally defined transport equations that govern the transition criteria. There have been numerous attempts over the past three decades to develop a locally defined turbulence model that accounts for transition, including Wilcox’s low-Reynolds number modification to his $k-\omega$ model [13] and the $k-k_L-\varepsilon$ model (later $k-k_L-\omega$) of Walters and Leylek [14,15]. A key quality of these models is that they may be regarded as CFD-compatible. Langtry [16,17] defines a model to be CFD-compatible if it meets the following criteria:

1. Allows the calibrated prediction of the onset and the length of transition
2. Allows the inclusion of different transition mechanisms
3. Is formulated locally (e.g. no search or line-integration operations)
4. Avoids multiple solutions (same solution for initially laminar or turbulent boundary layer)

5. Does not affect the underlying turbulence model in fully turbulent regimes
6. Allows a robust integration down to the wall with similar convergence as the underlying turbulence model
7. Is formulated independent of the coordinate system
8. Is applicable to three-dimensional boundary layers

These criteria have been echoed by DiPasquale et al. [18] and are intended to reflect a model's robustness, its ability to be effectively parallelized in modern domain decomposition methods, and its ability to be calibrated for specific applications while taking full advantage of the best eddy-viscosity turbulence models available.

The watershed event for this approach was the development and public release of the local-correlation (γ - Re_{θ_t}) model originally proposed by Menter [19] and more-fully developed by Langtry and Menter [20], which locally evaluates a transition criterion in a manner that is CFD-compatible. The model solves for two additional transported scalars: intermittency, γ , which defines if the flow is laminar or turbulent, and a transitional momentum-thickness Reynolds, Re_{θ_t} , which controls the transition correlations. The model interacts with Menter's popular and widely used k - ω SST eddy-viscosity model [21], which has contributed to its appeal and proliferation in CFD solvers since the transition and the turbulence models may be separately calibrated.

The original γ - Re_{θ_t} framework is capable of modeling transition due to Tollmien-Schlichting instabilities, bypass transition from high free-stream turbulence levels, and separation-induced transition from both laminar separation and shockwaves [20]. The framework has also been extended by Seyfert and Krumbein to include transition due to cross-flow instabilities [12]. The model of Langtry and Menter has been implemented in a variety of CFD solvers, including commercial codes (see, for example, Refs. [22-24]) and government codes [25,26], and thus represents the current state of the art in RANS CFD transition modeling using transport equations. The capabilities of this model have allowed for greatly improved

computational aerodynamic predictions at transitional Reynolds numbers, although there remains a deficiency in that the correlations are strictly local. There is no inclusion of physically relevant upstream flow history when evaluating the $Re_{\theta t}$ transition criteria. Although the correlations may be calibrated with tremendous accuracy for a specific type of problem (e.g. two-dimensional airfoils [27]), they are inherently an expression of a known database of solutions rather than an application of the underlying physics.

Accounting for the influence of upstream boundary-layer history on the transition location is typically done with linear stability theory, specifically using the e^N method of Smith and Gamberoni [28] and van Ingen [29]. Full application of this theory is by no means trivial given that the specific unstable frequency that first reaches critical amplification ratio must be identified. The calculations can be simplified through the use of databases [30], but it remains computationally intensive. The approximate envelope methods of Gleyzes et al. [31] and Drela and Giles [32] simplify the e^N method by tracking only the maximum amplitude of the most-amplified frequencies (i.e. the envelope of the frequencies) and assuming linear amplitude growth with locally self-similar boundary-layer development. Unfortunately, linear stability methods are not directly CFD compatible as they depend on either global or integral quantities to determine the instability amplification rates.

It is therefore desirable to bridge the gap between CFD-compatibility and linear stability theory. An appealing approach is to make use of the approximate envelope method, in which the only variable that needs to be tracked is the envelope amplification factor, $\tilde{n}$, as it develops in the boundary layer. While it is a simplification of the transition process, it is a reasonable simplification of the physics rather than an empirical curve fit of a set of transition data.

The research documented in this dissertation focuses on the development of a new, one-transport-equation transition model for CFD applications that is based on the approximate envelope method for Tollmien-Schlichting instabilities (which includes separation-induced

transition) and is formulated in such a manner that it requires only local and otherwise solution-invariant information. Development of the new model begins by exploring the physics of the transition process in which small disturbances in the flow field grow into large disturbances and, eventually, turbulence. Various approaches for predicting transition are discussed, including e^N theory, which is based on the linear stability theory that describes the growth of small disturbances, local-correlation methods that compare representative boundary-layer properties, and CFD-compatible PDE-based methods that require the solution of transport equations. Bridging the gap between linear stability theory and PDE methods requires the determination of the integral boundary-layer properties, in particular the shape factor. This challenge has been overcome through the introduction of a local shape factor, developed for this work, from which the integral properties can be estimated without any path integration. From here, the source term of a transport equation that describes the evolution of the approximate envelope amplification factor can be defined. Such a transport equation is a new contribution to computational aerodynamics. Methods are developed for the transported amplification factor to introduce transitional behavior to the widely used one-equation Spalart-Allmaras [33] and the two-equation Menter $k-\omega$ SST [21] eddy-viscosity models. Aerodynamic predictions using these combinations for a variety of two- and three-dimensional geometries show improvement over those made with the CFX-v-1.1 version of the PDE-based, four-equation Langtry-Menter model [20] and require less computational overhead due to fewer transport equations being necessary.

Chapter 2

Background

The transition process can be completely described by the Navier-Stokes equations with either Direct Numerical Simulation (DNS) or Large Eddy Simulation (LES) [34] necessary to sufficiently resolve the scales of the instabilities, but the computational overhead requirements for such simulations are not feasible at this time for geometries of practical interest. One projection of the future of computational capabilities predicts that it will not be until the year 2045 that LES will be possible for the wing of a commercial transport during the design process [35].

In the absence of direct resolution of the relevant flow scales in an aerodynamic simulation, the problem of predicting transition in the flow field becomes two-fold: 1) knowing how flow instabilities develop in a laminar boundary layer, and 2) knowing when the instabilities lead to the break down into turbulence.

2.1 Physics of Transition

Transition is the collection of processes through which a laminar flow becomes turbulent. It was first described by Reynolds [36], who characterized pipe flows using a non-dimensional quantity that has since been referred to as the Reynolds number. The transition process begins with the receptivity stage in which disturbances in the free stream, both vortical and acoustic, are “felt” in the boundary layer (or other shear layer) as velocity and pressure perturbations. In cases with relatively low levels of turbulence in the free stream (typically less than 1%), these perturbations are sufficiently small compared to the mean velocity field that their initial growth and decay as they move downstream behave in a manner that is predictable using linearized

theories. Some disturbances grow to sufficient amplitudes that secondary instabilities begin to dominate [7]. From there, turbulent spots begin to form [38] and ultimately merge into fully turbulent flow. This process is shown graphically in Fig. 2-1 (from Ref. [39]).

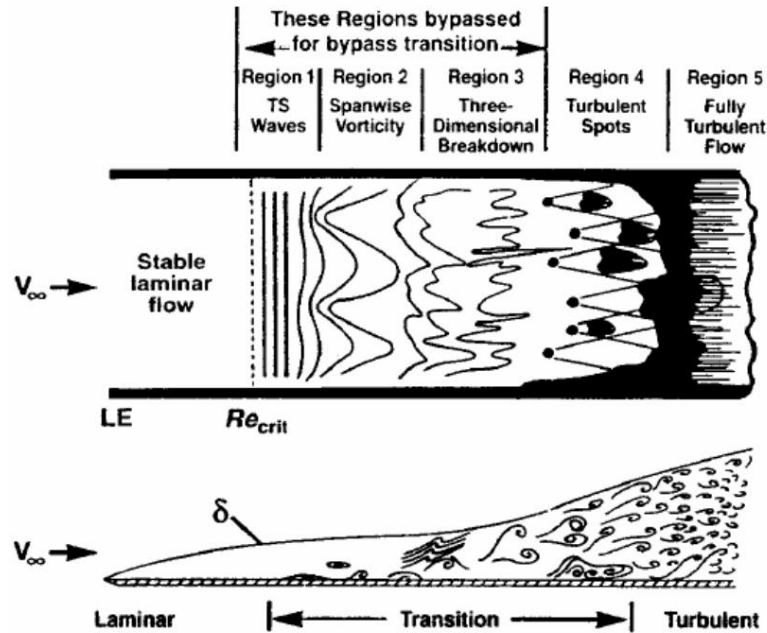


Figure 2-1. Graphical depiction of the natural transition process (from Ref. [39]).

While Prandtl was able to qualitatively describe the physics of instability growth in the boundary layer [40], his student Tollmien [41] solved the Orr-Sommerfeld equation for the Blasius boundary-layer profile by assuming locally parallel flow. Tollmien was able to determine the neutral-stability curve for the profile and the critical displacement-thickness Reynolds number, below which all instabilities decay, and Schlichting [42] was later able to calculate the growth rates of the instabilities. While these results were purely mathematical at the time, the existence of these Tollmien-Schlichting waves was eventually confirmed by the experiments of Schubauer and Skramstad [43].

Mayle [44] categorizes the transition process into three different modes: natural, bypass, and separated-flow transition. Natural transition occurs when there is very little turbulence in the

free stream (typically regarded as being less than 1% in intensity). In this situation the velocity perturbations initially present in the boundary layer permit the primary, linear growth mode. This growth can be in the form of Tollmien-Schlichting waves (in the case of primarily two-dimensional flow), cross-flow instabilities [45] on swept wings with three-dimensional boundary layers, Gortler instabilities [46] on concave surfaces, etc. Regardless of the form of the instability, its growth and decay can be accurately predicted with linear methods such as the Orr-Sommerfeld equation. Bypass transition [47] occurs when the free-stream turbulence is sufficiently intense that the velocity perturbations transferred into the boundary layer are too large to undergo linear growth. In this sense the primary growth mode is “bypassed” and the flow instabilities immediately experience secondary instabilities and form Emmons spots. Separated-flow transition [44] occurs in regions of separated flow, such as in a laminar-separation bubble or in a laminar wake, and may undergo either natural transition (e.g. Tollmien-Schlichting transition in a laminar-separation bubble) or bypass transition.

A phenomenon similar to bypass transition can occur for swept-wing aircraft where turbulent flow from the fuselage can propagate along the attachment line of the wing and prevent any laminar flow from occurring on the wing [48]. This is termed attachment-line contamination, and its influence is predictable using linear stability theory [49].

2.2 Linear Stability Theory

The problem of tracking the development of flow instabilities can be made tractable by introducing simplifying assumptions to the Navier-Stokes equations. In particular, the following sections focus on the set of simplifications that define linear stability theory. Although there are other, more complex theories that accurately describe the behavior of flow instabilities with less-

restrictive assumptions [50], linear stability theory represents the cornerstone of our understanding and is the current standard for transition prediction in aeronautical applications.

A set of assumptions applied to the Navier-Stokes equations to understand their stability characteristics are that the mean flow is steady, incompressible, fully developed, and parallel and that the velocity perturbations are relatively small. These assumptions were applied independently by Orr [51,52] and Sommerfeld [53] for two-dimensional flows. By defining the total velocity and static pressure as the sum of a mean-flow component and a perturbation component (denoted with a prime), the perturbation continuity and momentum equations can be expressed as

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = 0 \quad (2.1)$$

$$\frac{\partial u'}{\partial t} + U \frac{\partial u'}{\partial x} + v' \frac{\partial U}{\partial y} = -\frac{1}{\rho} \frac{\partial P'}{\partial x} + \nu \left(\frac{\partial^2 u'}{\partial x^2} + \frac{\partial^2 u'}{\partial y^2} \right) \quad (2.2)$$

$$\frac{\partial v'}{\partial t} + U \frac{\partial v'}{\partial x} = -\frac{1}{\rho} \frac{\partial P'}{\partial y} + \nu \left(\frac{\partial^2 v'}{\partial x^2} + \frac{\partial^2 v'}{\partial y^2} \right) \quad (2.3)$$

where the mean velocity is assumed to be known. The pressure perturbations may be removed by taking the curl of these momentum equations, yielding

$$\frac{\partial}{\partial t} \left(\frac{\partial u'}{\partial y} - \frac{\partial v'}{\partial x} \right) + U \frac{\partial^2 u'}{\partial x \partial y} + v' \frac{\partial^2 U}{\partial y^2} - U \frac{\partial^2 v'}{\partial x^2} = \nu \left(\frac{\partial^3 u'}{\partial x^2 \partial y} + \frac{\partial^3 u'}{\partial y^3} - \frac{\partial^3 v'}{\partial x^3} - \frac{\partial^3 v'}{\partial y^2 \partial x} \right) \quad (2.4)$$

As it is of most interest to understand the streamwise development of any disturbances, the velocity perturbations are represented by a perturbation streamfunction,

$$\psi = \phi(y) A_0 \exp[i(\alpha x - \omega t)] \quad (2.5)$$

where x is the streamwise coordinate, y is the wall-normal coordinate, and A_0 is an undetermined coefficient. The temporal component of the streamfunction appears in the interest of satisfying Galilean invariance. This form of the streamfunction has useful properties in that differential operators can be expressed as

$$\frac{\partial}{\partial t} = -i\omega \quad (2.6)$$

$$\frac{\partial}{\partial x} = i\alpha \quad (2.7)$$

The specific velocity components are

$$u' = \frac{d\phi}{dy} A_0 \exp[i(\alpha x - \omega t)] \quad (2.8)$$

$$v' = -i\alpha\phi(y) A_0 \exp[i(\alpha x - \omega t)] \quad (2.9)$$

At this point it is also useful to define, for simplicity,

$$\xi(x, t) = A_0 \exp[i(\alpha x - \omega t)] \quad (2.10)$$

Applying these definitions to Eq. 2.4 gives

$$-i\omega \left(\frac{d^2\phi}{dy^2} \xi - \alpha^2 \phi \xi \right) + i\alpha U \frac{d^2\phi}{dy^2} \xi - i\alpha \phi \xi \frac{\partial^2 U}{\partial y^2} - i\alpha^3 U \phi \xi = \nu \xi \left(\frac{d^4\phi}{dy^4} + \alpha^4 \phi - 2\alpha^2 \frac{d^2\phi}{dy^2} \right) \quad (2.11)$$

Every term of this equation is linear in ξ , which, therefore, may be eliminated from both sides.

Regrouping the terms gives the following equation:

$$\left(U - \frac{\omega}{\alpha} \right) \left(\frac{d^2}{dy^2} - \alpha^2 \right) \phi - \frac{\partial^2 U}{\partial y^2} \phi = \frac{\nu}{i\alpha} \left(\frac{d^2}{dy^2} - \alpha^2 \right)^2 \phi \quad (2.12)$$

This equation is commonly known as the Orr-Sommerfeld equation and is the basis for linear stability theory. It is a linear, fourth-order ordinary differential equation in ϕ for which α and ω are eigenvalues that allow the boundary conditions of the velocity perturbations to be satisfied.

Solutions to Eq. 2.12 for a specified velocity profile are often collected into stability diagrams that depict curves of constant amplification rate. The curve representing zero growth of instabilities is referred to as the neutral stability curve, or, in some resources (see, for example, Ref. [54]), as the “zarf”, which is the Turkish word for “envelope”. Examples of neutral stability curves for various Pohlhausen profiles and various Falkner-Skan profiles are included as Figs. 2-2

(from Ref. [39]) and 2-3 (from Ref. [28]), respectively. Inside any one of these curves, instabilities grow, and outside they decay. One of the most important features of the neutral stability curve is the lowest Reynolds number of its extent. This limit is the critical Reynolds number for this boundary-layer velocity profile, and all instabilities decay if the displacement-thickness Reynolds number is below this value.

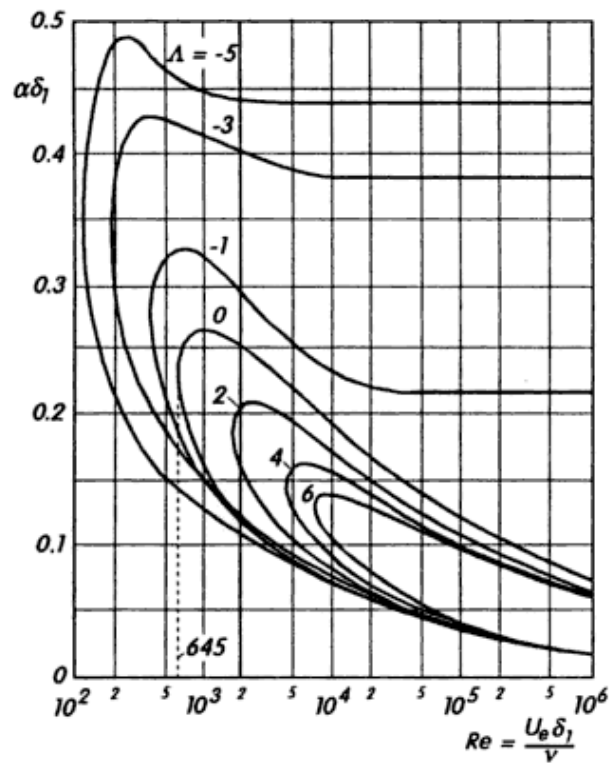


Figure 2-2. Neutral stability curves for Pohlhausen profiles of varying pressure-gradient parameter, A (from Ref. [39]).

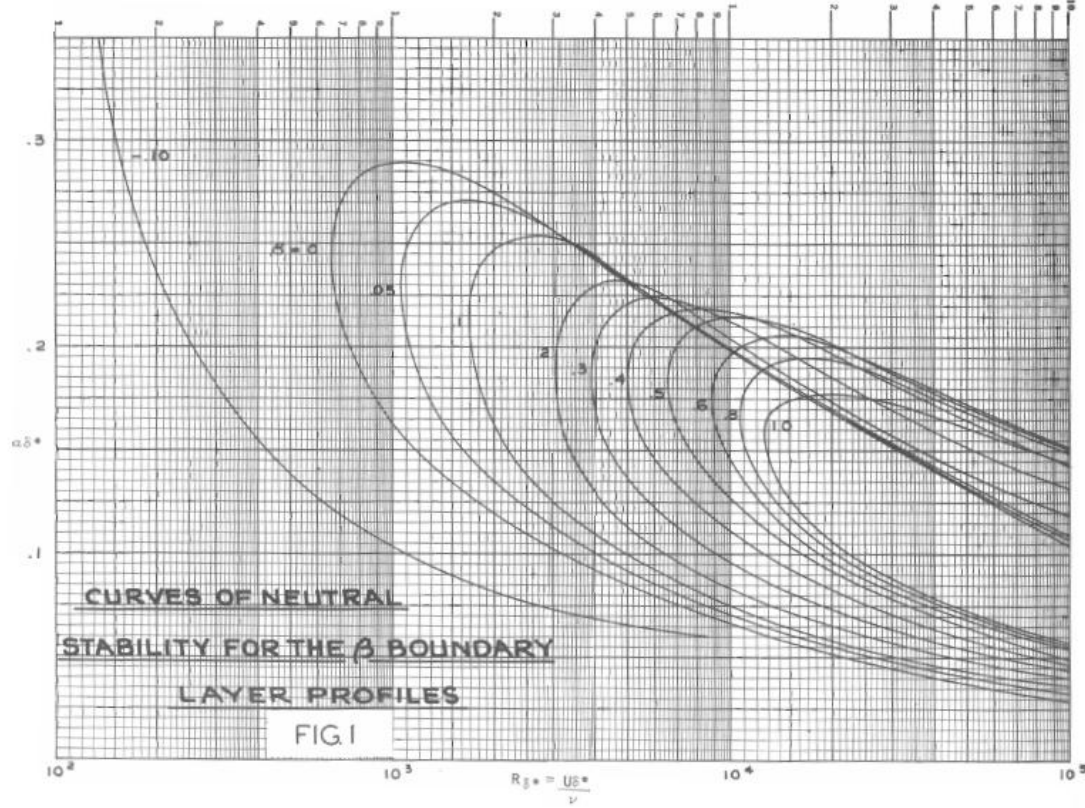


Figure 2-3. Neutral stability curves for Falkner-Skan profiles of varying pressure-gradient parameter, β (from Ref. [28]).

Linear stability theory can also be extended to account for the influence of a non-parallel flow as well as other streamwise variations in the flowfield. One method of doing this is by using the Parabolized Stability Equations (PSE) of Herbert and Bertolotti [55], in which the two-dimensional disturbance streamfunction is slightly redefined as

$$\psi = A_0 \phi(x, y) \xi(x, t) \quad (2.13)$$

where

$$\xi(x, t) = \exp[i(\theta(x) - \omega t)] \quad (2.14)$$

The function $\theta(x)$ is a generalization of the αx term in Eq. 2.5 such that

$$\alpha = \frac{d\theta}{dx} \quad (2.15)$$

This yields the partial differential equation given by (from Ref. [50])

$$(L + L')\phi - i \frac{\partial L}{\partial \alpha} \frac{\partial \phi}{\partial x} - \frac{i}{2} \frac{\partial^2 L}{\partial \alpha^2} \frac{d\alpha}{dx} \phi = 0 \quad (2.16)$$

where L is the Orr-Sommerfeld operator (i.e. if Eq. 2.12 were expressed as $L\phi = 0$). Equation 2.16 is largely parabolic in nature and can be solved using a marching procedure in the streamwise direction. This equation also recovers the Orr-Sommerfeld equation as a limiting case with parallel flow and constant α . The benefits of the PSE methodology are that it is not limited to non-parallel, streamwise varying flows, and it can also be readily extended to three-dimensional flows/disturbances and compressible flows. The equation is not fundamentally restricted to a homogeneous right-hand side, which may be replaced with the otherwise-neglected non-linear terms. Thus, the parabolized stability equations can also predict the behavior of non-linear instabilities.

2.3 Predicting Transition

2.3.1 e^N Theory

During the early 1940s, Liepmann [56] postulated that transition occurs in a boundary layer when the magnitude of the Reynolds stresses due to velocity perturbations becomes equal in magnitude to the maximum shear stress present in the boundary layer. He provided the following expression to quantify this relationship:

$$\frac{\tau_T}{\tau_L} = -\frac{2}{c_{f,L}} \left[kb \left(\frac{u'}{U_e} \right)_0^2 A^2(x) \right]_{\max} \quad (2.17)$$

In this equation, $A(x)$ is the amplification ratio at streamwise location x of the original velocity perturbations that are present in the boundary layer. The quantity k is a correlation coefficient

relating the Reynolds stresses to the mean flow, b is a measure of the anisotropy of the velocity perturbations, and c_{fL} is the laminar skin-friction coefficient. The left-hand side is the ratio of the turbulent shear stress, τ_T , to the laminar shear stress, τ_L . This expression for transition still requires knowledge of the initial disturbances in the boundary layer and a calculation of the amplification factors.

Smith and Gamberoni [28] set out to solve and evaluate Liepmann's hypothesis for airfoil flows and, in the process, discovered that transition consistently occurred at the location where any frequency first obtains an amplification ratio approximately equal to e^9 (referred to as the critical amplification ratio). Simultaneously and independently, van Ingen [29] came to a similar conclusion upon analyzing the Schubauer and Skramstad data [43], in which transition occurs over a range beginning when the maximum amplification is $e^{7.8}$ and ending when the maximum amplification is $e^{10.0}$ (these values have since been revised by van Ingen [57] to $e^{8.22}$ and $e^{10.30}$, respectively). The two results are very similar and consistent with each other. The slight discrepancies are regarded as being due to Smith and Gamberoni looking primarily looking at airfoil flows where transition happens rapidly, while van Ingen started from the Schubauer and Skramstad flat-plate experiment [43] where transition happens much more gradually and the secondary-growth region is of significant length. In the methods of Smith and Gamberoni and van Ingen, the exponent of the transitional amplification ratio is called the critical N -factor, and the method itself is subsequently referred to as the e^N method. Van Ingen notes that the N -vs- x curve is very steep near transition on an airfoil, and thus it is not surprising that Smith and Gamberoni determined a mean value of $N = 9$ [28] for transition despite the process occurring over a finite length.

The tracking of all frequencies that may grow in a boundary layer, typically referred to as the “full” e^N method, can be a computationally intensive task as the integral

$$N = - \int_{s_0}^s \delta_2 \alpha_i \left(\frac{\omega \delta_2}{U_e}, H_{12}, R_{\delta_2} \right) ds \quad (2.18)$$

must be evaluated for all possible frequencies. In this integral, α_i is the imaginary part of the wave number from Eq. 2.12 and is a function of the instability frequency and boundary-layer shape factor, displacement thickness, and edge velocity. Integration begins where the frequency first crosses neutral stability (subscript ‘0’) and begins to grow. As integration continues downstream of this, the collection of amplification factors from the most-amplified frequency at each point, referred to as the “envelope”, can be defined. Transition is predicted to occur once the amplification envelope reaches the critical amplification factor.

Simplifications to the method have been developed since its inception. For example, the growth and decay of the instabilities can be calculated from a database of non-dimensionalized stability diagrams [30,57], and the first frequency to reach to the critical N -factor can be determined through the use of a Newton solver [58]. Another simplification is to track only the most unstable frequencies at each integration step, but this no longer represents a full e^N method. That simplification can be taken further by tracking only the envelope of maximum amplification [31,32]. In this approximate envelope method, the amplification rates along the envelope of maximum amplification can be linearly approximated for self-similar flows, such as is depicted in Fig. 2-4 (from Ref. [32]). The envelope amplification factor, $\tilde{n}$, can be evaluated using

$$\tilde{n} = \int_{s_0}^s \frac{d\tilde{n}}{dR_{\delta_2}} \frac{dR_{\delta_2}}{ds} ds \quad (2.19)$$

The envelope growth rate, $d\tilde{n}/dR_{\delta_2}$, depends only on the boundary-layer shape factor, and the boundary-layer growth rate, dR_{δ_2}/ds , depends on the shape factor and the momentum thickness. There is not a dependency on the specific instability frequency in the calculation procedure, as the integrand of Eq. 2.19 depends only on the local integral boundary-layer properties. Transition is regarded as occurring when $\tilde{n}$ first reaches the critical N -factor.

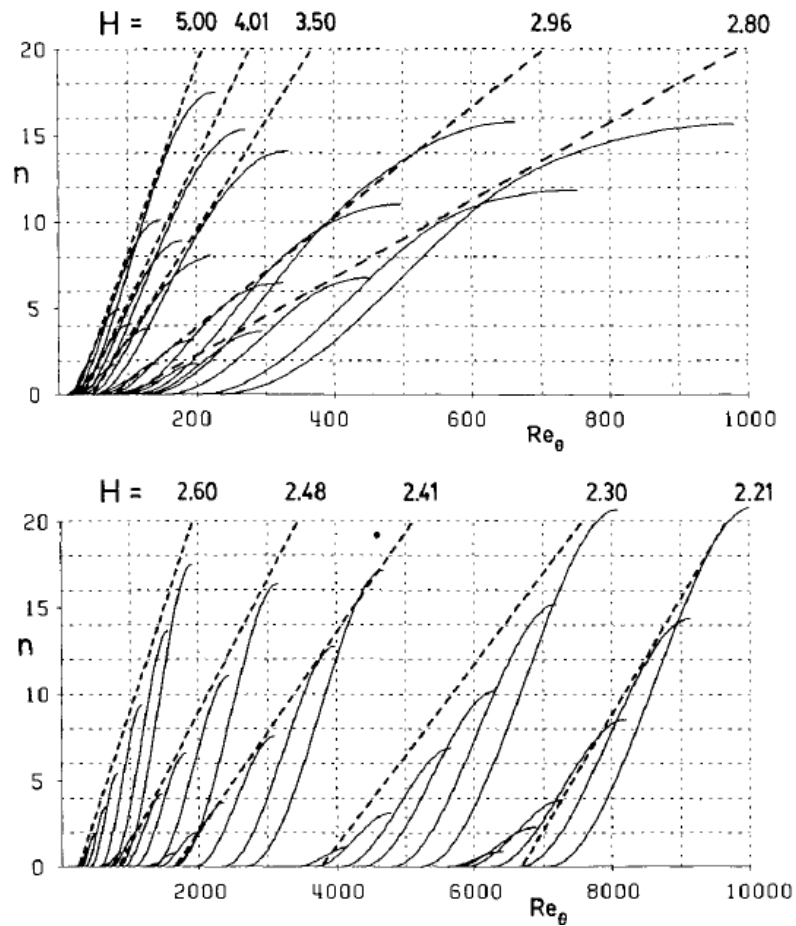


Figure 2-4. Amplification envelopes and corresponding linear approximations for various constant-shape-factor boundary layers (from Ref. [32]).

2.3.2 Local-Correlation Methods

A group of transition prediction methods that are separate from, but related to, the stability-theory-based methods are the local-correlation methods. These correlations use information that is known at a given streamwise location in the boundary layer, such as a characteristic Reynolds number, a pressure-gradient parameter or shape factor, and/or the free-stream turbulence intensity, and compare this information to an empirical correlation to those

values causing transition. At least one quantity that characterizes the boundary-layer profile is required in such correlations. These methods are completely local and do not explicitly account for upstream flow history, but their effectiveness stems from the ability to calibrate the correlation to match a certain class of problem.

Michel Criterion

One of the oldest local-correlation methods is the one proposed by Michel [59], which compares the local momentum-thickness Reynolds number, $R_{\delta 2}$, to the local running Reynolds number, R_x , using the relationship

$$R_{\delta 2} = 1.174 \left[1 + \frac{22400}{R_x} \right] R_x^{0.46} \quad (2.20)$$

Transition is predicted to occur at the first point where the momentum-thickness Reynolds number exceeds the value predicted by Eq. 2.20. Michel's criterion is most accurate in regions with strong adverse gradients where the momentum thickness grows rapidly relative to the streamwise distance. Its accuracy decreases in weak gradients where both sides of the equation grow at similar rates, since such scenarios cause small inaccuracies in the calibration to be realized as relatively large distances.

Granville Criterion

The method of Granville [60] was an early attempt to incorporate linear stability theory into the prediction methodology of transition. Granville recognized that the boundary-layer development downstream of the neutral stability location was the primary influence on the

transition location. In particular, the Granville method focuses on the development of the Thwaites parameter, Λ_2 , given by

$$\Lambda_2 = \frac{\delta_2^2}{\nu} \frac{dU_e}{ds} \quad (2.21)$$

The running average of Λ_2 ,

$$\overline{\Lambda_2} = \frac{\int_{s_{cr}}^s \frac{\delta_2^2}{\nu} \frac{dU_e}{ds} ds}{\int_{s_{cr}}^s ds} \quad (2.22)$$

is used to evaluate the change in momentum-thickness Reynolds number from neutral stability that is necessary for transition to occur. Reference [60] does not provide an equation for the momentum-thickness Reynolds number relations, but graphs of the relation are included. While this method is rooted in stability theory and attempts to account for the total boundary-layer development, it is still a local correlation method in that a local Reynolds number is compared to a transitional Reynolds number predicted by a representative pressure-gradient parameter and it predates the formal application of linear stability theory to predicting transition.

Van Driest and Blumer Criterion

Van Driest and Blumer [61] proposed a transition prediction method rooted in the notion that transition occurs when the vorticity Reynolds number,

$$R_v = \frac{\left| \frac{du}{dy} \right| y^2}{\nu} \quad (2.23)$$

reaches a limiting value. The influence of pressure gradient and turbulence intensity are included by considering the Pohlhausen fourth-order velocity profile and the fluctuations of the boundary-layer edge velocity. The criterion can be expressed as

$$\frac{9860}{R_\delta} = 1 - 0.485\Lambda + 3.36R_\delta \left(\frac{u'}{U_e} \right)^2 \quad (2.24)$$

where δ is the boundary-layer thickness and Λ is the Pohlhausen parameter,

$$\Lambda = \frac{\delta^2}{\nu} \frac{dU_e}{ds} \quad (2.25)$$

which characterizes the pressure gradient.

Eppler Criterion

The transition correlation of Eppler was developed empirically for use in airfoil design and analysis [27]. The method uses the local H_{32} shape factor as a representative pressure-gradient parameter to calculate the transitional momentum-thickness Reynolds number. This correlation is given as (from Ref. [27])

$$\ln R_{\delta 2} \geq -21.74 + 18.4H_{32} + 125(H_{32} - 1.573) \quad (2.25)$$

Unlike other local-correlation methods, Eppler's method is based on matching experimentally measured profile-drag coefficients of airfoils rather than measured transition locations. This method is intended for low-turbulence applications and is unable to account for varying free-stream turbulence intensity.

Abu-Ghannam and Shaw Correlation

The Abu-Ghannam and Shaw transition correlation [62] predicts the transitional momentum-thickness Reynolds number as a function of the local Thwaites parameter and the free-stream turbulence intensity. The correlation for transition onset, given by

$$R_{\delta 2, onset} = 163 + \exp \left[F(\Lambda_2) - \frac{F(\Lambda_2)}{6.91} Tu(\%) \right] \quad (2.26)$$

where

$$F(\Lambda_2) = \begin{cases} 6.91 + 12.75\Lambda_2 + 63.34\Lambda_2^2 & (\Lambda_2 \leq 0) \\ 6.91 + 2.48\Lambda_2 - 12.27\Lambda_2^2 & (\Lambda_2 > 0) \end{cases} \quad (2.27)$$

was developed primarily for turbomachinery flows and is well-calibrated for high free-stream turbulence intensities where bypass transition is the dominant mode of transition.

2.3.3 PDE-Based Methods

Transition models based on the solution of transport-type partial differential equations are of particular interest for CFD methods. Being in the form of a transport equation makes these models compatible with the solution techniques available in CFD solvers. In tandem with that quality is that they can be defined locally, making the models amenable to massive parallelization. Two such transition models will be discussed in this section: the three-equation model of Walters and Leylek [14] and the four-equation transitional SST model of Langtry and Menter [20]. Although this is far from being a complete survey of PDE-based transition prediction models, these two models are representative of current modeling techniques and have been implemented into commercial CFD solvers.

Walters and Leylek Transitional Turbulence Model

The model of Walters and Leylek is based on the notion of “laminar kinetic energy” [14], which is the energy contained in the streamwise velocity fluctuations of instability waves. The physical velocity fluctuations that are being modeled are considered to be laminar in that the

cascade of energy from large scales to small scales is not present. Nevertheless, the model treats the laminar kinetic energy analogously to the turbulent kinetic energy, with the laminar kinetic energy essentially being that of the large scale fluctuations. The laminar energy grows via the “splat” mechanism, where the presence of the wall redirects wall-normal fluctuations into streamwise fluctuations. This model is capable of predicting both natural and bypass transition. The original form of the model [14] was based on the low-Reynolds number k - ε equations, expressed as

$$\frac{Dk_T}{Dt} = P_T + R + R_{NAT} - \varepsilon - D_T + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\alpha_T}{\sigma_k} \right) \frac{\partial k_T}{\partial x_j} \right] \quad (2.28)$$

$$\frac{Dk_L}{Dt} = P_L - R - R_{NAT} - D_L + \frac{\partial}{\partial x_j} \left[\nu \frac{\partial k_T}{\partial x_j} \right] \quad (2.29)$$

$$\frac{D\varepsilon}{Dt} = C_{\varepsilon 1} \frac{\varepsilon}{k_T} (P_T + R_{NAT}) + C_{\varepsilon R} R \frac{\varepsilon}{\sqrt{k_T k_L}} - C_{\varepsilon 2} \frac{\varepsilon^2}{k_T} - \frac{\varepsilon}{k_T} D_T + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\alpha_T}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] \quad (2.30)$$

The two kinetic energy equations can be summed to yield a standard turbulent kinetic energy equation, reflecting the splitting of fluctuation length scales in this model.

The transition model has since been adapted by Walters and Cokljat [15] to interact with a k - ω -type eddy-viscosity model, though it is conceptually the same. This form of the model is implemented in the commercial ANSYS FLUENT solver [23], which is a widely used commercial CFD solver. Examples of the predictive capabilities of the k - k_L - ω variant for the T3A-, T3A, and T3B zero-pressure-gradient flat plate cases (which have inlet turbulent intensities of 0.874%, 3.3%, and 6.5%, respectively) [63] are included as Fig. 2-5 (from Ref. [15]). Despite this quality of agreement, the major drawback of this model is that the transition criteria are strongly coupled to the eddy-viscosity equations, preventing the two parts of the model from being calibrated independently.

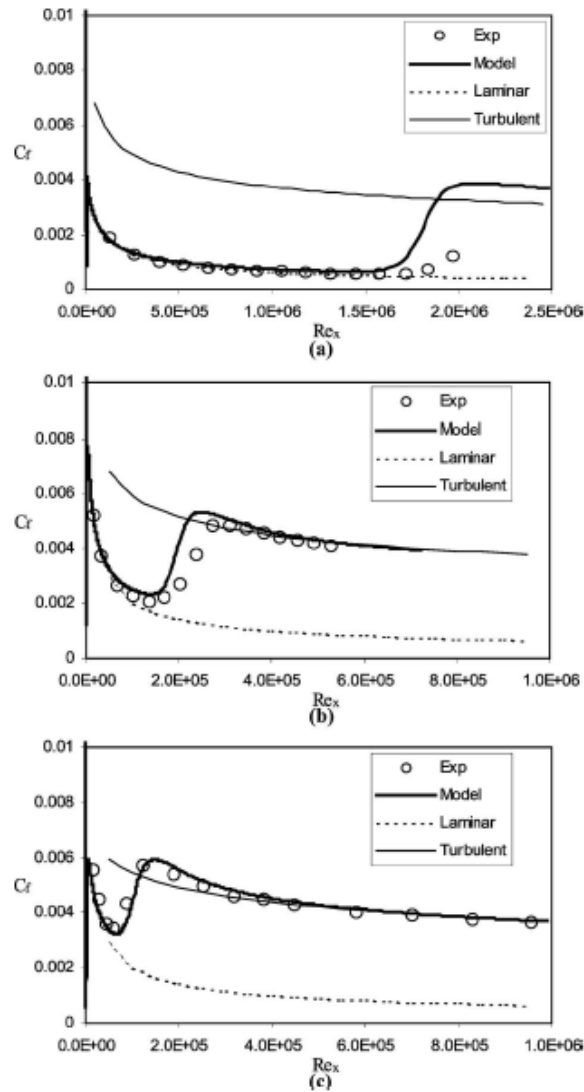


Figure 2-5. Skin-friction distributions predicted using $k-k_L-\omega$ model for the (a) T3A-, (b) T3A, and (c) T3B zero-pressure-gradient flat plate cases (from Ref [15]).

Langtry-Menter Local Correlation Transition Model

The transition method proposed by Menter [19] is a PDE implementation of local-correlation models and has a philosophy much different from that of Walters et al. [14,15] First, the transition equations are largely decoupled from the eddy viscosity equations to allow the benefits of the underlying turbulence model to be realized after transition. Second, the transition

correlations are empirical and based on quantities well known in traditional boundary-layer analyses. This is accomplished by solving two transport equations in addition to the Menter SST model, which serves as the underlying eddy-viscosity model. These two extra equations are

$$\frac{\partial(\rho\gamma)}{\partial t} + \frac{\partial(\rho u_j \gamma)}{\partial x_j} = P_\gamma - E_\gamma + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_r}{\sigma_f} \right) \frac{\partial \gamma}{\partial x_j} \right] \quad (2.31)$$

$$\frac{\partial(\rho \tilde{Re}_{\theta t})}{\partial t} + \frac{\partial(\rho u_j \tilde{Re}_{\theta t})}{\partial x_j} = P_{\theta t} - + \frac{\partial}{\partial x_j} \left[\sigma_{\theta t} (\mu + \mu_t) \frac{\partial \tilde{Re}_{\theta t}}{\partial x_j} \right] \quad (2.32)$$

The first equation is for the intermittency and tracks the upstream transition history. In laminar regions, γ is zero, and it is ramped to one when a transition criterion is satisfied. The intermittency is subsequently advected downstream to maintain turbulent flow. The second equation is for the transitional momentum-thickness Reynolds number, which controls the transition criteria used by the intermittency-equation source term. Both equations are phenomenological in nature.

The methodology was further refined by Langtry and Menter [20] in their development of the CFX-v-1.1 version of the γ - $Re_{\theta t}$ transitional SST model. The key to the Langtry-Menter model's predictive capabilities is that it uses a separately calibrated transition correlation that depends on the Thwaites parameter (for pressure gradient information) and the turbulence intensity at the edge of the boundary layer. Even though the Thwaites parameter depends on the integral boundary-layer momentum thickness, which is unavailable in most solvers, the Langtry-Menter model approximates the value with a locally defined transitional momentum-thickness Reynolds number, which is, in turn, a function of the Thwaites parameter. This closure method is exact only at the transition location, but this does not hinder the model's capabilities. The transition correlation is defined for a wide range of turbulence intensities and accounts for both natural and bypass transition. Additional correlations are included in the intermittency model to predict separation-induced transition. Examples of the predictive capabilities are included as Figs.

2-6 and 2-7, both of which are from Ref. [20]. The predictions in Fig. 2-6 show the T3-series of zero-pressure-gradient flat plate cases as well as the Schubauer-Klebanoff case [64]. It should be noted that the Schubauer-Klebanoff experiment was performed with a free-stream turbulence intensity of 0.03%; however, it was found that acoustic disturbances were primarily responsible for transition rather than free-stream turbulence. Accordingly, this experiment only weakly influenced the transition correlation that was described in Ref. [20], and the comparison with experiment was run using an effective free-stream turbulence intensity rather than the actual intensity [65]. The predictions included in Fig. 2-7 show the capability of the model to respond to varying pressure gradients and capture transition in favorable, adverse, and locally neutral pressure gradients, as well as in a separated boundary layer.

Development of this model was sponsored by ANSYS and is implemented in their solvers [22,23]. The model (and others of the same type) has also been implemented into OVERFLOW [26], CFL3D [25], Metacomp CFD++ [24], and CD-adapco Star-CCM+ [66].

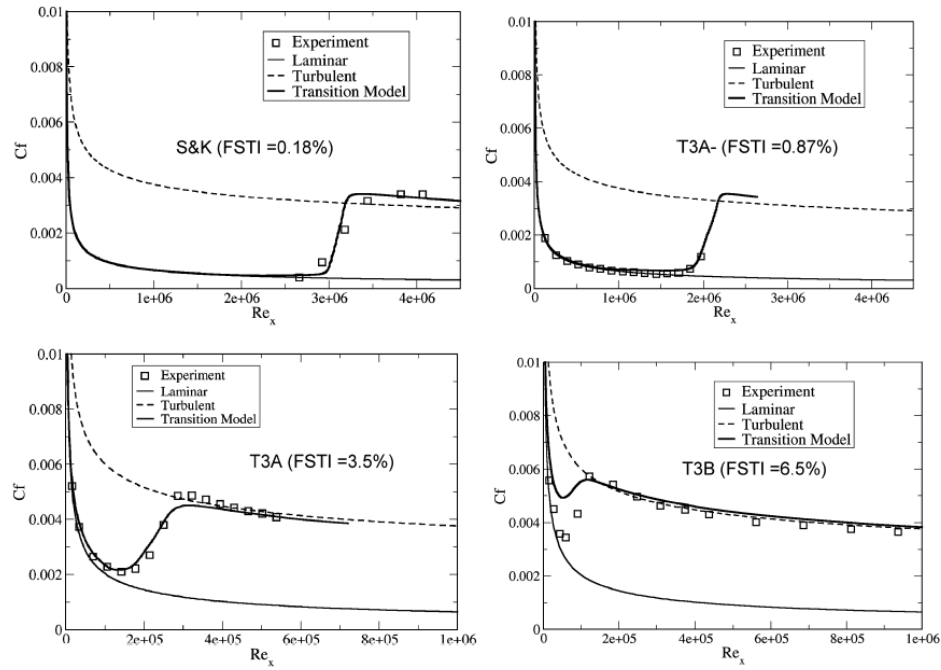


Figure 2-6. Skin-friction distributions predicted using the Langtry-Menter model for zero-pressure-gradient flat plate cases (from Ref [20]).

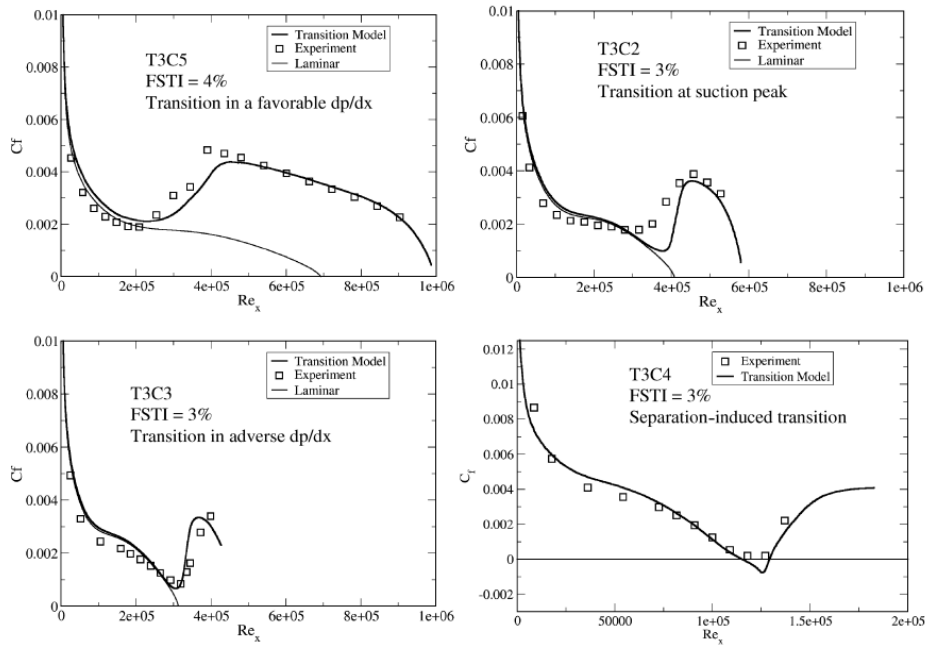


Figure 2-7. Skin-friction distributions predicted using the Langtry-Menter model for flat-plate cases with varying pressure gradients (from Ref. [20]).

2.3.4 Summary of Methods

The transition prediction methods described in this section are summarized in chronological order below in Table 2-1. Listed for each model is its type (i.e. the way the physics are modeled), whether or not the model accounts for upstream flow history, and the formulation of the transition criteria. These models represent a wide variety of transition prediction methods, and this summary is not intended to be an exhaustive listing.

Table 2-1. Summary of transition models

<u>Method</u>	<u>Year</u>	<u>Type</u>	<u>Upstream Flow History</u>	<u>Formulation</u>
Michel	1951	Local Correlation	No	$R_{\delta_2}-R_x$
Granville	1953	Local Correlation	Yes	$R_{\delta_2}-A_2$
e^N (Smith and Gamberoni, van Ingen)	1956	Linear Stability	Yes	$N(\omega,s)$
Van Driest and Blumer	1963	Local Correlation	No	$R_\delta-A_2$
Eppler	1980	Local Correlation	No	$R_{\delta_2}-H_{32}$
Abu-Ghannam and Shaw	1980	Local Correlation	No	$R_{\delta_2}-A_2$
Approximate Envelope (Drela and Giles)	1987	Linear Stability	Yes	$\tilde{n}(s)$
Walters-Leylek	2004	Laminar Kinetic Energy	Yes	k_L PDE
Langtry-Menter	2009	Local Correlation	No	Re_{θ_t} PDE

Chapter 3

Amplification Factor Transport Equation

As it is computationally prohibitive to directly resolve the relevant velocity and pressure perturbations involved in the laminar-turbulent transition process in RANS CFD simulations, the process must instead be modeled if the influence of transition is to be captured in the solution. A goal of any physical model is to include as much physics as possible in as few equations as possible. For this purpose, the development of a one-equation transition model based on linear-stability theory for use in CFD solvers is described here.

Computational fluid dynamics solvers are designed to solve equations that can be expressed in terms of a transport equation, either integral or differential. The conservative, differential form of such equations may be written as

$$\frac{\partial(\rho\phi)}{\partial t} + \frac{\partial(\rho u_j \phi)}{\partial x_j} = \text{Production} - \text{Destruction} + \text{Diffusion} \quad (3.1)$$

where ϕ is an arbitrary scalar quantity. The left-hand side of this equation is equivalent to the material derivative of the scalar quantity, where the momentum and velocity are brought inside the differential operators due to conservation of mass. The production and destruction terms on the right-hand side are often collectively referred as the source terms. The diffusion term may be linear or non-linear and typically involves both the molecular viscosity and the eddy viscosity.

The notion that a model is fully compatible with a CFD solver implies that it is of the form indicated by Eq. 3.1 and that all terms on the right-hand side can be evaluated using solution data available to a given node/cell and its immediate neighbors. In other words, evaluation of the production and destruction terms must not require any integration paths or global searches. The motivation for this is to preserve the ability of a CFD solver to be parallelized such that a given

subset of the computational domain requires only the corresponding subset of solution data to be available rather than the solution data for the entire domain.

While it is, in theory, possible to model the growth of specific instability frequencies using a transport equation, the frequency that first reaches critical amplification is never known *a priori*. Tracking multiple frequencies with the hope, but not the guarantee, that one of them is the critical frequency requires a separate transport equation for each one. Thus, such an approach quickly becomes cumbersome and loses its practicality. The approximate envelope method of Drela and Giles (Eq. 2.19) [32], however, provides an appealing alternative in its simplification of e^N theory to a single scalar variable and thereby serves as the basis for the amplification factor transport equation.

3.1 Advection Terms

Following the form of Eq. 3.1, the convection terms of an amplification factor transport equation come from substituting the envelope amplification factor $\tilde{n}$ for the arbitrary scalar ϕ . Thus,

$$\frac{\partial(\rho\tilde{n})}{\partial t} + \frac{\partial(\rho u_j \tilde{n})}{\partial x_j} = \text{Production} - \text{Destruction} + \text{Diffusion} \quad (3.2)$$

The left-hand side of this equation is, in effect, a conservative, Galilean-invariant streamwise derivative. Applying this concept allows Eq. 3.2 to be rewritten, without loss of generality, as

$$\frac{\partial(\rho\tilde{n})}{\partial t} + \frac{\partial(\rho u_j \tilde{n})}{\partial x_j} = \rho U \frac{d\tilde{n}}{ds} \quad (3.3)$$

where U is the local velocity magnitude (not to be confused with the edge velocity).

This is an incredibly useful result, as an expression for the streamwise derivative on the right-hand side already exists as part of the approximate envelope method; however, the physical

implications and limitations of this expression must be considered as the source and diffusion terms are defined.

3.2 Source Term

The approximate envelope method of Drela and Giles (Eq. 2.19) [32] is rooted in integral boundary-layer theory, in which the boundary-layer properties (such as displacement and momentum thickness) are effectively only defined on the surface. The growth rate of the amplification factor is dependent only on the integral parameters for the given streamwise location. This creates an immediate conflict with CFD methods. It is not important to modern CFD solvers whether a given point is in the boundary layer, as both the boundary layer and the outer, inviscid flow are governed by the same transport equations. It is possible for a model to determine *if* a given point is in a viscous region of the flow field, but it is impossible to determine *where* in the viscous region the point lies.

Therefore, the only way to evaluate the right-hand side of Eq. 3.3 is to locally estimate the shape factor, H_{12} , and the integral momentum thickness, δ_2 , as they cannot be exactly determined locally in a CFD solution. Estimating the integral shape factor begins with calculating a local shape factor or pressure-gradient parameter. The key qualities of a suitable local shape factor are that it must be 1) non-dimensional, 2) invariant of Reynolds number, and 3) meaningfully monotonic with respect to the integral shape factor for a family of laminar boundary-layer profiles. A local shape factor that exhibits these qualities is,

$$H_L = \frac{\Omega d}{U_e} \quad (3.4)$$

where Ω is the local vorticity magnitude, d is the distance to the nearest wall, and U_e is the velocity at the edge of the boundary layer. There is an immediate concern with using U_e as a

velocity scale, as it has a non-local interpretation; however, the velocity in the inviscid outer flow is directly related to the pressure in the absence of shockwaves, and boundary-layer theory assumes that the wall-normal pressure gradient is negligible in the boundary layer. Thus, the velocity U_e may be calculated using a Galilean-invariant form of the total energy for a calorically perfect ideal gas,

$$\left(\frac{\gamma}{\gamma-1}\right)\frac{P_\infty}{\rho_\infty} + \frac{U_\infty^2}{2} = \left(\frac{\gamma}{\gamma-1}\right)\frac{P}{\rho_e} + \frac{U_e^2}{2} \quad (3.5)$$

where, assuming isentropic outer flow,

$$\rho_e = \left[\frac{\rho_\infty^\gamma}{P_\infty} P \right]^{\frac{1}{\gamma}} \quad (3.6)$$

In the case of incompressible flow where the density is constant, Bernoulli's equation may be used instead,

$$P_\infty + \frac{1}{2}\rho_\infty U_\infty^2 = P + \frac{1}{2}\rho_\infty U_e^2 \quad (3.7)$$

For any non-dimensional velocity profile, the wall-normal distribution of H_L is independent of Reynolds number. The next question is whether any meaningful relationship can be developed between the local shape factor and the integral shape factor for a family of laminar profiles. For this, the Falkner-Skan (or Hartree) profiles are promising candidates, as they are used with much success in integral boundary-layer methods and for describing the growth of instabilities as a function of shape factor [58,67]. These profiles satisfy the third-order ordinary differential equation,

$$f''' + ff'' + \beta(1 - f'^2) = 0 \quad (3.8)$$

The representative distribution of H_L for favorable, neutral, adverse, and separation pressure gradients is shown in Fig. 3-1.

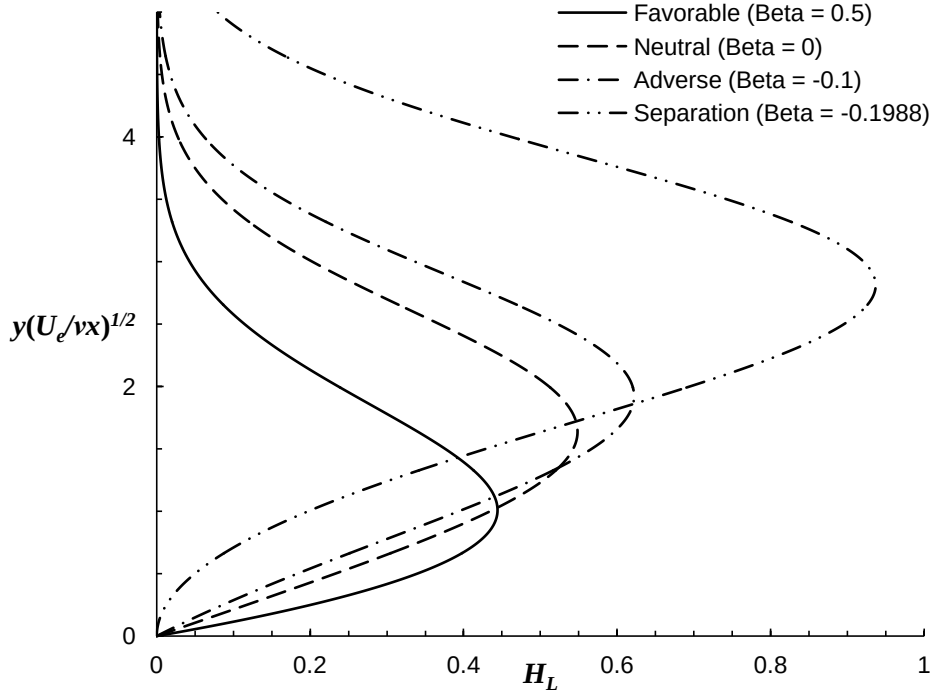


Figure 3-1. Distribution of local shape factor for Falkner-Skan profiles with different values of β .

For all types of profiles, the local shape factor is zero at the wall and in the outer-flow limit with a distinct peak at approximately 70% of the boundary-layer height. Moreover, the location of the peak is very close to the location of maximum vorticity Reynolds number, which is where van Driest and Blumer hypothesized transition to originate [61]. The peak values also yield the desired monotonic relationship as their magnitude increases with the integral shape factor. A further benefit of this behavior is that lower values of H_L correspond to pressure gradients that are more favorable and thus have slower growing amplification factors. In the context of locality in a CFD solution, this property helps prevent unwanted growth of the amplification factor away from the maxima of H_L .

Based on these arguments, a curve fit relating H_{12} to H_L for attached and separated Falkner-Skan profiles was developed for this dissertation and is given by

$$H_{12} = 13.9766H_L^4 - 22.9166H_L^3 + 13.7227H_L^2 - 1.0023H_L + 1.6778 \quad (3.9)$$

This curve fit is depicted graphically in Fig. 3-2. This equation applies everywhere in the flowfield despite being developed for a locus of H_L values. For numerical stability when calculating H_{12} , the local shape factor must be limited such that $H_L \geq 0.328$.

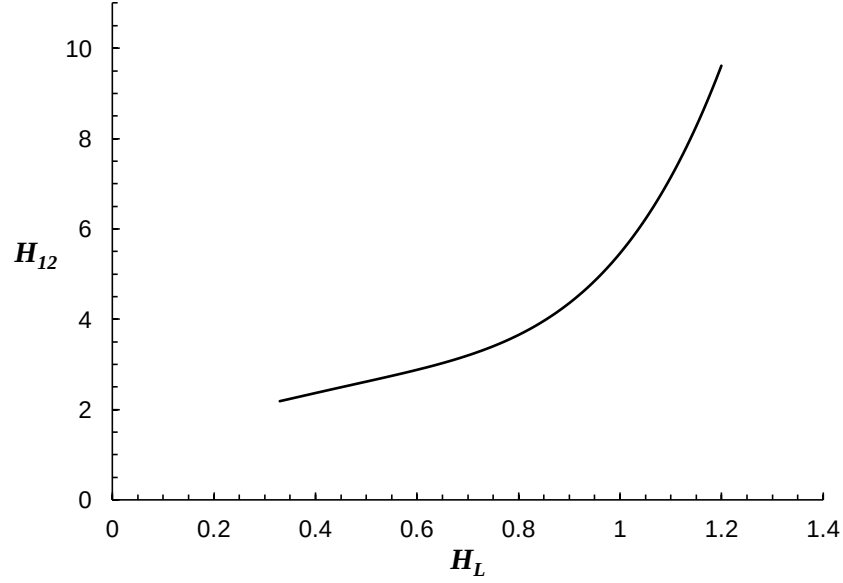


Figure 3-2. Variation of estimated integral, H_{12} shape factor with local, H_L shape factor.

The growth of the boundary layer is described by the dimensional quantity $U(dR_{\delta_2}/ds)$.

Using relations from Drela and Giles [32], this quantity may be written as

$$U \frac{dR_{\delta_2}}{ds} = \frac{U}{\delta_2} \frac{(1 + m(H_{12}))l(H_{12})}{2} \quad (3.10)$$

where

$$m(H_{12}) = \left(0.058 \frac{(H_{12} - 4)^2}{H_{12} - 1} - 0.068 \right) \frac{1}{l(H_{12})} \quad (3.11)$$

and

$$l(H_{12}) = \frac{6.54H_{12} - 14.07}{H_{12}^2} \quad (3.12)$$

The next step is to treat the presence of the momentum thickness, which is a non-local quantity and must be locally estimated. For a non-dimensional velocity profile, the wall distance is directly proportional to the momentum thickness for a given non-dimensional wall distance with a constant of proportionality that is independent of Reynolds number. By rewriting the influence of velocity and momentum thickness as

$$\frac{U}{\delta_2} = \Omega \left(\frac{U}{\Omega y} \right) \frac{y}{\delta_2} \quad (3.13)$$

two quantities that are functions of the shape factor become apparent. The first is the quantity in the parentheses, which is a local shape factor akin to H_L , and the second is the ratio of the wall distance to the momentum thickness. These may be treated similarly to the relation between H_L and H_{12} by taking the values of these quantities at the location where H_L is at its maximum. A combined curve fit, termed $D(H_{12})$, was generated based on this assumption for the two quantities and is given by

$$D(H_{12}) = \frac{0.0616H_{12}^2 + 0.2339H_{12} + 3.4298}{0.0047H_{12}^3 - 0.1056H_{12}^2 + 0.9350H_{12} - 1.2071} \quad (3.14)$$

This leads to a combined term defining the laminar boundary-layer growth as a function of the locally-estimated value of H_{12} . This term, F_{growth} , is defined as

$$F_{growth} = D(H_{12}) \frac{(1 + m(H_{12}))l(H_{12})}{2} \quad (3.15)$$

The proper dimensionality is introduced by multiplying F_{growth} by the local vorticity magnitude per Eq. 3.13.

The non-dimensional envelope amplification factor growth rate, $d\tilde{n}/dR_{\delta 2}$, and critical momentum-thickness Reynolds number, $R_{\delta 2,0}$, may be calculated using (from Ref. [68])

$$\frac{d\tilde{n}}{dR_{\delta 2}} = 0.028(H_{12} - 1) - 0.0345 \exp \left(- \left[\frac{3.87}{H_{12}} - 2.52 \right]^2 \right) \quad (3.16)$$

and

$$\log_{10}(R_{\delta 2,0}) = 0.7 \tanh\left(\frac{14}{H_{12}-1} - 9.24\right) + \frac{2.492}{(H_{12}-1)^{0.43}} + 0.62 \quad (3.17)$$

The influence of the critical momentum-thickness Reynolds number on the source term may be incorporated into it through the use of a switching function; however, this (or any other method) requires an estimate of the momentum-thickness Reynolds number, as the exact value cannot be determined locally. For a given non-dimensional velocity profile, any local Reynolds number is directly proportional to the momentum-thickness Reynolds number where the constant of proportionality is a function only of the non-dimensional distance to the wall. A possible candidate for this local Reynolds number is the vorticity Reynolds number, but its quality of having a peak value actually compounds the errors introduced by using H_L , which also exhibits a peak value. A more suitable candidate is a local wall-distance-based Reynolds number, R_y ,

$$R_y = \frac{\rho U d}{\mu} \quad (3.18)$$

which grows with wall distance. This leads to a definition of a switching function, F_{crit} , to determine if the boundary layer is capable of containing unstable modes. This function is defined as

$$F_{crit} = \begin{cases} 0, & R_y < R_{y,0} \\ 1, & R_y \geq R_{y,0} \end{cases} \quad (3.19)$$

where

$$R_{y,0} = k_y R_{\delta 2,0} \quad (3.20)$$

and

$$k_y = -0.00315 H_{12}^3 + 0.0986 H_{12}^2 - 0.242 H_{12} + 3.739 \quad (3.21)$$

The function k_y is based on the Falkner-Skan profiles and describes the proportionality of the two Reynolds numbers, R_y and $R_{\delta 2}$, where H_L is maximum in the profile.

Combined, the source term of the amplification factor transport equation is

$$Production - Destruction = \rho \Omega F_{crit} F_{growth} \frac{d\tilde{n}}{dR_{\delta 2}} \quad (3.22)$$

Preliminary studies with the amplification factor transport equation used the strain-rate magnitude rather than the vorticity magnitude [69]. This version produced the correct behavior in the boundary layer, but it also permitted the amplification factor to grow outside of the boundary layer near the stagnation point, which does not make physical sense. Allmaras [70] suggested using the vorticity magnitude instead to produce a cleaner, more physical distribution of the transported amplification factor.

3.3 Diffusion Term

Direct transformation of the approximate envelope method into a transport equation does not yield any terms resembling a diffusion term. The reasons for this are both physical and mathematical. First and foremost, the physical envelope amplification factor is a global measure of the instabilities and is constant across the boundary layer for a given streamwise station. The local value of the amplification factor is determined solely from upstream information with no influence of the downstream values for a given pressure field. There is no physical relevance regarding the “total” amplification in the boundary layer, which would be the amplification factor integrated with respect to wall distance. Outside of the boundary layer, the amplification factor is not defined. Thus, there are no physical conservation laws that govern a redistribution of the amplification factor across streamlines due to viscosity. Mathematically and numerically, no diffusion term is necessary in the approximate envelope method because the governing equations are one-dimensional along a single streamline and completely parabolic in nature.

Nevertheless, the qualities of the governing equation that permit it to be non-diffusive are lost when the approximate envelope method is cast as a transport equation. Most importantly, the effective one-dimensionality is lost since the boundary layer is resolved in the wall-normal direction. Thus, the boundary layer can no longer be regarded as existing along a single streamline. As the assumptions used above in generating the source term lead to non-constant growth of the transported amplification factor across the boundary layer, the amplification itself will therefore be non-constant as well. If the location of maximum amplification factor growth, which corresponds to maximum H_L , occurs on the same streamline for all downstream locations, this would not be a concern. However, this is generally not the case. It therefore becomes necessary to redistribute the transported amplification factors across streamlines so downstream growth can occur for the maximum amplification factor in the boundary layer at each streamwise location.

It is proposed that a gradient diffusion term of the form

$$Diffusion = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_n} \right) \frac{\partial \tilde{n}}{\partial x_j} \right] \quad (3.23)$$

should be sufficient for the amplification factor transport. This type of diffusion is very common for transport equations and appears in the Navier-Stokes equations. The use of the molecular viscosity is necessary, as diffusion is required when the boundary layer is laminar and the eddy viscosity is zero. The eddy viscosity is included in the diffusion term to aid the numerical behavior of the transport equation. The calibration constant σ_n is assumed to be of order unity based on existing transport-equation-based models [14,15,20]. As will be confirmed by the results shown in Chapter 5, setting $\sigma_n = 1.0$ produces satisfactory behavior.

3.4 Combining the Terms

Combining the convection, source, and diffusion terms developed and discussed above gives the amplification factor transport equation:

$$\frac{\partial(\rho\tilde{n})}{\partial t} + \frac{\partial(\rho u_j \tilde{n})}{\partial x_j} = \rho \Omega F_{growth} F_{crit} \frac{d\tilde{n}}{dR_{\delta 2}} + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_n} \right) \frac{\partial \tilde{n}}{\partial x_j} \right] \quad (3.24)$$

The advection and diffusion terms of this equation are of the same form as those already present in CFD solvers and the same discretization schemes can be applied. Evaluating the source term of this equation for a given point in the flow field begins by calculating the boundary-layer edge velocity U_e using Eqs. 3.5-3.7. With this value, the distance to the nearest wall, and the local vorticity magnitude, the H_L shape factor is evaluated (Eq. 3.4). Subsequently, H_{12} is calculated with Eq. 3.9. With this value of H_{12} , the $D(H_{12})$, $m(H_{12})$, and $l(H_{12})$ quantities can be evaluated using Eqs. 3.14, 3.11, and 3.12, respectively. These comprise the F_{growth} function, which is defined by Eq. 3.15. The envelope growth rate, $d\tilde{n}/dR_{\delta 2}$, comes from evaluating Eq. 3.16 with the local value of H_{12} . Determining the local value of F_{crit} begins by using H_{12} to calculate $R_{\delta 2,0}$ from Eq. 3.17 and k_y from Eq. 3.21. The values of R_y and $R_{y,0}$ are calculated using Eqs. 3.18 and 3.20, respectively, and are then substituted into Eq. 3.19.

3.5 Boundary Conditions

The final data necessary to solve Eq. 3.24 are the boundary conditions in the free stream and on solid walls. In the free stream, disturbances are assumed to be unamplified. Thus, the free-stream amplification factor should be set to zero. This allows the flow to attach with no amplification of disturbances and for the free-stream value to persist until the critical momentum-thickness Reynolds number is met and the amplification factor is allowed to grow.

Near solid walls, the nature of the physical amplification factor must be considered. The amplification factor is interpreted as being constant across the boundary layer, as it is a global measure. The value at the wall therefore takes on whatever value is present in the boundary layer, leading to a zero-normal-gradient condition.

Stated mathematically, these two boundary conditions are

$$\tilde{n}_{\infty} = 0 \quad (\text{Free-stream condition}) \quad (3.25)$$

and

$$\frac{\partial \tilde{n}}{\partial y} = 0 \quad (\text{Solid-wall condition}) \quad (3.26)$$

where y is the local wall-normal coordinate direction.

Chapter 4

Interaction with Eddy-Viscosity Models

Being able to predict the growth of instabilities is necessary, but not sufficient for effective transition modeling in the RANS CFD environment. It is desirable for the amplification factor transport equation to be applicable to both one- and two-equation turbulence models with the same added computational expense. Two specific criteria need to be met for an effective interaction of the transition prediction with the eddy viscosity model: true laminar flow upstream of the transition location must be maintained as a stable solution, and the fully turbulent behavior needs to be reproduced downstream of transition.

In the following discussion, two eddy-viscosity models will be considered: the one-equation Spalart-Allmaras model [33] and the two-equation Menter Shear Stress Transport model [21]. Although there are a wide variety of other eddy-viscosity models, these two models are the “workhorses” of industrial CFD analysis, and taking advantage of their behavior in a combined transition/turbulence framework is of great practical interest. Therefore, any transition model of industrial interest must be compatible with these eddy-viscosity models in a unified and consistent manner.

4.1 Eddy-Viscosity Models

4.1.1 Spalart-Allmaras Model

The Spalart-Allmaras (SA) model is a transport equation for the eddy viscosity that is based on empirical arguments with Galilean-invariant quantities of the proper dimensionality

[33]. The terms of the model were developed hierarchically, starting with a “basic” high-Reynolds number version of the model that yields the production and non-linear diffusion terms. An additional near-wall term was introduced that is based on the law-of-the-wall (or logarithmic-layer) behaviors of the turbulent eddy viscosity and the vorticity. Since the log-layer still behaves as a high-Reynolds number region, the behavior of the viscous sublayer is introduced by redefining the model in terms of a modified eddy viscosity that retains the log-layer behavior all the way to the wall. Accordingly, a damping function is necessary to convert the modified eddy viscosity to the actual eddy viscosity. The standard form of the Spalart-Allmaras model is given as

$$\frac{D\tilde{\nu}}{Dt} = c_{b1}\tilde{S}\tilde{\nu}(1 - f_{t2}) - \left[c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2} \right] \left(\frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right] \quad (4.1)$$

The eddy viscosity is defined as

$$\mu_t = \rho \tilde{\nu} f_{v1} \quad (4.2)$$

where

$$f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3} \quad (4.3)$$

and

$$\chi = \frac{\tilde{\nu}}{\nu} \quad (4.4)$$

The other functional relationships are

$$\tilde{S} = \Omega + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v2} \quad (4.5)$$

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad (4.6)$$

$$f_w = g \left[\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right]^{\frac{1}{6}} \quad (4.7)$$

$$g = r + c_{w2} (r^6 - r) \quad (4.8)$$

$$r = \min \left[\frac{\tilde{\nu}}{\tilde{S} \kappa^2 d^2}, 10 \right] \quad (4.9)$$

The f_{t2} function is defined as

$$f_{t2} = c_{t3} \exp(-c_{t4} \chi^2) \quad (4.10)$$

and is used to allow user-specified transition locations with the model (more will be discussed on this function later in this chapter). The calibration constants for the model are $c_{b1} = 0.1355$, $c_{b2} = 0.622$, $\sigma = 2/3$, $\kappa = 0.41$, $c_{w2} = 0.3$, $c_{w3} = 2$, $c_{v1} = 7.1$, $c_{t3} = 1.2$, $c_{t4} = 0.5$. The c_{w1} constant is calculated as

$$c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma} \quad (4.11)$$

to enforce the proper value of the von Karman constant, $\kappa = 0.41$, in the logarithmic layer.

4.1.2 Menter SST Model

The two-equation, $k-\omega$ Shear Stress Transport (SST) model was developed by Menter [21] by blending the behaviors of the Jones-Launder $k-\varepsilon$ [71] and the Wilcox $k-\omega$ [72] two-equation models. The motivation for this stemmed from two key observations. The first is that the $k-\varepsilon$ model performs very well in the outer part of the boundary layer and in the outer flow, but has very poor near-wall behavior that necessitates the use of wall functions. The second is that the Wilcox $k-\omega$ model is better behaved near the wall, but that its behavior depends very strongly on the free-stream value of the specific dissipation rate. The SST model takes advantage of these behaviors by blending the two models such that the Wilcox model is active near the wall and that

the Jones-Launder model (transformed from k - ε to k - ω) is active elsewhere. The two transport equations of the model are

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = \mu_t S^2 - \beta^* \rho \omega k + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right] \quad (4.12)$$

$$\frac{\partial(\rho \omega)}{\partial t} + \frac{\partial(\rho u_j \omega)}{\partial x_j} = \gamma \rho S^2 - \beta \rho \omega^2 + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_\omega \mu_t) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \frac{\rho \sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (4.13)$$

The calibration constants γ , β , σ_k , and σ_ω are a blending of the inner (1) and outer (2) constants using the F_1 blending function as

$$\phi = F_1 \phi_1 + (1 - F_1) \phi_2 \quad (4.14)$$

where

$$F_1 = \tanh\left(\arg_1^4\right) \quad (4.15)$$

$$\arg_1 = \min \left[\max \left(\frac{\sqrt{k}}{\beta^* \omega d}, \frac{500\nu}{d^2 \omega} \right), \frac{4\rho \sigma_{\omega 2} k}{CD_{k\omega} d^2} \right] \quad (4.16)$$

$$CD_{k\omega} = \max \left(2 \frac{\rho \sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j}, 10^{-20} \right) \quad (4.17)$$

The eddy viscosity is calculated as

$$\mu_t = \frac{a_1 \rho k}{\max(a_1 \omega, \Omega F_2)} \quad (4.18)$$

This differs from the standard eddy-viscosity definitions for two-equation models by the use of a vorticity-based “realizability” constraint, which comes from Bradshaw’s relationship that the principle turbulent stress is proportional to the turbulent kinetic energy in certain regions of the flow field [73]. The F_2 function is used to suppress this constraint in other regions of the flow field (such as wakes) where this proportionality does not hold and is given as

$$F_2 = \tanh\left(\arg_2^2\right) \quad (4.19)$$

$$arg_2 = \max\left(\frac{2\sqrt{k}}{\beta^* \omega d}, \frac{500\nu}{d^2 \omega}\right) \quad (4.20)$$

The calibration constants for this model are $\beta^* = 0.09$, $\sigma_{k1} = 0.85$, $\sigma_{k2} = 1.0$, $\sigma_{w1} = 0.5$, $\sigma_{w2} = 0.856$, $\beta_1 = 0.075$, $\beta_2 = 0.0828$, and $a_1 = 0.31$. The constant γ is defined as

$$\gamma = \frac{\beta}{\beta^*} - \frac{\sigma_\omega \kappa^2}{\sqrt{\beta^*}} \quad (4.21)$$

to enforce the proper value of the von Karman constant, which is $\kappa = 0.41$.

4.2 Enforcing Laminar Flow

4.2.1 Zonal Approach

The most conceptually simple manner to implement laminar-turbulent transition is to define zones in the computational domain where the turbulence model is suppressed and where it is active. While this is very effective and has been used with great success [6,7,11,74], it requires global (non-local) knowledge of the computational domain and flow field to implement. For cases where the user directly specifies where transition is to occur, this is not a detriment. Knowing the transition locations *a priori*, however, is impractical. In these implementations, transition is automatically predicted in an effective “post-processing” step that occurs between iterations. The transition calculations in these methods already require non-local solution information, so a zonal approach to enforcing laminar flow is not inconsistent. An algebraic intermittency function may also be employed to blend between the laminar and turbulent regions; however, the parameters of the intermittency function in such applications are still non-local in nature.

4.2.2 Intermittency Transport Equation

One possible CFD-compatible method for determining whether the flow is locally laminar or turbulent is through the introduction of an intermittency transport equation. True intermittency is a physical measure of the fraction of time for which a given point in the flow field exhibits turbulent behavior over a given time interval [39]. In the laminar boundary layer, the intermittency is zero, and once the boundary layer has completely transitioned, the intermittency is one. In the transition region, where instabilities become of significant amplitude and Emmons spots [38] begin to form, the intermittency blends from zero to one in a manner such as is shown in Fig. 4-1 (from Ref. [64]). The distribution of intermittency is well-characterized as a Gaussian cumulative distribution function that is a function of downstream distance in the boundary layer. As this is essentially a distributed Heaviside function, intermittency may therefore be regarded as a pseudo-binary characterization of the flow turbulence.

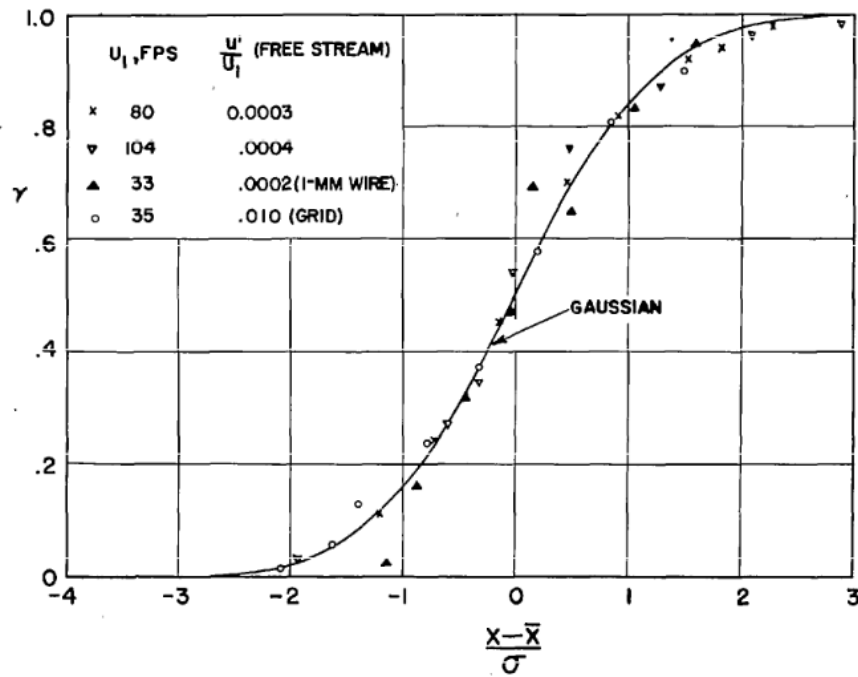


Figure 4-1. Measured intermittency factor, γ , in transition regions with Gaussian integral curve matched at $\gamma = 0.5$ (from Ref. [64]).

In the context of the present work, intermittency is not to be confused with phenomena such as wall damping where the eddy viscosity goes to zero in the boundary layer. Such near-wall behavior is accounted for in any practical, fully turbulent eddy-viscosity model and thus would be described as having an intermittency of one. Various transport models exist to describe the intermittency distribution in the transition region and typically require knowledge of the free-stream turbulence and how far a point is from the prescribed transition onset location. These models may be placed in two categories: ones that modify the mean-flow equation, and ones that modify the turbulence-model equations.

Models of the first category include those of Steelant and Dick [75] and Suzen and Huang [76]. The Steelant-Dick model produces a nearly-uniform intermittency distribution across the turbulent boundary layer [75], which is then applied to conditioned versions of the mean-flow equations. Although the intermittency distribution allows the desired qualities of the corresponding turbulence model to be realized, the use of conditioned mean-flow equations makes the model undesirable for general-purpose CFD codes. The Suzen-Huang model sought to improve upon the Steelant-Dick model by predicting realistic cross-stream intermittency distributions [76], even though this degrades the model's quality as these effects are already accounted for in the turbulence model calibrations. Furthermore, the interaction between the intermittency and the mean flow is much simpler than the Steelant-Dick model in that the intermittency multiplies the eddy viscosity as

$$\mu_{t,eff} = \gamma \mu_t \quad (4.22)$$

where $\mu_{t,eff}$ is used in the mean-flow equations and μ_t is that predicted by the eddy-viscosity model. While it is less rigorous than the conditioning used by Steelant and Dick, it heuristically represents a Reynolds averaging of an intermittent eddy viscosity.

The second category is typified by the approach proposed by Menter [17,19,20] in which the production and destruction terms of the turbulent kinetic energy equation are modified to suppress eddy viscosity production in laminar regions. These modifications are

$$\tilde{P}_k = \gamma_{eff} P_{k,orig} \quad (4.23)$$

$$\tilde{D}_k = \gamma_{lim} D_{k,orig} \quad (4.24)$$

where

$$\gamma_{lim} = \min \left[\max(\gamma_{eff}, 0.1), 1.0 \right] \quad (4.25)$$

The specific dissipation equation and eddy viscosity definition remain unaltered. Laminar flow is achieved by Eqs. 4.23-4.24 predicting near-zero turbulent kinetic energy when the intermittency is nearly zero. This approach is used with the transitional SST model of Langtry and Menter [20].

The original version of Menter's intermittency equation was not without problems as it tended to over-produce intermittency near the stagnation point, thereby contaminating the rest of the boundary layer. It also had difficulty reproducing the desired transition length [16]. Langtry improved this intermittency equation by addressing these specific deficiencies, ultimately giving it the form that is now typically used with this approach. Even with the improvements, a notable deficiency remains in that the intermittency is excessively destroyed in the viscous sublayer. This behavior necessitates the attenuation of the destruction term by the intermittency, as seen in Eq. 4.25, to balance the reduced production in the viscous sublayer and approximately preserve the proper near-wall behavior. Even though the approach was developed specifically for Menter's $k-\omega$ SST model [21] (and, by extension, the Wilcox 1988 $k-\omega$ model), it is applicable to all eddy-viscosity models that use a turbulent kinetic energy equation.

Although intermittency/turbulence models in this category may achieve the same behavior as those in the first category, modifying the turbulence model equations is a philosophically different interpretation of the physics. Menter's method addresses the deficiencies

of the Steelant-Dick model by not requiring the mean-flow equations to be conditioned and it also addresses the deficiencies of the Suzen-Huang model by better accounting for the influence of near-wall intermittency behavior. It may also be inferred from the proliferation of Menter's approach in commercial and research CFD solvers [22-26,66] that it is the intermittency-based transition method preferred by the community, likely due to its completely local nature being readily compatible with RANS solvers.

4.2.3 The f_{t2} Function

An alternative to the intermittency transport equation is the algebraic f_{t2} function developed by Spalart and Allmaras for their one-equation turbulence model [33]. The f_{t2} function, given above as Eq. 4.10 as part of the standard version of the turbulence model, is a suppression function used to ensure that a stable, true laminar solution is possible and primarily modifies the production term of the model. This leads to negative eddy-viscosity production when the modified eddy viscosity satisfies

$$\frac{\tilde{\nu}}{\nu} \leq \left[\frac{1}{c_{t4}} \ln(c_{t3}) \right]^{\frac{1}{2}} \quad (4.26)$$

The f_{t2} function also appears in the destruction term of the model (Eq. 4.1) to analytically balance the modified production in the viscous sublayer when the boundary layer is fully turbulent.

In the case of user-specified transition locations, an additional term of the form $f_{t1}\Delta U^2$ can be added to Eq. 4.1. The f_{t1} function permits a sudden increase in the eddy viscosity near a user-specified wall location and allows the model to overcome the basin of attraction to laminar flow created by f_{t2} . The fully turbulent variant of the Spalart-Allmaras turbulence model (SA-noft2) omits all terms containing f_{t1} and f_{t2} , as they are intended to be passive in such cases.

4.2.4 Mathematical Stability of the Laminar Flow Solutions

With the intermittency transport equation and f_{t2} function, it is necessary to consider whether laminar flow ($\mu_t \rightarrow 0$) is both a valid and a stable solution to the eddy-viscosity models. Rumsey investigated the pseudo-transition behavior of the standard versions of the Spalart-Allmaras and Menter SST models that appears as grids are refined [77].

Even though intentional transition modeling was not considered in Rumsey's investigation, the f_{t2} function appears in the standard version of the Spalart-Allmaras model and was thus included in the analysis [77]. By neglecting the influence of diffusion terms, it was found that zero eddy viscosity satisfies source-term equilibrium, meaning that laminar flow is a valid solution. By fully differentiating each term with respect to the modified eddy viscosity and substituting the values of the functions in the zero-eddy-viscosity limit, it is found that

$$\left. \frac{\partial R}{\partial \tilde{\nu}} \right|_{\tilde{\nu}=0} \rightarrow \Omega_{C_{b1}} (1 - c_{t3}) \quad (4.27)$$

where R are the source terms of the model. As c_{t3} is greater than zero, zero eddy viscosity is not only a valid solution but it is also a stable solution in that small perturbations return to the laminar solution rather than continuing to grow.

Similarly, the analysis of the Menter SST model begins by neglecting the diffusion terms of both the turbulent kinetic energy and specific dissipation equations. By again satisfying source-term equilibrium, it was found that laminar flow is a valid solution and that the eigenvalues of the source terms represent a saddle point in the limit of zero eddy viscosity.

Following the same procedure as Rumsey [77], the source terms of the intermittency-sensitized SST model in the zero-eddy-viscosity limit may be rewritten as

$$R_k = \gamma_{eff} a_1 k S - \gamma_{lim} \beta^* k \omega \quad (4.28)$$

$$R_\omega = \gamma_1 S^2 - \beta_1 \omega^2 \quad (4.29)$$

which may be regarded as simplified forms of the model for which a steady-state solution is the homogeneous form of the equations. From inspection, the laminar-flow solution of $k = 0$ satisfies the homogenous turbulent kinetic energy equation, implying that it is a valid steady-state solution. The specific dissipation equation has a homogenous solution of $\omega = S(\gamma_1/\beta_1)^{1/2}$. The Jacobian matrix of this system of equations is

$$J = \begin{bmatrix} \gamma_{\text{eff}} a_1 S - \gamma_{\text{lim}} \beta^* \omega & -\gamma_{\text{lim}} \beta^* k \\ 0 & -2\beta_1 \omega \end{bmatrix} \quad (4.30)$$

Substituting the laminar-flow values of k and ω yields two negative eigenvalues to the Jacobian matrix. Thus, the laminar-flow solution is a stable one in the presence of intermittency sensitization, which contrasts with the saddle-point stability of the fully turbulent model [77].

For models where laminar flow is enforced by Eq. 4.22, this analysis is not applicable as a valid laminar flow solution to the eddy-viscosity model is not a necessary condition for laminar flow to exist in the solution.

4.3 Unifying the Interaction with Eddy-Viscosity Models

At this point, it is apparent that there is, at present, a discrepancy in the capabilities of modeling transition available for one-equation and two-equation turbulence models. In the case of two-equation models, an intermittency transport equation seems to be required, whereas an algebraic method is available for one-equation models. Therefore, the primary questions that must be addressed in developing a method to interface the amplification-factor transition model with popular eddy-viscosity models are: how should an intermittency transport equation interact with a one-equation turbulence model, and is there an analog to f_{t2} that is applicable to two-equation models? In the following discussion, the specific turbulence models being considered are the Spalart-Allmaras one-equation model [33] and the Menter SST two-equation model [21].

Applying the intermittency transport equation to the Spalart-Allmaras turbulence model in a manner consistent with the transition Menter SST model requires behavioral connections to be made between the terms of both models. An ad hoc connection was used by Medida and Baeder by treating the source terms of the two models as being directly analogous [78]; however, their approach does not allow for a stable laminar solution. Instead, a procedure used by Menter [79] to transform the standard k - ε equations into a one-equation model will be applied to the transition-sensitized Menter SST model.

The Spalart-Allmaras model cannot be derived directly from any two-equation models, but it nevertheless has the same general form as other one-equation models that are derived from the k - ε equations. Even though the Menter SST model is a k - ω model rather than a k - ε model, the models are strongly related, and the same transformation procedures may be applied. The process begins with the turbulent kinetic energy and specific dissipation equations of the transitional SST model. The transitional turbulent kinetic energy equation is

$$\frac{Dk}{Dt} = \gamma_{\text{eff}} \nu_t S^2 - \gamma_{\text{lim}} \beta^* k \omega + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial x_j} \right] \quad (4.31)$$

while the specific dissipation is unaltered from Eq. 4.13. The eddy viscosity and the calibration constants are still defined in the same way as the fully turbulent form of the SST model. The transformation begins with Bradshaw's approximation for equilibrium flows [73], which states

$$\mu_t S = a_1 k \quad (4.32)$$

where the constant a_1 is the square root of β^* from Eq. 4.31. While the definition of the eddy viscosity in the SST model is more complex than for other two-equation models [71,72], it does not hinder the transformation as Eq. 4.32 is the realizability constraint used in the eddy viscosity (Eq. 4.18). Such an assumption is reasonable in the law-of-the-wall region of the turbulent boundary layer where $y^+ \rightarrow \infty$, but not in the viscous sublayer where $y^+ \rightarrow 0$. However, the Spalart-Allmaras model uses a modified eddy viscosity designed to recover the law-of-the-wall

behavior the entire way to the wall. Since the purpose of this exercise is to understand how the intermittency should be applied to a one-equation model, the assumption used in the transformation remains valid.

Under this transformation, the turbulent quantities k and ω may be expressed in terms of the eddy viscosity and strain rate as

$$k = \frac{\nu_t S}{a_1} \quad (4.33)$$

$$\omega = \frac{S}{a_1} \quad (4.34)$$

Thus, a transport equation for the eddy viscosity may be written as

$$\frac{D\nu_t}{Dt} = \frac{1}{\omega} \frac{Dk}{Dt} - \frac{k}{\omega^2} \frac{D\omega}{Dt} = \frac{a_1}{S} \left(\frac{Dk}{Dt} - \nu_t \frac{D\omega}{Dt} \right) \quad (4.35)$$

Substituting Eqs. 4.33 and 4.34 into Eqs. 4.13 and 4.31 yield

$$\frac{Dk}{Dt} = \nu_t S^2 \left(\gamma_{eff} - \gamma_{lim} \frac{\beta^*}{a_1^2} \right) + \frac{1}{a_1} \frac{\partial}{\partial x_j} \left[\left(\nu + \sigma_k \nu_t \right) \left(S \frac{\partial \nu_t}{\partial x_j} + \nu_t \frac{\partial S}{\partial x_j} \right) \right] \quad (4.36)$$

$$\frac{D\omega}{Dt} = S^2 \left(\alpha - \frac{\beta}{a_1^2} \right) + \frac{1}{a_1} \frac{\partial}{\partial x_j} \left[\left(\nu + \sigma_\omega \nu_t \right) \frac{\partial S}{\partial x_j} \right] + \frac{2(1-F_1)\sigma_{w2}}{a_1 S} \left(S \frac{\partial \nu_t}{\partial x_j} + \nu_t \frac{\partial S}{\partial x_j} \right) \frac{\partial S}{\partial x_j} \quad (4.37)$$

respectively. It is apparent that, in the case of fully turbulent flow where the intermittency is exactly one, the source terms of the turbulent kinetic energy equation would exactly cancel in the law-of-the-wall region under Bradshaw's approximation. Special consideration must also be given to the diffusion coefficients that are subject to differentiation. Although they vary spatially, the variation is zonal in that they are constant in the boundary layer and near wake and constant in the outer flow. Thus, for the purpose of understanding how the terms translate to a one-equation model, the diffusion coefficients may be regarded as being locally constant. Further expanding the diffusion terms in Eqs. 4.36 and 4.37 gives

$$\begin{aligned} \frac{Dk}{Dt} = & \nu_t S^2 \left(\gamma_{eff} - \gamma_{lim} \frac{\beta^*}{a_1^2} \right) \\ & + \frac{1}{a_1} \left[\sigma_k \frac{\partial \nu_t}{\partial x_j} \left(S \frac{\partial \nu_t}{\partial x_j} + \nu_t \frac{\partial S}{\partial x_j} \right) + (\nu + \sigma_k \nu_t) \left(S \frac{\partial^2 \nu_t}{\partial x_j^2} + \nu_t \frac{\partial^2 S}{\partial x_j^2} + 2 \frac{\partial S}{\partial x_j} \frac{\partial \nu_t}{\partial x_j} \right) \right] \end{aligned} \quad (4.38)$$

$$\begin{aligned} \frac{D\omega}{Dt} = & S^2 \left(\alpha - \frac{\beta}{a_1^2} \right) + \frac{1}{a_1} \left[\sigma_\omega \frac{\partial \nu_t}{\partial x_j} \frac{\partial S}{\partial x_j} + (\nu + \sigma_\omega \nu_t) \frac{\partial^2 S}{\partial x_j^2} \right] \\ & + \frac{2(1-F_1)\sigma_{\omega 2}}{a_1} \left[\frac{\partial \nu_t}{\partial x_j} \frac{\partial S}{\partial x_j} + \frac{\nu_t}{S} \left(\frac{\partial S}{\partial x_j} \right)^2 \right] \end{aligned} \quad (4.39)$$

Next, substituting Eqs. 4.38 and 4.39 into Eq. 4.35 and regrouping gives the general form of a one-equation turbulence model [79],

$$\begin{aligned} \frac{D\nu_t}{Dt} = & a_1 \left(\frac{\beta}{a_1^2} - \alpha + \gamma_{eff} - \gamma_{lim} \frac{\beta^*}{a_1^2} \right) S \nu_t - 2(1-F_1)\sigma_{\omega 2} \left(\frac{\partial S}{\partial x_j} \right)^2 \frac{\nu_t^2}{S^2} + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_t) \frac{\partial \nu_t}{\partial x_j} \right] \\ & + \frac{\nu + (3\sigma_k - \sigma_\omega - 2(1-F_1)\sigma_{\omega 2})\nu_t}{S} \frac{\partial \nu_t}{\partial x_j} \frac{\partial S}{\partial x_j} + \frac{(\sigma_k - \sigma_\omega)}{S} \frac{\partial^2 S}{\partial x_j^2} \nu_t^2 \end{aligned} \quad (4.40)$$

The first and third terms of the right-hand side are the production and diffusion terms, respectively, and have direct analogs in the Spalart-Allmaras model. The production term takes the same general form in both models. The diffusion in the transformed SST model is linear whereas the Spalart-Allmaras utilizes non-linear diffusion. The second and fourth terms are the destruction contributions for the two branches of the SST model. The second term of Eq. 4.40 is active only for the $k-\varepsilon$ branch of the model, which applies in the outer portion and outside of the boundary layer where $F_1 = 0$. This term does not explicitly depend on the wall distance, but instead features the von Karman length scale. In the logarithmic layer of the turbulent boundary layer, this length scale is directly proportional to the wall distance; however, the $k-\varepsilon$ form of the destruction is not active in the lower portion of the boundary layer. Rather, the fourth term is the destruction term in the boundary layer when the $k-\omega$ branch of the model is active. This is otherwise negligible in the $k-\varepsilon$ branch since σ_k and σ_ω are approximately equal [79].

The intermittency appears only in the production term of the transformed transitional SST model. Relating this to the Spalart-Allmaras model, the intermittency should act to attenuate the c_{b1} constant such that a stable laminar solution is possible. The intermittency-sensitized form of the Spalart-Allmaras model that is consistent with the intermittency-sensitized SST model is

$$\frac{D\tilde{\nu}}{Dt} = \left[c_{b1} + a_1 (\gamma_{eff} - \gamma_{lim}) \right] \tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right] \quad (4.41)$$

The influence of the intermittency on the production term is nearly identical to the influence of the f_{t2} function. This result is very encouraging as the f_{t2} function behaves very similarly to the converged Langtry-Menter intermittency transport equation. However, several problems quickly become evident. First, the value of γ_{lim} in laminar flow is 0.1 with the transitional SST model. In the Spalart-Allmaras model, this is not sufficient to allow a stable laminar solution. Instead, a limiting value of at least 0.472 is necessary, and to match the strength of the attractor with that of f_{t2} , the limiting value should be 0.562. Second, there is no intermittency present in the destruction term to balance the near-wall behavior of the intermittency. Therefore, achieving consistent intermittency sensitization between the SST and Spalart-Allmaras models is not possible without sacrificing the near-wall behavior of the Spalart-Allmaras model. As this is unnecessarily detrimental to the performance of the model, the following intermittency sensitization should be used with the Spalart-Allmaras model:

$$\begin{aligned} \frac{D\tilde{\nu}}{Dt} = & \left[c_{b1} + a_1 (\gamma_{eff} - \gamma_{lim}) \right] \tilde{S}\tilde{\nu} - \left[c_{w1} f_w + \frac{a_1 (\gamma_{eff} - \gamma_{lim})}{\kappa^2} \right] \left(\frac{\tilde{\nu}}{d} \right)^2 \\ & + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right] \end{aligned} \quad (4.42)$$

where κ is the von Karman constant.

The next goal is to work this analysis in reverse to find a suitable method of applying the algebraic f_{t2} function to the SST model in a manner consistent with the Spalart-Allmaras model. It

is evident from the preceding discussion that the intermittency should interact very similarly with the Spalart-Allmaras model as the f_{i2} function, although accounting for the near-wall behavior is not straightforward. Bradshaw's approximation breaks down in the sublayer and the same transformation between one- and two-equations cannot be used. Lacking a suitable assumption to transform the equations, it becomes useful to consider the turbulent kinetic energy budget in the sublayer,

$$\gamma_{eff} \nu_t S^2 - \gamma_{lim} \beta^* k \omega + \frac{\partial}{\partial y} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial y} \right] = 0 \quad (4.43)$$

where y is the local wall-normal coordinate. In the viscous sublayer, the SST turbulence model is nearly identical to the Wilcox 1988 k - ω model [21,72]. Accordingly, the turbulent quantities scale such that $k \sim y^{3.23}$ and $\omega \sim 1/y^2$ [72]. Consequently, the eddy viscosity scales as $y^{5.23}$. At initial inspection, this seems appealing since the destruction and molecular diffusion terms are of the same leading order with the production term being higher order (given that intermittency increases with wall distance in the sublayer). This implies that the asymptotic near-wall behavior is not influenced by the intermittency in the production term and that γ_{lim} should always be set to one (i.e. the destruction term remains unaltered from the original SST model).

Given that the transitional SST model of Langtry and Menter [20] was developed with significant industrial experience, it is reasonable to assume that there is a benefit to the form of γ_{lim} used in the model compared to the seemingly obvious asymptotic near-wall behavior. Initial numerical experimentation on geometries of practical interest revealed that fixing γ_{lim} to 1.0 produces very negative behavior in the boundary-layer development. The sensitivity to the lower limit of γ_{lim} near its published value was subsequently considered, but it is difficult to both quantify and qualify the changes in behavior about a practical geometry. Therefore, this sensitivity was studied using post-transition solutions of the transitional SST model for fully developed, zero-pressure-gradient flow with γ_{lim} lower limits of 0.05, 0.1, and 0.2. The

transitional momentum-thickness Reynolds number was set to $Re_{\theta_t} = 1000$ for all solutions rather than being solved for from a transport equation. This simplification comes from the transport equation predicting essentially constant value across the boundary layer and that $Re_{\theta_t} = 1000$ is, for order of magnitude purposes, representative of typical zero-pressure-gradient values. In addition to these solutions, a solution to the original form of the SST model was generated to provide a better context for the effect of intermittency. The predicted velocity profiles are plotted in Fig. 4-2 and show that the primary influence of the γ_{lim} lower limit is on the offset constant, B , in the logarithmic layer. The fully turbulent solution shows the log-layer offset constant to be very close to the intended value of 5.1 [72]. The intermittency-sensitized solutions show a surprisingly strong sensitivity to the value of the γ_{lim} lower limit, and the published limiting value of 0.1 produces results in very close agreement with the fully turbulent predictions.

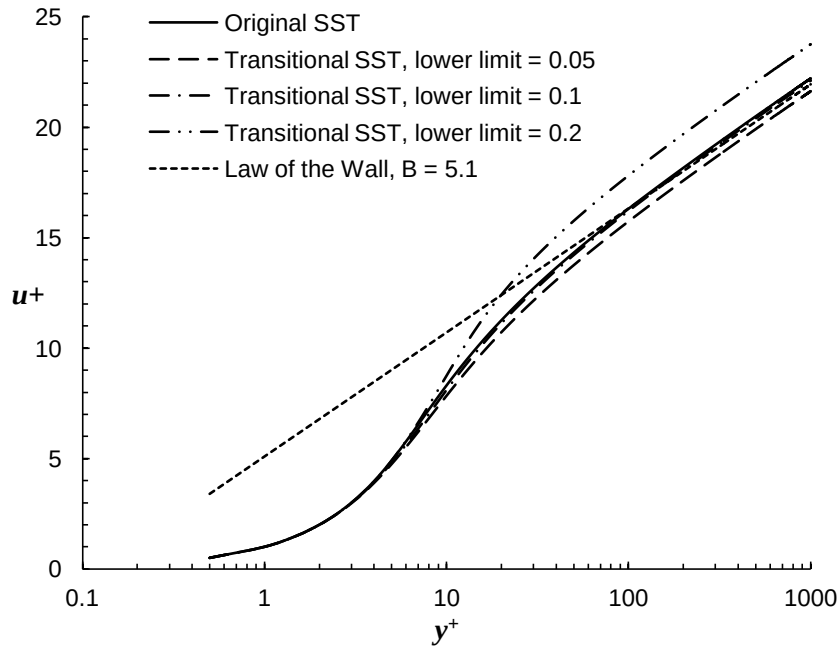


Figure 4-2. Turbulent velocity profiles for the original and transitional SST model with varying γ_{lim} lower limits.

It is therefore prudent to yield to the industrial experience used to develop the transitional SST model and incorporate the f_{t2} function accordingly. This modifies the turbulent kinetic energy equation to

$$\frac{Dk}{Dt} = \nu_t S^2 \left(1 - \frac{f_{t2}}{1.2} \right) - \beta^* k \omega \max \left(1 - \frac{f_{t2}}{1.2}, 0.1 \right) + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial x_j} \right] \quad (4.44)$$

where f_{t2} is defined in the same way as it is with the Spalart-Allmaras model, and χ is the solution to

$$\frac{\chi^4}{\chi^3 + 7.1^3} - \frac{\nu_t}{\nu} = 0 \quad (4.45)$$

The factor of 1.2 in the denominator of Eq. 4.44 is to allow for a consistent definition of f_{t2} between turbulence models.

4.4 Introducing the Amplification Factor to the f_{t2} Function

Determining how the amplification factor should interact with the f_{t2} function begins by considering the growth of the Reynolds stresses in the laminar boundary layer, which can be expressed as [56]

$$\tau \sim (u')_0^2 A^2(x) \quad (4.46)$$

The key feature of this relation is the dependency on the square of the amplification ratio. Normalizing Eq. 4.46 by its value at transition yields

$$\frac{\tau}{\tau_{crit}} = \left(\frac{A}{A_{crit}} \right)^2 \quad (4.47)$$

From the definition of the amplification factor, the right-hand side of this equation may also be written as $\exp(2(\tilde{n} - N_{crit}))$. For $\tilde{n} \ll N_{crit}$, this ratio is essentially zero, and when $\tilde{n} = N_{crit}$ it is one.

Any modification of f_{t2} needs to have the opposite behavior where f_{t2} should be fully active when $\tilde{n} \ll N_{crit}$ and zero (or negative) when $\tilde{n} \approx N_{crit}$. A suitable modification to f_{t2} takes the form

$$f_{t2} = \left[1 - \exp\left(2(\tilde{n} - N_{crit})\right)\right] c_{t3} \exp\left(-c_{t4} \chi^2\right) \quad (4.48)$$

In the laminar boundary layer, the modified f_{t2} function behaves in the same manner as the original f_{t2} function. Downstream of transition where $\tilde{n}$ is a passive variable, the modifications to the f_{t2} function serve only to attenuate the c_{t3} constant. As the exact choice of c_{t3} is unimportant (assuming it is of order one) in the fully turbulent boundary layer, the modifications do not affect the fully turbulent behavior.

With the amplification factor transport equation, the value of c_{t3} is kept at 1.2, which is consistent with modern implementations of the Spalart-Allmaras model, while the value of c_{t4} is 0.05, which is lower than the modern standard value of 0.5. The f_{t2} function, both original and modified, has a limited basin of attraction to laminar flow, and excessive values of free-stream eddy viscosity may overwhelm the function. Ideally one would use zero free-stream eddy viscosity with the model, but this can be numerically problematic. Spalart and Allmaras [33] suggest that the free-stream eddy viscosity be set such that

$$\frac{\tilde{\nu}}{\nu} \leq 0.1 \quad (4.49)$$

Numerical experimentation has shown that the upper limit of 0.1 is suitable for use with the amplification factor transport equation.

In an e^N -type model, transition is regarded as occurring when the local amplification factor exceeds a pre-defined critical value. For aerodynamic flows, this value is often set to $N_{crit} = 9$; however, it is often desirable to account for the influence of varying levels of free-stream turbulence intensity. This behavior may be modeled through the use of Mack's relationship [80], given by

$$N_{crit} = -8.43 - 2.4 \ln \left(\frac{Tu(\%)}{100} \right) \quad (4.50)$$

It is recommended that N_{crit} be limited to 9 in general aeronautic applications for both numerical reasons and for the fact that $N_{crit} = 9$ works well for engineering purposes when there is very low free-stream turbulence [28]. Mack’s relationship is able to support arbitrarily low turbulence levels and is reasonably accurate if the acoustic disturbance levels are sufficiently low. Nevertheless, one must always be judicious about the applicability of the N_{crit} value used for a given simulation. Care must be taken to ensure that free-stream turbulence levels used with the model are not so high that Eq. 4.50 would predict a negative critical amplification factor.

Whether N_{crit} is hard-coded, an independent user-specified parameter, or a function of the user-specified free-stream turbulence levels (e.g. using Eq. 4.50), the parameter should be constant throughout the entire flow domain. As such, the transition model behavior is not dependent on domain size for a given set of free-stream conditions. The linearity of the transported amplification factor provides some accounting of varying N_{crit} for complex configurations, though one must also consider the behavior of the eddy-viscosity model for these cases as well.

4.5 Summary

Methods have been explored to allow the amplification factor transport equation to interact with the one-equation Spalart-Allmaras and the two-equation Menter SST turbulence models, yielding the “SA-AFT” and “SST-AFT” transition/turbulence models, respectively. In both cases the f_{t2} function is used to provide an algebraic intermittency closure method as

$$f_{t2} = \left[1 - \exp \left(2(\tilde{n} - N_{crit}) \right) \right] c_{t3} \exp \left(-c_{t4} \chi^2 \right) \quad (4.51)$$

With the Spalart-Allmaras model [33], the f_{t2} function interacts with the source terms of the transport equation in the exact same manner as in the standard form of the model, which is

$$\frac{D\tilde{\nu}}{Dt} = c_{b1}\tilde{S}\tilde{\nu}(1 - f_{t2}) - \left[c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2} \right] \left(\frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right] \quad (4.52)$$

with

$$\chi = \frac{\tilde{\nu}}{\nu} \quad (4.53)$$

For the Menter SST model [21], the f_{t2} function is introduced in a manner analogous to that of the transported intermittency equation developed by Langtry and Menter [20] by modifying the turbulent kinetic energy equation to be

$$\frac{Dk}{Dt} = \nu_t S^2 \left(1 - \frac{f_{t2}}{1.2} \right) - \beta^* k \omega \max \left(1 - \frac{f_{t2}}{1.2}, 0.1 \right) + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial x_j} \right] \quad (4.54)$$

To evaluate the f_{t2} function, χ may be calculated as being the solution to

$$\frac{\chi^4}{\chi^3 + 7.1^3} - \frac{\nu_t}{\nu} = 0 \quad (4.55)$$

The specific dissipation equation remains unaltered from that of the fully turbulent SST model.

Although this approach does not produce the ideal near-wall behavior in the fully turbulent regime, it was deemed nevertheless prudent to defer to the industrial experience to develop the approach.

Chapter 5

Implementation, Verification, and Validation

5.1 Implementation

The amplification factor transport equation transition model has been implemented into a copy of OVERFLOW 2.2f maintained by the author. OVERFLOW is an implicit, structured, overset Reynolds-averaged Navier-Stokes solver written in the FORTRAN language, and is developed and maintained by NASA [26]. The solver is capable of analyzing two- and three-dimensional geometries in either time-accurate or steady-state (non-time-accurate) simulations using a variety of discretization schemes and time-marching algorithms.

A key feature of OVERFLOW 2.2f that facilitates implementation of the transition model is that it contains an implementation of the Langtry-Menter transition model in both its original form applied to the SST turbulence model as well as to the Spalart-Allmaras turbulence model [78]. Since the amplification factor transport equation and both equations of the Langtry-Menter model are in the form of Eq. 3.1, programming the new transition model required minimal modifications to the code. Unfortunately, the existing transition model subroutines in the solver are hard coded to two equations, whereas the new model requires only one. For simplicity, both equations were overwritten to solve the amplification factor transport equation identically to each other. While this is not computationally efficient, it nevertheless permits benchmarking of relative solution times and it prevents the unintended influence of a different, superfluous transported scalar. Modifying the diffusion terms required only the diffusion coefficients to be changed to that of the new model. The free-stream values of intermittency and momentum-thickness Reynolds number were redefined to be identically zero (Eq. 3.25). No changes were necessary for

the solid-wall boundary conditions, as the existing equations already used the zero-normal-gradient condition (Eq. 3.26). The subroutine that evaluates the source terms of the transition model equations was rewritten to evaluate that of the amplification factor transport equation. While the free-stream flow conditions, which are necessary for evaluating the new source term, are not passed to the source-term subroutines, these quantities are readily available through a COMMON block. The flow solver expects the Jacobian of the source term to be specified in the subroutine as well. Since the growth of the amplification factor does not depend on the amplification itself, the Jacobian is thereby set to zero. The flow solution update limiter in the code needed also to be revised to reflect the different physical limits of the amplification factor compared to the transported intermittency and momentum-thickness Reynolds number. Finally, the subroutines that evaluate the source terms of the eddy-viscosity models needed to be changed to reflect the new model.

The implementation of the transition model does not place any restrictions on which discretization schemes may be used for the mean-flow equations. All steady-state solutions presented in the following discussion were run non-time-accurate using a third-order-accurate Roe scheme [81] and a scalar, pentadiagonal approximate factorization algorithm [82] with solver-recommended artificial dissipation until force and moment convergence was achieved (which typically corresponded with residuals steadily approaching machine zero). Convergence of the solutions is aided by introducing a negative limit to the modified f_{t2} function to prevent excessive growth of the modified eddy viscosity while solution transients damp out. With this, the residuals of both the amplification factor transport equation and the eddy-viscosity models show good convergence to machine zero in tandem with the velocity field.

5.2 Two-Dimensional “Bump-in-Channel” Verification

With the modifications to the underlying eddy-viscosity models that are required to use them with a transition model, it is necessary to confirm that the true fully turbulent behavior is recovered after transition. For this, the turbulence model verification cases available through the NASA Langley Research Center “Turbulence Modeling Resource” website [83] are used. These cases are intended to provide a consistent set of grids and prescribed free-stream conditions for CFD developers to verify the implementation of a turbulence model. These cases are not intended to assess the accuracy of the model itself (which would constitute ‘validation’), but rather to assess if the model equations are being solved correctly.

The website provides predictions and grid-convergence information from the FUN3D [84] and CFL3D [25] solvers (both maintained by NASA) for the cases using popular turbulence models. Data for the Spalart-Allmaras and Menter SST models are included for every case as these are the most popular of the eddy-viscosity models. For some cases, predictions from OVERFLOW are included on the website as the turbulence model implementations in the release versions of OVERFLOW are all considered to be verified.

The “2D Bump-in-channel” case is one of particular interest due to its inclusion in both the 5th AIAA CFD Drag Prediction Workshop (DPW-5) [85] and the 2nd AIAA CFD High-Lift Prediction Workshop (HiLiftPW-2) [86]. The case is two-dimensional and features a no-slip region extending from $x = 0.0$ to $x = 1.5$ in non-dimensional units, within which is a sinusoid bump that lies from $x = 0.3$ to $x = 1.2$ and has a non-dimensional height of 0.05. The bump is shown graphically in Fig. 5-1 (originally from Ref. [83]). Upstream and downstream of the no-slip region are symmetry boundary conditions. Opposite the bump is also a symmetry condition. The inflow conditions correspond to a free-stream Mach number of 0.2 and a Reynolds number of 3 million per non-dimensional unit length.

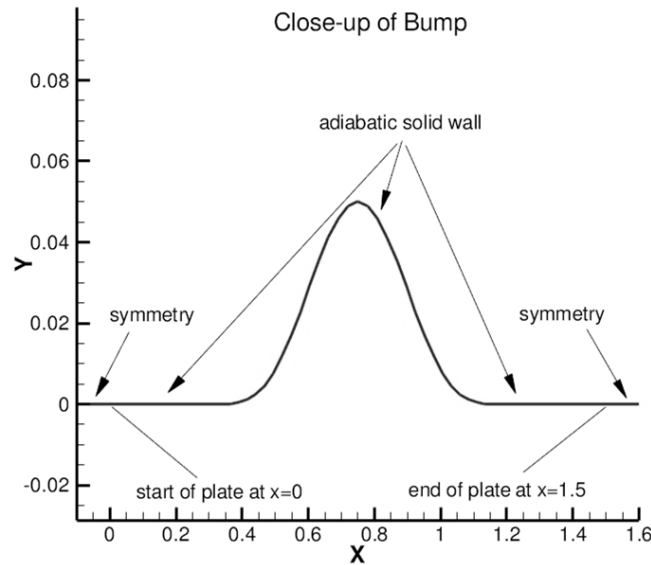


Figure 5-1. Graphical depiction of two-dimensional bump in channel [83].

Five turbulence model variants are considered with this verification case, three of which are variants of the Spalart-Allmaras model and two of which are variants of the Menter SST model. The Spalart-Allmaras variants are the standard model (SA), which has $c_{t3} = 1.2$ and $c_{t4} = 0.5$, the SA-noft2 variant, which has $c_{t3} = 0$, and a modification of the standard model where c_{t4} is changed to 0.05 (SA-ft2mod), which is the value used with the SA-AFT model. The variants of the Menter SST model are the standard version of the model (SST) and one in which the f_{t2} function is introduced in accordance with the discussion of Chapter 4 (SST-ft2) but with $c_{t4} = 0.05$. For all turbulence model variants using the f_{t2} function, the original form of the function is employed [33] with the same relationship between the modified eddy viscosity and the actual eddy viscosity applied. For all models, the free-stream eddy viscosity ratio was set to 5 and the free-stream turbulence intensity was set to 1% to force the turbulence models to experience fully turbulent behavior [77,83].

The skin-friction distributions predicted by the Spalart-Allmaras variants are plotted in Fig. 5-2, although the distributions are indistinguishable from each other. The SA-ft2mod result shows a small dip in the skin friction near the leading edge of the no-slip region corresponding to pseudo-transition. A similar region does not appear with the standard SA variant (which also has the f_{t2} function) since the free-stream eddy viscosity is sufficiently high. Nevertheless, once the boundary layer is fully turbulent, the variants behave the same. This confirms that lowering the value of c_{t4} to 0.05 does not degrade the behavior of the turbulence model in the presence of pressure gradients.

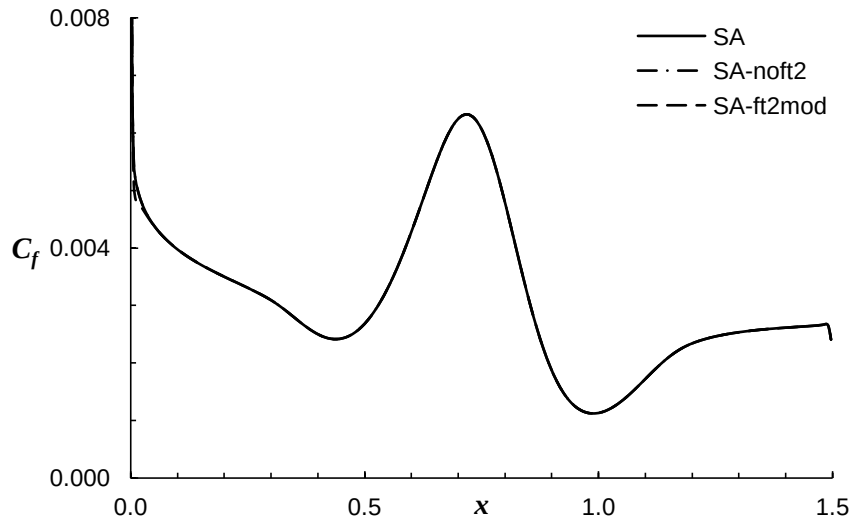


Figure 5-2. Predicted skin-friction distribution along 2D bump using Spalart-Allmaras model variants.

With the Menter SST model variants, the same agreement quality is not present in the skin friction distributions, which are shown in Fig. 5-3. Upstream of the bump's peak, there is a distinct shift in the skin friction where the f_{t2} -sensitized variant predicts higher values. Along the bump itself where the pressure gradients, plotted in Fig. 5-4, are the strongest, the skin friction is in the closest agreement between the model variants. There is still, however, an overshoot in the skin friction at the peak by the f_{t2} -sensitized variant. The skin-friction discrepancies are a direct

result of modifying the destruction term near the wall to accommodate the f_{t2} function, and should be kept in mind whenever such transition-sensitized versions of the Menter SST model are used.

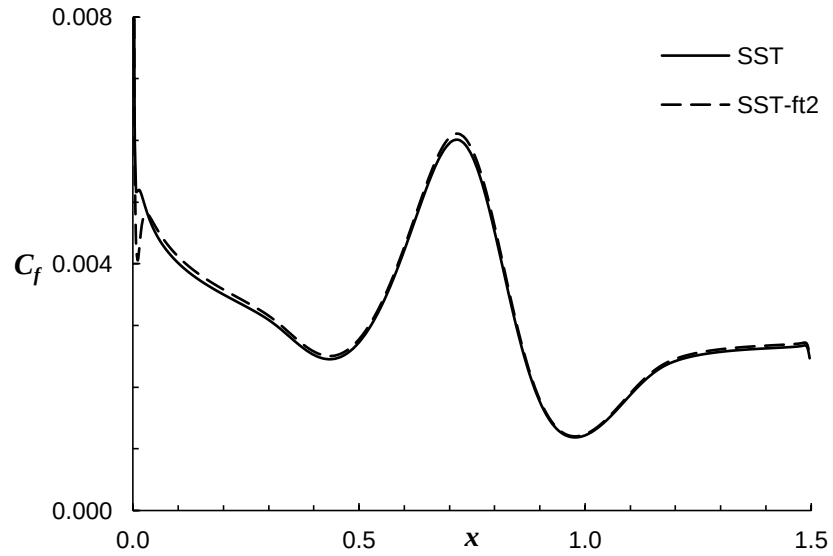


Figure 5-3. Predicted skin-friction distribution along 2D bump using Menter SST model variants.

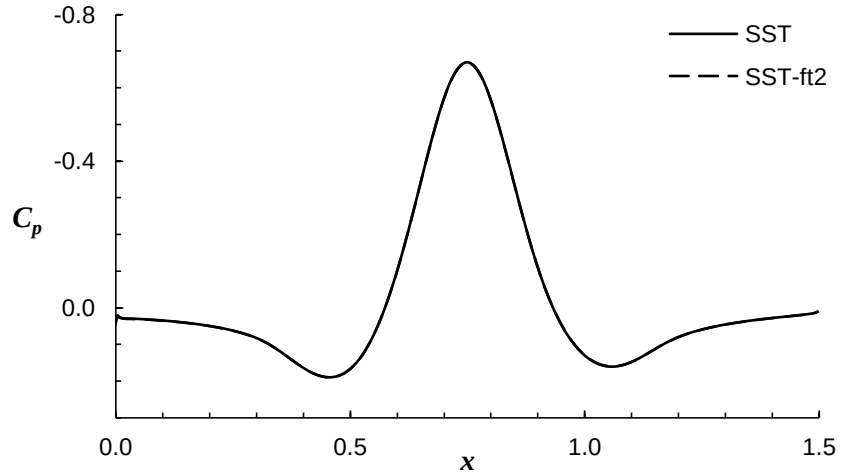
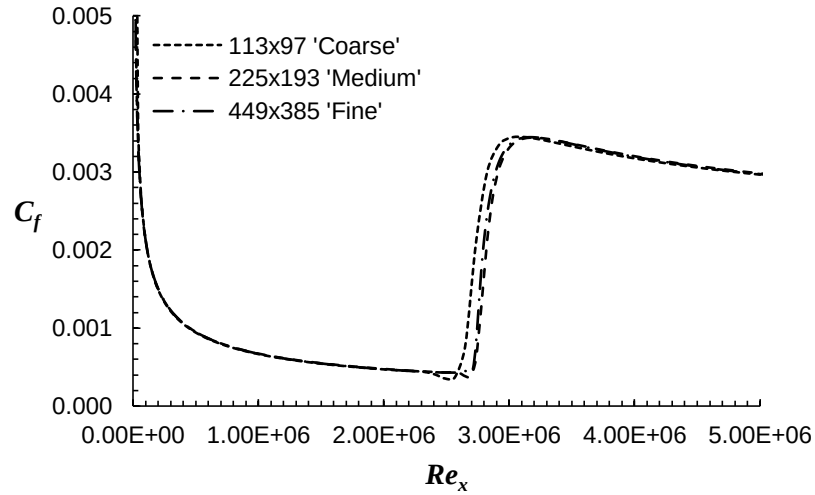


Figure 5-4. Predicted surface pressure distribution along 2D bump using Menter SST model variants.

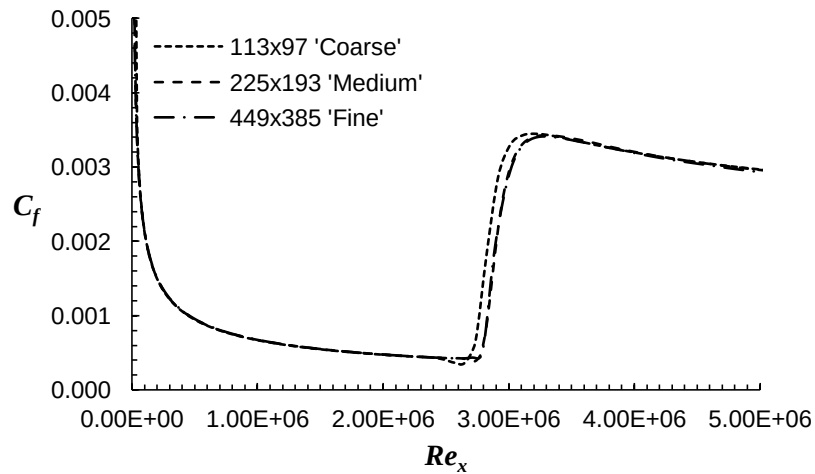
5.3 Zero-Pressure-Gradient Flat-Plate Study

The zero-pressure-gradient flat plate is a benchmark solution for boundary-layer flows and serves as a useful test case for the new transition model. Three cases are included here: a grid refinement study, a verification study on the development of $\tilde{n}$ and the influence of N_{crit} , and a validation study using the skin-friction data of Schubauer and Klebanoff [64].

The grid-refinement study was performed on a common sequence of grids to demonstrate the robustness of the transition model. For this study, N_{crit} was set to 9 to reflect how the model would be used in a typical aerodynamic analysis. The baseline grid is a 449-by-385 point mesh, which is referred to as the “fine” grid and is consistent with the finest grid resolution provided for the Flat Plate Verification Case from Ref. [83]. “Medium” and “coarse” grids are generated by successively removing every other point in both computational directions. The development of the skin friction for the three grids is plotted in Fig. 5-5(a) for the SA-AFT transition/turbulence model and Fig. 5-5(b) for the SST-AFT model. These data show only a small difference in the transition location between the coarse and medium grids and almost no difference between the medium and fine grids. The medium grid is therefore regarded as producing grid-converged solutions. The lack of significant variation between the medium- and fine-grid solutions shows that true grid convergence is possible with this transition model and that mesh refinement does not deteriorate the solution quality.



(a) SA-AFT



(b) SST-AFT

Figure 5-5. Predicted skin-friction distribution along a zero-pressure-gradient flat plate for successively refined grids using the new transition model.

The results of a computational study of the influence of N_{crit} on the behavior of the model on the medium grid with the Spalart-Allmaras model are presented in Fig. 5-6. In this figure, both the maximum amplification factor in the boundary layer and the wall skin friction are plotted with running Reynolds number for different values of N_{crit} . Also plotted is an “analytic” growth of $\tilde{n}$ predicted using the relations in Eqs. 3.15 and 3.16 and assuming the Blasius boundary-layer solution. The CFD transition model shows the desired behavior when the specified critical amplification factor is reached, and the CFD-predicted growth of the amplification factor shows

very good agreement with what would be expected from the analytic case. There is a slight offset between the curves, but this is due primarily to the flow at the leading edge in the CFD solution not obeying the Blasius solution.

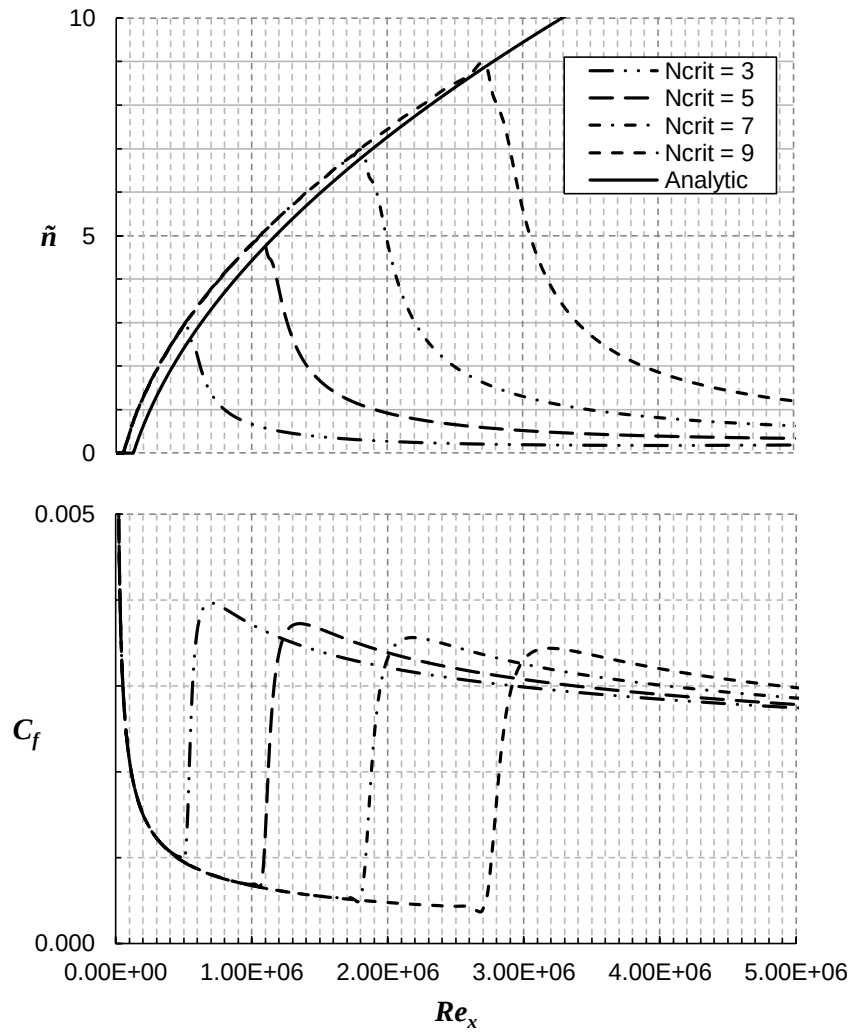


Figure 5-6. Flat-plate skin-friction distributions and maximum amplification factors for varying N_{crit} .

Additional confidence in the model has been gained by validating its predictions against the Schubauer and Klebanoff low-turbulence flat-plate data [64]. Their experiment is reported to have been performed with a free-stream turbulence intensity of 0.03%, which corresponds to N_{crit}

= 11.04 using Mack's relationship. For this specific experiment, however, acoustic disturbances were the primary cause of transition rather than free-stream turbulence levels, and thus Mack's relationship does not directly apply [57,80]. Van Ingen analyzed this case using linear stability theory and reports that the boundary layer has fully transitioned from laminar to turbulent at an N-factor of 10.30 [57]. Given that the transition model does not explicitly account for transition length, this was the value of N_{crit} specified for the computations. A comparison of the computational and experimental skin frictions is shown in Fig. 5-7, along with fully laminar and fully turbulent predictions. Both SA-AFT and SST-AFT are used for this case. There is a slight discrepancy between the predictions these two models that is a result of different eddy-viscosity growth rates once the critical amplification factor is reached. The CFD-predicted transition begins further downstream than was measured in the experiment. More importantly, however, the transitional CFD solutions agree very well with the experiment in the skin friction values once transition is completed. Predicting the boundary-layer development downstream of transition is crucial for CFD to provide accurate drag predictions, and this is not possible without accurate boundary-layer properties at the beginning of the turbulent regime.

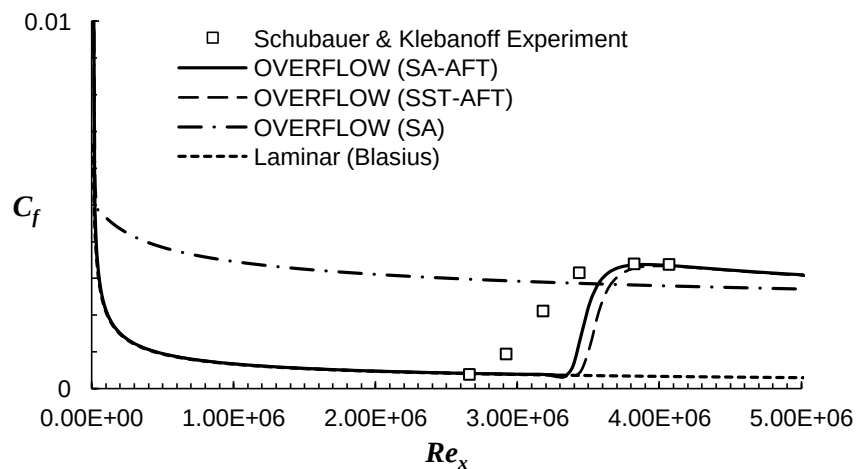


Figure 5-7. Predicted and measured skin-friction distributions for the Schubauer-Klebanoff experiment [64].

5.4 Two-Dimensional Airfoil Verification and Validation

Verifying and validating the amplification factor transport transition model continues with two-dimensional, airfoil test cases, which are appealing for a variety of reasons. Airfoils are the basis of all aerodynamic lifting surfaces, and the surface pressure gradients that drive the boundary-layer development are representative (if not identical) to those found on wings. In this case, the two-dimensionality simplifies the problem while maintaining realism. There is a wealth of experimental low-turbulence aerodynamic section data at a wide range of Reynolds numbers available for airfoil geometries against which model predictions can be compared. Finally, there are well-validated specialized analysis tools for airfoils that can provide insights into quantities that may not be measured in an experiment.

For the subsequent airfoil analyses, the amplification factor transport equation coupled with the Spalart-Allmaras eddy-viscosity model (SA-AFT) as implemented in OVERFLOW 2.2f will be used. Unless otherwise specified, the free-stream Mach number is set to 0.2 (to model low-speed, incompressible tests) and the free-stream turbulence intensity is set to 0.045% (corresponding to $N_{crit} = 10.07$). This turbulence intensity is representative of low-turbulence wind tunnels such as the Penn State University Low-Speed, Low-Turbulence Wind Tunnel (LSLTT) [87,88] and the Delft University Low-Speed Wind Tunnel (LSWT) [89]. With the Spalart-Allmaras eddy-viscosity model, the free-stream eddy-viscosity ratio is set to 2.8×10^{-7} . For some cases, the new transition model is used with the SST eddy-viscosity model (SST-AFT) with a free-stream eddy-viscosity ratio of 0.1 (which is the default value for OVERFLOW) in order to demonstrate its capabilities. All grids were generated using Chimera Grid Tools [90] following best practices in grid generation [91].

For all cases, the predictions with the new transition model are compared to those using the Langtry-Menter transitional SST model (referred to as “SST-LM”) for the same grid and

discretization schemes in OVERFLOW 2.2f. The free-stream turbulence conditions are $Tu = 0.045\%$ and an eddy-viscosity ratio of 10. The discrepancy between the models in the free-stream eddy-viscosity is due to inherent differences in their behaviors. The Spalart-Allmaras model ideally uses a true laminar free-stream, but a very small non-zero value is beneficial for numerical behavior. On the other hand, the Menter SST model tends to favor higher values of eddy viscosity to mitigate the free-stream decay of the turbulence. It would therefore be desirable to use a value higher than 0.1 with the amplification factor transport equation, but high values can overwhelm the f_{t2} function, leading to premature transition.

Also compared to the CFD-predicted section characteristics are those predicted using XFOIL 6.96 [68] (for single-element airfoils) and MSES 3.05 [92] (for multi-element airfoils). Both solvers were developed primarily by Mark Drela of MIT to analyze airfoils by coupling an inviscid flow solver with an integral boundary-layer method. XFOIL uses a panel-method-based potential-flow solution for the outer, inviscid flow. While the method is incompressible, compressibility effects are introduced through the Karman-Tsien correction. MSES, alternatively, uses an upwind Euler solver that is inherently compressible and can provide accurate predictions through the transonic regime. Both codes solve the same integral boundary-layer equations and predict transition using the approximate envelope method [32]. A full e^N method is available in MSES; however, it has issues with numerical convergence and is not recommended for practical use [93].

5.4.1 PSU 94-097 Winglet Airfoil

The PSU 94-097 airfoil was designed specifically for low-speed winglet applications and has been tested extensively in the Penn State Low-Speed, Low-Turbulence Wind Tunnel (LSLTT) [94]. The Reynolds numbers of the tests ranged from 0.24×10^6 to 1.0×10^6 , for which

section aerodynamic characteristics and surface pressure distributions were measured. For validating the new transition model, the lowest and highest Reynolds numbers are considered.

The section aerodynamic characteristics for $R = 0.24 \times 10^6$ predicted with OVERFLOW using the new transition model (SA-AFT) are plotted in Fig. 5-8 along with fully turbulent predictions, XFOIL predictions, and experimentally measured characteristics. In Fig. 5-9, OVERFLOW predictions using the Langtry-Menter model and the SST-AFT variant of the new transition model are also compared. All OVERFLOW predictions agree very well with each other in the linear range of the lift curve, including the shift in apparent zero-lift angle of attack between $\alpha = -2^\circ$ and $\alpha = -1^\circ$ due to the upper-surface laminar separation bubble. XFOIL and the OVERFLOW SA-AFT predictions agree for the upper limit of the low-drag lift-coefficient range (the “laminar bucket”), which is at a higher lift coefficient than was measured experimentally. The SST-based predictions with both the new transition model and the Langtry-Menter model predict this to occur at a lower lift coefficient. This is due to the onset of trailing-edge separation rather than an influence of the predicted transition location. For this case, transition on the upper surface occurs through a laminar separation bubble for all lift coefficients, which is why the various methods are in good agreement for the corners of the laminar bucket. The SST-LM and SST-AFT predictions show a discrepancy in the drag values in the laminar bucket, with the SST-LM predictions being the highest. Even though transition occurs through a laminar separation bubble, differences in the predicted length of the bubble and the corresponding viscous-inviscid interactions can lead to very different predictions. This is further illustrated in Fig. 5-10, in which the predicted transition locations from the SA-AFT, SST-LM, and XFOIL data are plotted. Even though the predicted chordwise location of transition is very different between the methods at low lift coefficients, they all agree in the upstream movement of transition at higher lift coefficients. The lower-surface, OVERFLOW-predicted transition locations agree very well with each other, but differ significantly from those predicted by XFOIL. As transition is by laminar separation

bubbles in all cases, the likely cause of this is the differences in the viscous-inviscid interactions between the methods.

Section aerodynamic characteristics predicted with OVERFLOW for $R = 1.0 \times 10^6$, both fully turbulent (SA-noft2) and using the new transition model (SA-AFT), and are presented in Fig. 5-11 along with experimental data and XFOIL predictions. The aerodynamic coefficients predicted with the new transition model are also compared in Fig. 5-12 to those predicted using the SST-AFT variant, the Langtry-Menter model, and those predicted with XFOIL. All theoretical predictions (CFD and XFOIL) agree very well with each other and with experiment for the lift and pitching-moment curves, though they all over-predict the maximum lift coefficient. The new transition model performs very well in the low-drag lift-coefficient range, and both sets of predictions using the model predict the profile drag very well. The new transition model predicts the lift-coefficient range of the laminar bucket with much better agreement to experiment than does the Langtry-Menter model, which is a key difference in their performance. The predicted transition locations for both models are plotted in Fig. 5-13 with respect to lift coefficient along with the transition locations predicted by XFOIL. These data show that the new model agrees well with XFOIL in capturing the upstream movement of the upper-surface transition location, whereas the Langtry-Menter model lags. This is due to the new model more accurately predicting natural transition in an adverse pressure gradient while the Langtry-Menter model tends to permit laminar separation bubbles before natural transition.

The behavior of the amplification factor transport equation and how it affects the underlying eddy-viscosity model has been investigated by exploring the development of both the transported amplification factor and the eddy-viscosity ratio in the boundary layer. Contours of both amplification factor and the Spalart-Allmaras modified-eddy-viscosity are shown in Fig. 5-14 for the same $\alpha = 2^\circ$ solution. For this angle of attack, transition is predicted to occur on the upper surface through a laminar separation bubble (LSB). This transition region is indicated in

the top portion of Fig. 5-14 and is characterized by a local maximum in the transported amplification factor. The amplification factor decays after transition due to significantly decreased production in the turbulent boundary layer, which allows diffusion to dominate. The contours of modified eddy viscosity plotted in the bottom portion of Fig. 5-14 show the fully turbulent behavior of the boundary layer downstream of transition with laminar flow existing everywhere else. On the lower surface, the amplification factor grows toward the trailing edge but does not become large enough to cause transition.

The predicted LSB transition on this airfoil has been confirmed by comparing the theoretical and experimental pressure distributions. Examples of such comparisons are included as Figs. 5-15 and 5-16 for the airfoil operating at $\alpha = 2^\circ$. The comparison in Fig. 5-15 shows the two variants of the new transition model (SA-AFT and SST-AFT) with experiment, while the comparison in Fig. 5-16 shows the SA-AFT predictions, the Langtry-Menter model predictions, and fully turbulent (SA-noft2) predictions. As expected, the fully turbulent solution predicts no laminar separation on the upper surface and thus does not exhibit the plateau in the pressure distribution beginning at mid-chord. The solution using the Langtry-Menter model has a laminar-separation bubble, but the length of the bubble is over-predicted resulting in turbulent reattachment further aft than was measured. The new transition model shows improved agreement with experiment over the other two CFD methods: the laminar separation bubble is present, and its length is more accurately predicted.

This $\alpha = 2^\circ$ solution is very well behaved numerically for all turbulence and transition models, making it a good case to benchmark the iterative convergence qualities of the model as well as their respective computational. The L^2 -norm residual convergence of the right-hand side of the mean-flow equations is plotted in Fig. 5-17 for the SA-noft2, SST, SA-AFT, SST-AFT, and SST-LM models using identical solutions for each. The fully turbulent models show similar convergence to each other and reach machine zero by approximately 15,000 iterations. The

transitional models converge more slowly, but are nevertheless approaching machine zero and the demonstrated force and moment convergence. Both AFT-based transitional solutions show very similar convergence to each other, which is consistent with that of the fully turbulent solutions. The SST-LM solution converges more slowly than the AFT solutions, and its L^2 norm exhibits some oscillatory behavior beyond 20,000 iterations even though the mean is approaching zero. In contrast, the SST-AFT shows monotonic convergence in this range.

OVERFLOW monitors the wall-clock time required for all of its processes and outputs a breakdown of the times at the end of the solution. The most prominent of these is timer for the actual flow-solver processes (to distinguish it from other routines such as reading the grid, calculating grid connectivity, etc), which is further broken down into categories that include evaluating the right-hand side, inverting the left-hand side, solving the turbulence model (which includes transition), checking convergence, and other bookkeeping tasks. The flow solver timers for the benchmark solutions are categorized in Table 5-1 as the requirements for solving the mean flow equations, solving the turbulence and transition equations, and all other flow-solver tasks. Also included in Table 5-1 is a normalization of the turbulence model requirements with respect to the mean-flow. This shows the relative cost of solving the additional turbulence/transition equations for a given iteration of the mean flow, which is essentially constant across the models. Based on the difference between the SA-noft2 and SA-AFT timers, and that the implementation of the new transition model has duplicate, identical instances of the equations, the time required to advance the amplification factor transport equation one iteration is 45.85% that of advancing all of the mean-flow equations. Following similar arguments, the two Langtry-Menter transition equations require a total of 109.5% of the time of the mean-flow equations, which is 2.4 times that of the new transition model. Considering the combined turbulence and transition models, an iteration of SST-LM takes 1.95 times longer to solve than one of SA-AFT (again accounting for the duplication in the implementation). It must be noted that the addition of AFT to the SST

model shows a much increased computational requirement compared to using the SA turbulence model. The reason for this is the evaluation of the Eq. 4.45 requiring Newton-Raphson iteration at each point in the flow field.

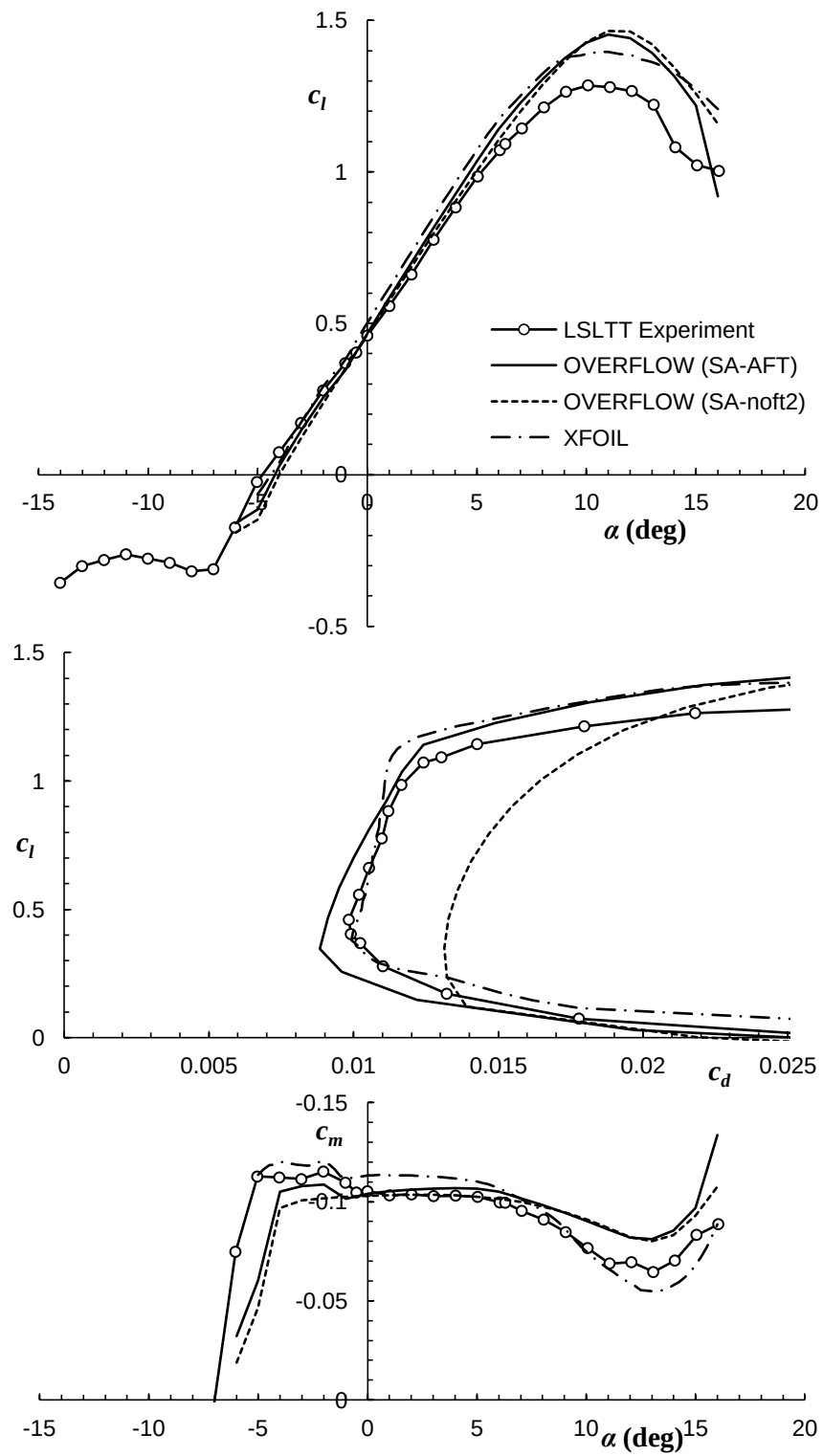


Figure 5-8. Predicted and measured [94] section aerodynamic characteristics of the PSU 94-097 airfoil, $R = 0.24 \times 10^6$.

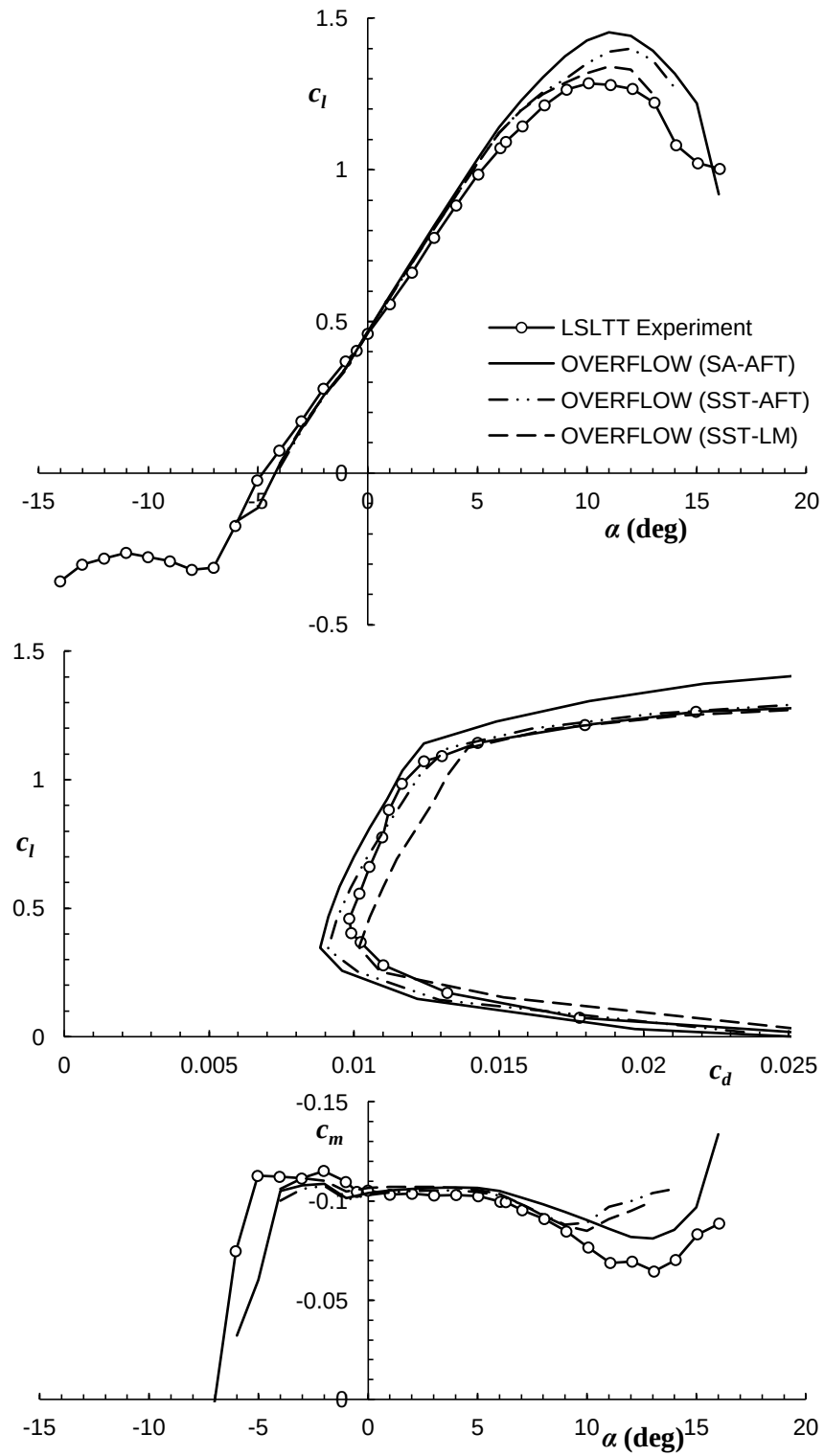


Figure 5-9. Predicted and measured [94] section aerodynamic characteristics of the PSU 94-097 airfoil, $R = 0.24 \times 10^6$.

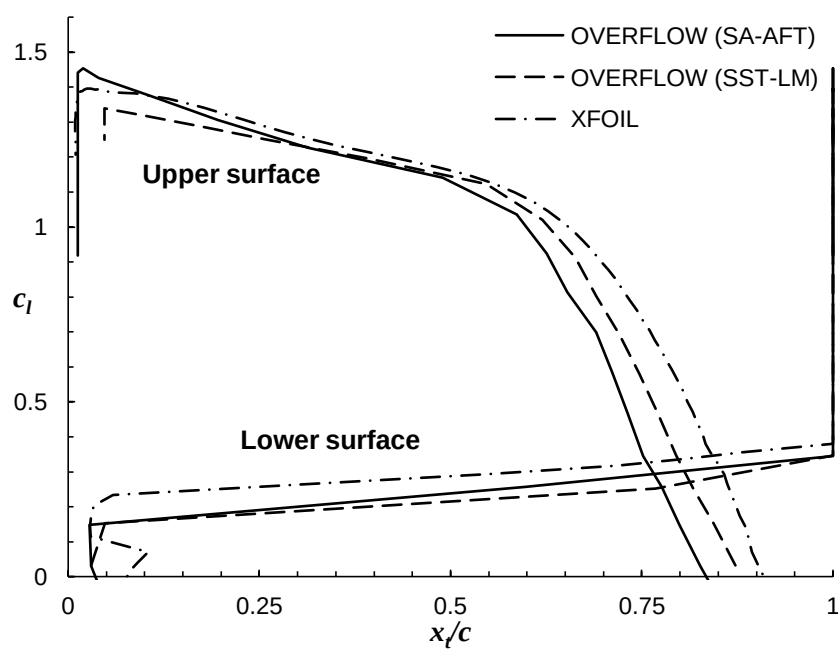


Figure 5-10. Predicted transition locations of the PSU 94-097 airfoil, $R = 0.24 \times 10^6$.

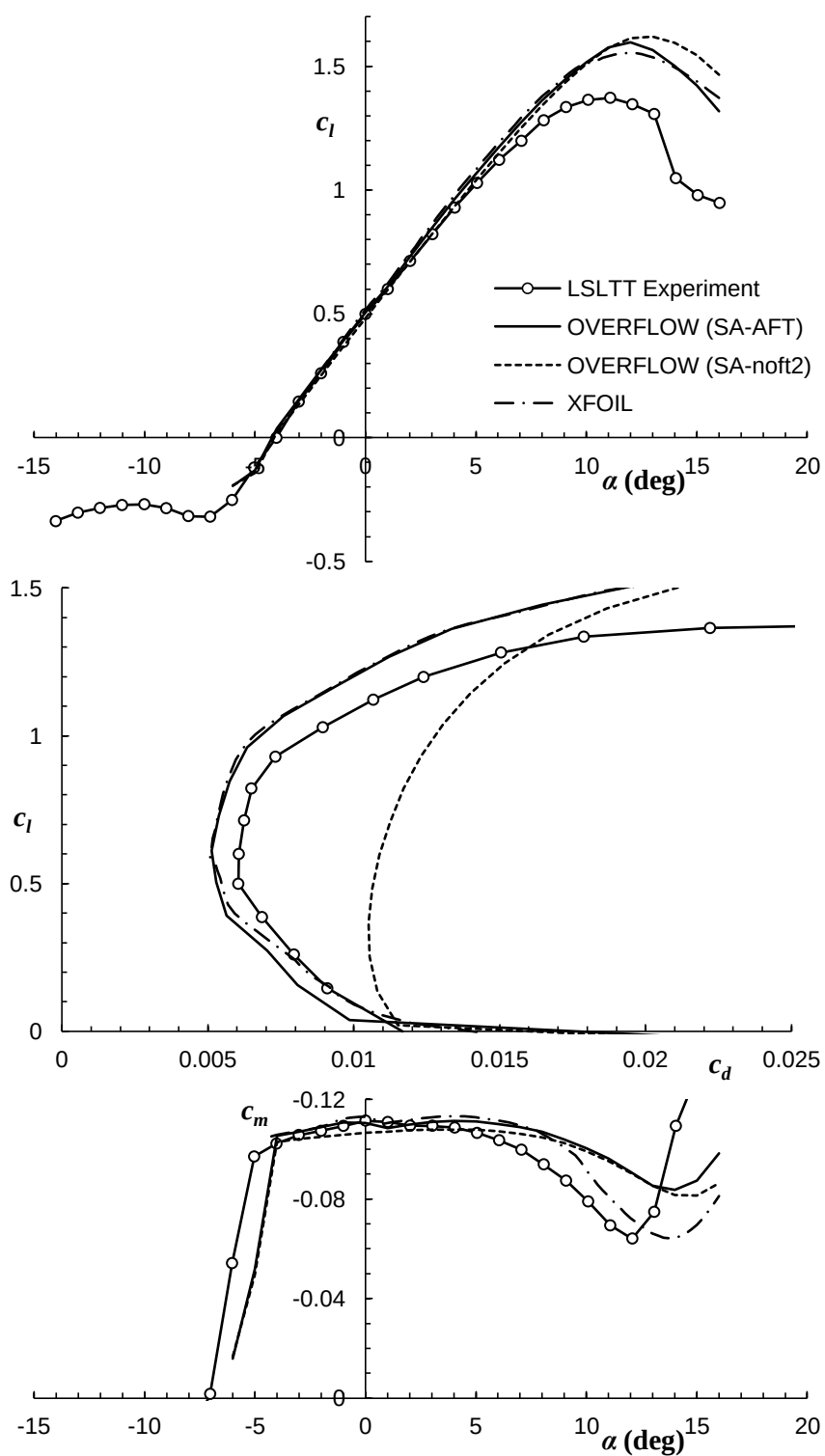


Figure 5-11. Predicted and measured [94] section aerodynamic characteristics of the PSU 94-097 airfoil, $R = 1.0 \times 10^6$.

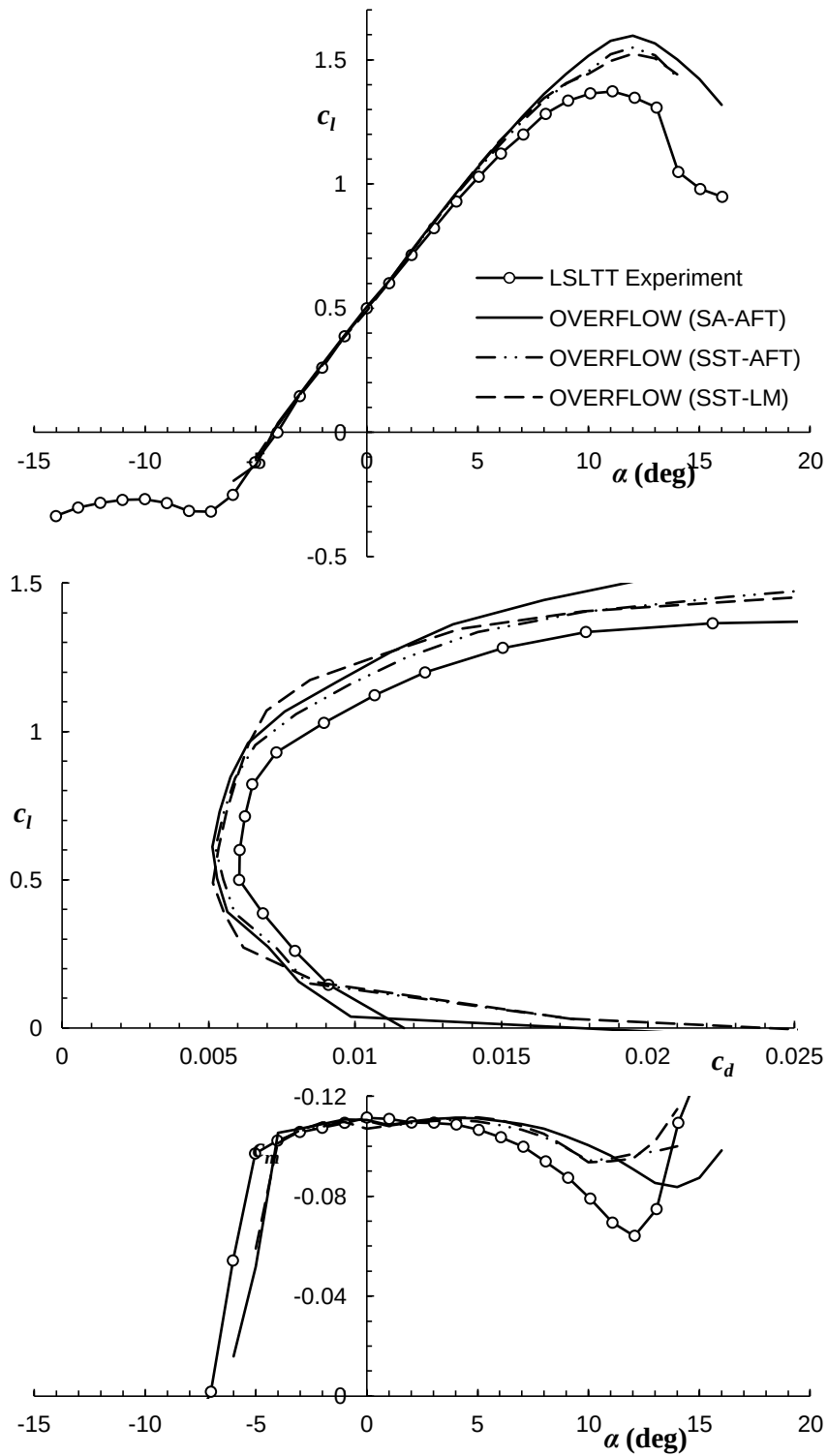


Figure 5-12. Predicted and measured [94] section aerodynamic characteristics of the PSU 94-097 airfoil, $R = 1.0 \times 10^6$.

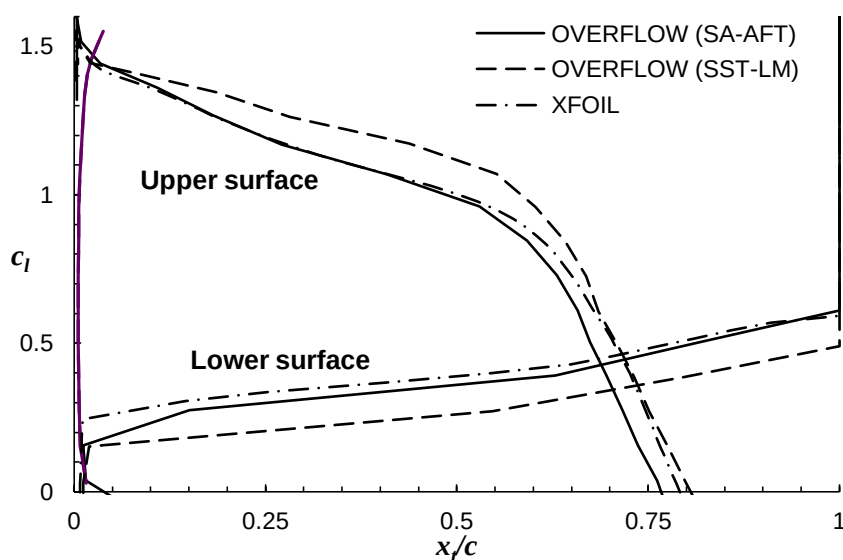
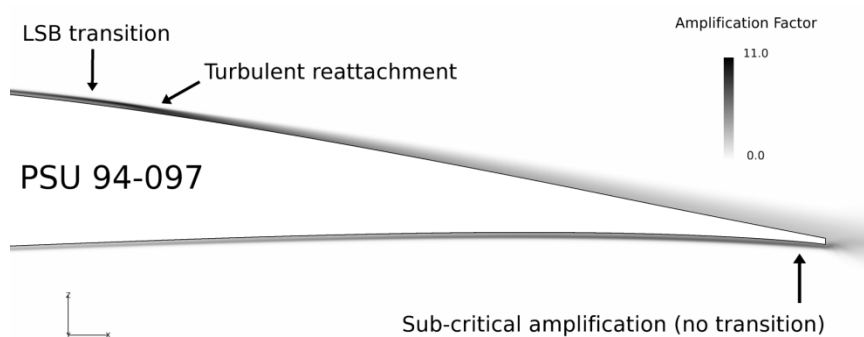
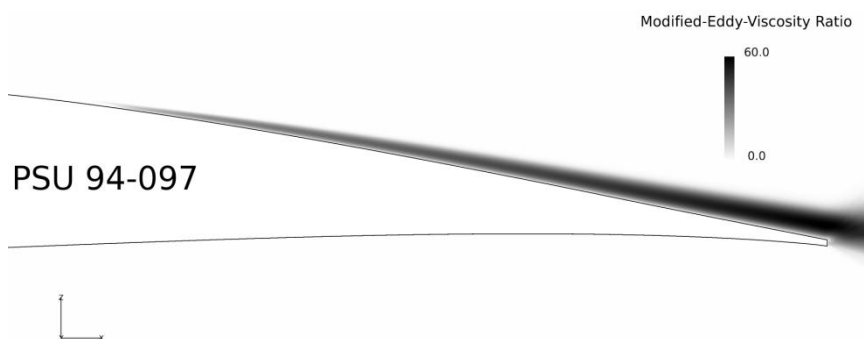


Figure 5-13. Predicted transition locations of the PSU 94-097 airfoil, $R = 1.0 \times 10^6$.



(a) Transported Amplification Factor, $\tilde{n}$



(b) Modified-Eddy-Viscosity Ratio

Figure 5-14. Contours of amplification factor and modified eddy-viscosity ratio around the PSU 94-097 airfoil, $\alpha = 2^\circ$ and $R = 1.0 \times 10^6$.

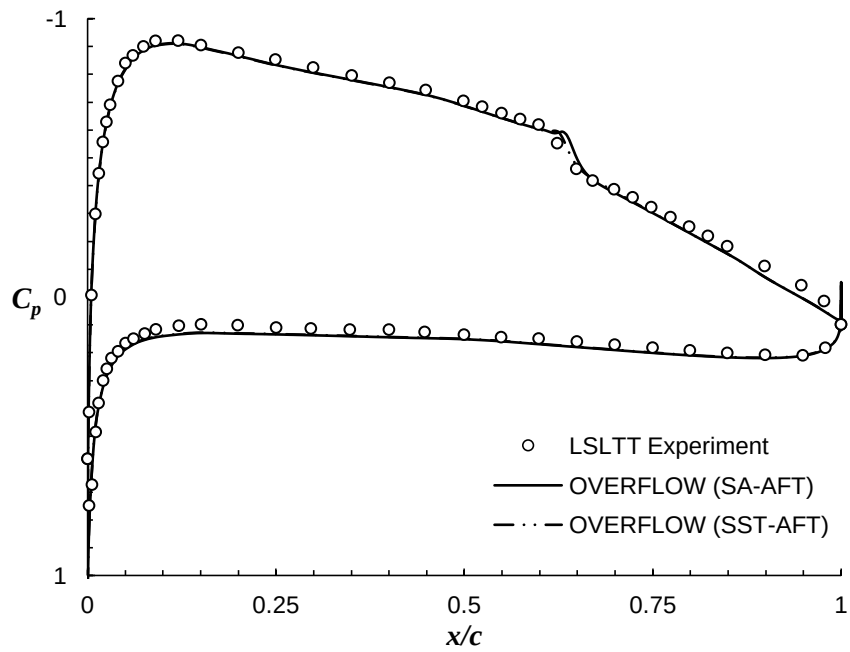


Figure 5-15. Theoretical and experimental [94] pressure distributions of the PSU 94-097 airfoil, $\alpha = 2^\circ$ and $R = 1.0 \times 10^6$.

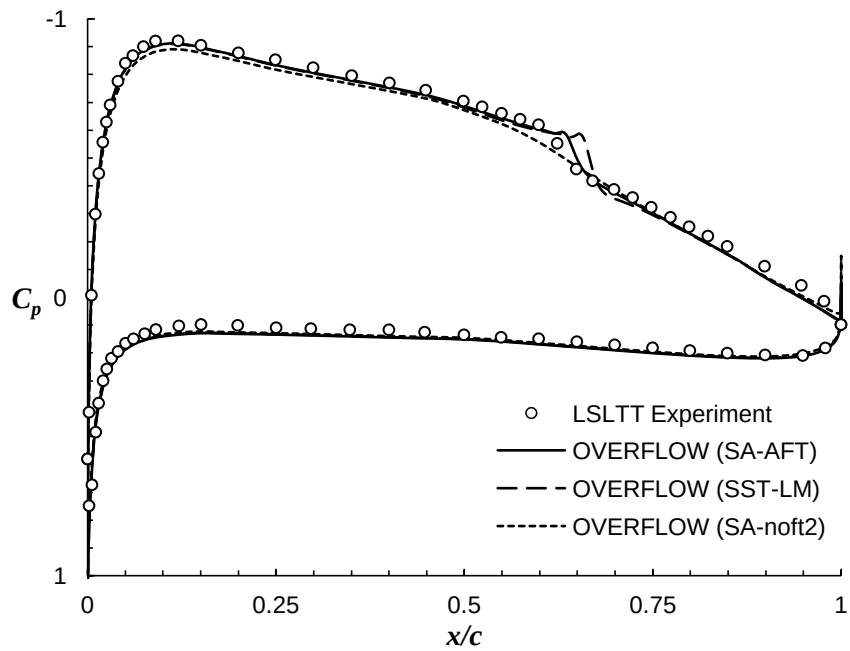


Figure 5-16. Theoretical and experimental [94] pressure distributions of the PSU 94-097 airfoil, $\alpha = 2^\circ$ and $R = 1.0 \times 10^6$.

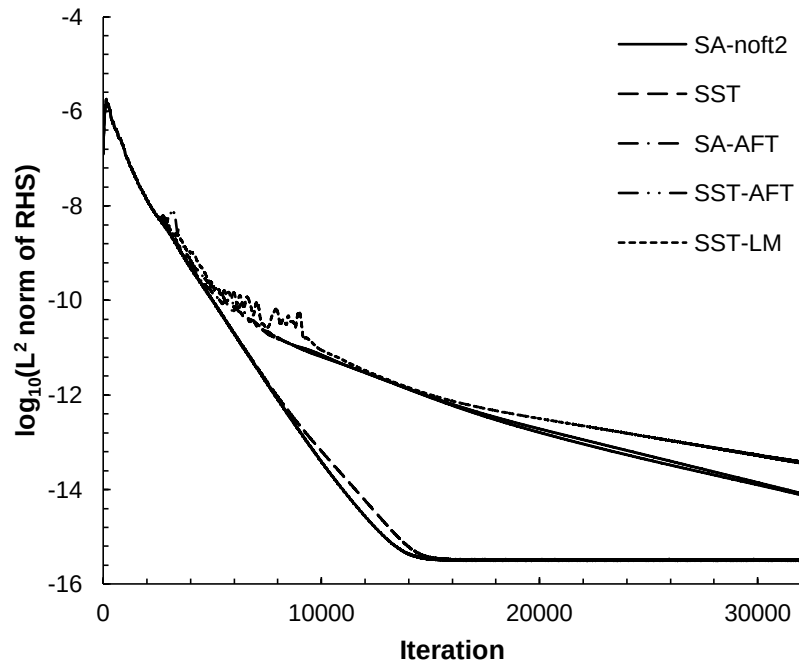


Figure 5-17. Iterative L^2 -norm convergence for different turbulence and transition models.

Table 5-1. Summary of OVERFLOW flow solver timers for benchmark case.

<u>Model</u>	<u>Actual Output Timers</u>			<u>Normalized Timers</u>	
	<u>Mean Flow</u>	<u>Turbulence</u>	<u>Other</u>	<u>Mean Flow</u>	<u>Turbulence</u>
SA-noft2	53.53%	39.42%	7.05%	1.000	0.736
SST	42.25%	52.26%	5.49%	1.000	1.237
SA-AFT	35.95%	59.42%	4.63%	1.000	1.653
SST-AFT	26.68%	69.90%	3.42%	1.000	2.620
SST-LM	28.88%	67.35%	3.77%	1.000	2.332

5.4.2 S805 Wind-Turbine Airfoil

The S805 airfoil was designed for stall-regulated, horizontal-axis wind turbines with the primary design objectives being a restrained, roughness-independent maximum lift coefficient and low profile drag [95]. Additional constraints applied during the design required that the zero-lift pitching-moment coefficient be no more negative (nose-down) than -0.05 and that the maximum airfoil thickness fall within a specific range. This airfoil was originally tested in the Delft University of Technology Low-Speed Wind Tunnel [95], but it has since been tested in the Penn State University LSLTT [96] for Reynolds numbers ranging from 0.5×10^6 to 1.5×10^6 .

The aerodynamic characteristics of this airfoil as measured in the Penn State wind tunnel for $R = 0.5 \times 10^6$ are compared with OVERFLOW predictions using the new transition model, the Langtry-Menter model, and with XFOIL predictions in Fig. 5-18. The predicted lift curves agree very well with each other and with the experiment, though with varying agreement in maximum lift. Both OVERFLOW lift curves and the XFOIL lift curve predict the zero-lift-angle-of-attack shift, which is due to the upper-surface laminar separation bubble, to occur at a lift coefficient lower than was observed experimentally. The experimental profile-drag measurements for this airfoil further indicate that the section aerodynamics are strongly influenced by the upper-surface bubble for this Reynolds number. This is most evident in the drag polar, which shows a sharp increase in the drag immediately above the lower corner of the laminar bucket. XFOIL predicts the drag polar very well in that the drag increase is accurately captured, even though XFOIL over-predicts the drag reduction towards the upper corner of the bucket. OVERFLOW results using the new transition model agree well with experiment at the lower corner of the bucket, but do not capture the drag increase with either eddy-viscosity model. The OVERFLOW results using the Langtry-Menter model captures the drag levels better on average, but is not consistently more accurate in any specific region of the drag polar. In particular, the drag at the lower corner of the

bucket is predicted to be much higher than predicted by the other methods, though some of the drag increase is captured. There is also an unexpected behavior in that it predicts the upper corner of the bucket to be at a lower lift coefficient than the new transition model predicts. A study of the individual flow solutions indicates this to be the result of trailing-edge separation and viscous-inviscid interactions influencing the pressure distributions. In any case, the OVERFLOW solutions do not consistently or accurately predict the drag rise due to the laminar separation bubbles, implying that more research is necessary into the behavior of eddy-viscosity models in separation bubbles. Finally, there is weak agreement in the pitching-moment curves with no method standing out as agreeing best with experiment. Given the differences between the OVERFLOW and XFOIL predictions, it may be inferred that this too is strongly influenced by the upper-surface laminar separation bubble.

Aerodynamic predictions using OVERFLOW with the new transition model for $R = 1.5 \times 10^6$ are compared to experiment in Fig. 5-19 along with predictions using OVERFLOW with the Langtry-Menter model and using XFOIL. All predictions agree very well with each other and with experiment for the lift coefficients, except that the SA-AFT solutions over-predict maximum lift while XFOIL and the SST-LM solutions agree much better with experiment. The pitching-moment coefficients show very good agreement as well. The experimental profile-drag polar for this Reynolds number shows significantly less influence from upper-surface laminar separation bubbles, though the separation bubbles are still present. The OVERFLOW solutions under-predict the lift coefficient for the lower corner of the drag bucket, though the new transition model predicts the upper corner more accurately than does the Langtry-Menter model. The strong agreement between both the new transition model and the Langtry-Menter model at the lower corner is due to both models predicting lower-surface transition to occur near the trailing edge via a laminar separation bubble. For the same angle of attack (and lower), XFOIL predicts transition

to occur by the same mechanism near the leading edge. Thus, it is reasonable to infer that there is a very strong influence from the viscous-inviscid interactions in this angle-of-attack range.

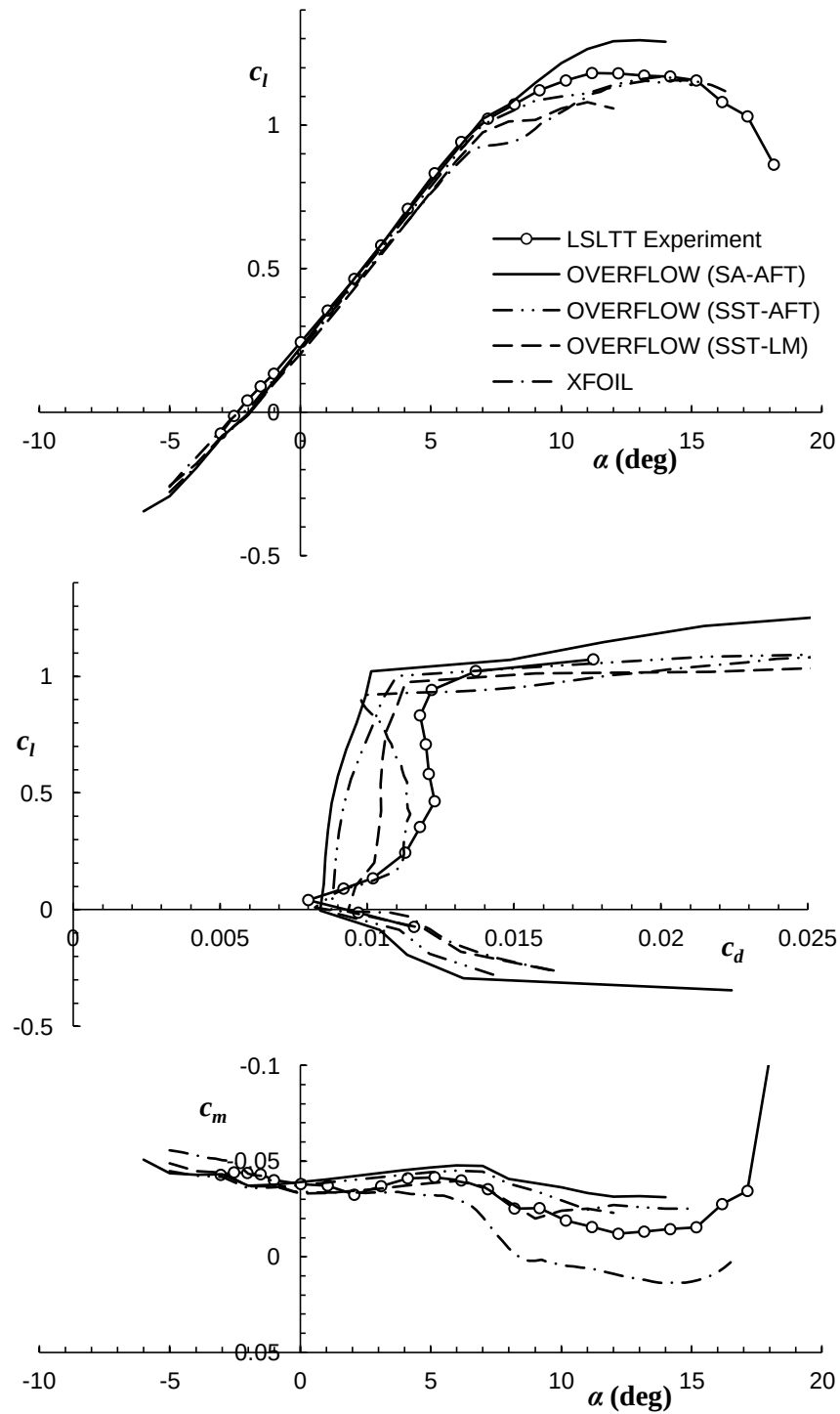


Figure 5-18. Predicted and measured [96] section aerodynamic characteristics of the S805 airfoil, $R = 0.5 \times 10^6$.

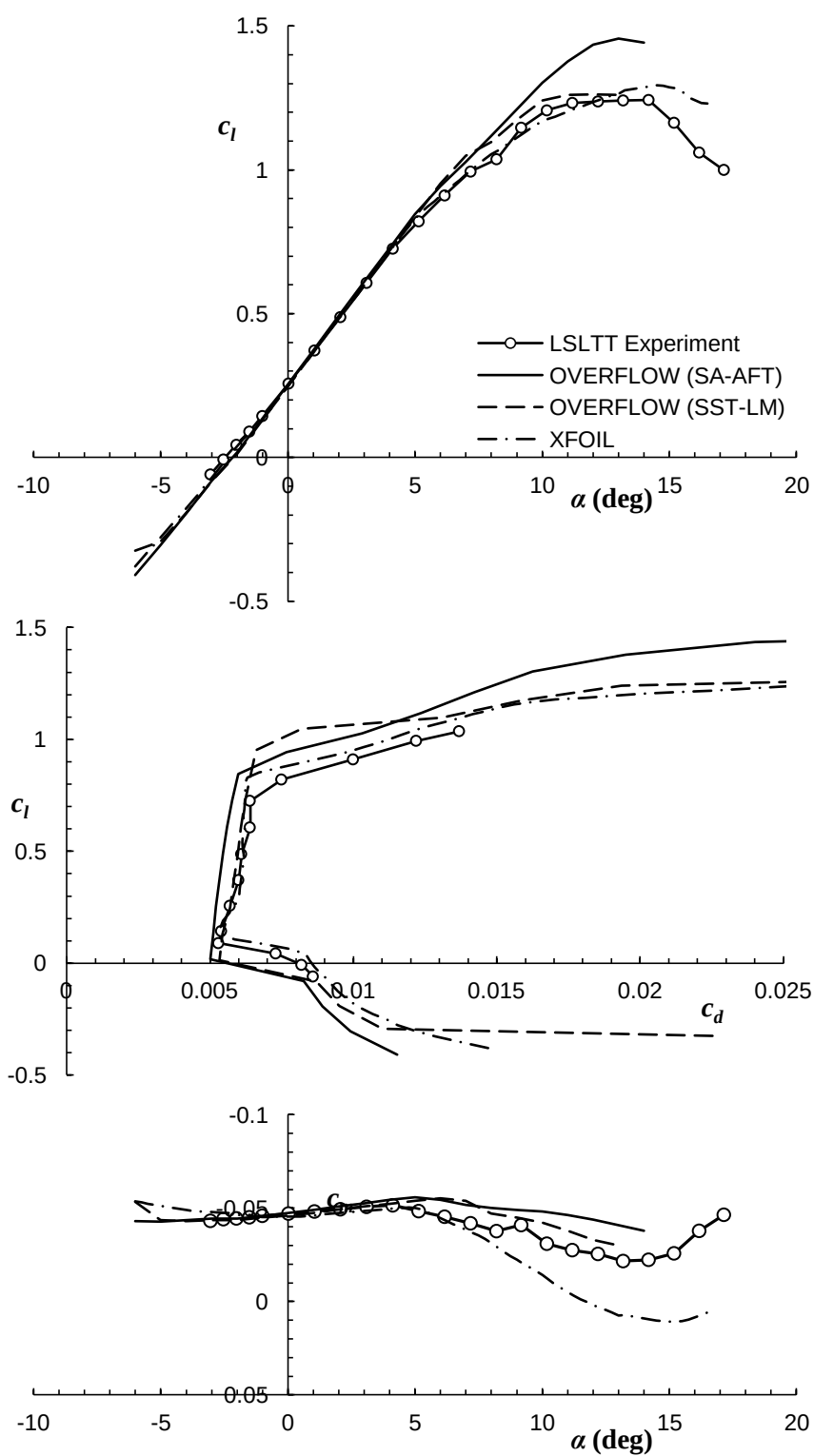


Figure 5-19. Predicted and measured [96] section aerodynamic characteristics of the S805 airfoil, $R = 1.5 \times 10^6$.

5.4.3 S414 Slotted, Natural-Laminar-Flow Airfoil

The new transition model has also been tested on the S414 slotted, natural-laminar-flow airfoil. The slotted, natural-laminar-flow airfoil concept [97,98] is a two-element configuration that has been found to achieve lower profile-drag coefficients and higher maximum-lift coefficients than single-element natural-laminar-flow airfoils designed to similar specifications. The aft element eliminates the requirement that the pressure at the trailing edge of the fore element recover to the free-stream value, allowing favorable pressure gradients to extend farther aft on the fore element. As a result, the fore element is entirely laminar in the low-drag range. The S414 airfoil, designed for the rotor of a small helicopter, has been tested in the Penn State LSLTT [99] for Reynolds numbers ranging from 0.5×10^6 to 1.5×10^6 . The increased complexity of this geometry, being two elements instead of one, provides a good test case for the capabilities of the new transition model.

OVERFLOW, with the new transition model, and MSES were used to predict the section aerodynamic characteristics for $R = 1.5 \times 10^6$. These data are shown in Fig. 5-20 along with the experimental data for comparison. The MSES predictions agree well with experiment for the drag levels in the laminar bucket and the lift coefficient of the lower corner of the bucket. MSES predicts the upper corner to occur at a higher lift coefficient than shown experimentally. The OVERFLOW predictions are consistent with those of MSES, particularly for the upper corner of the drag bucket, but the lower corner is predicted at too low of a lift coefficient. As was the case with the S805 data, this is the result of viscous-inviscid interactions causing the lower-surface laminar separation bubble to occur too far aft. The OVERFLOW predictions with the Langtry-Menter model have the same error in lower corner of the bucket as with the new transition model, but the Langtry-Menter model also over-predicts the lift coefficient for the upper corner of the drag bucket.

Further inspection of the solutions for each angle of attack has confirmed that the amplification factor transport equation behaves as intended and correctly modifies the behavior of the eddy-viscosity model for this case. An example of this is included as Fig. 5-21, in which contours of the Spalart-Allmaras modified-eddy-viscosity ratio are plotted for the $c_l = 1.043$ solution for $R = 1.5 \times 10^6$. Transition is predicted to occur on the upper surface of both elements, while the lower surfaces remain entirely laminar for this angle of attack. Downstream of the transition location, the eddy viscosity grows and the boundary layer is fully turbulent.

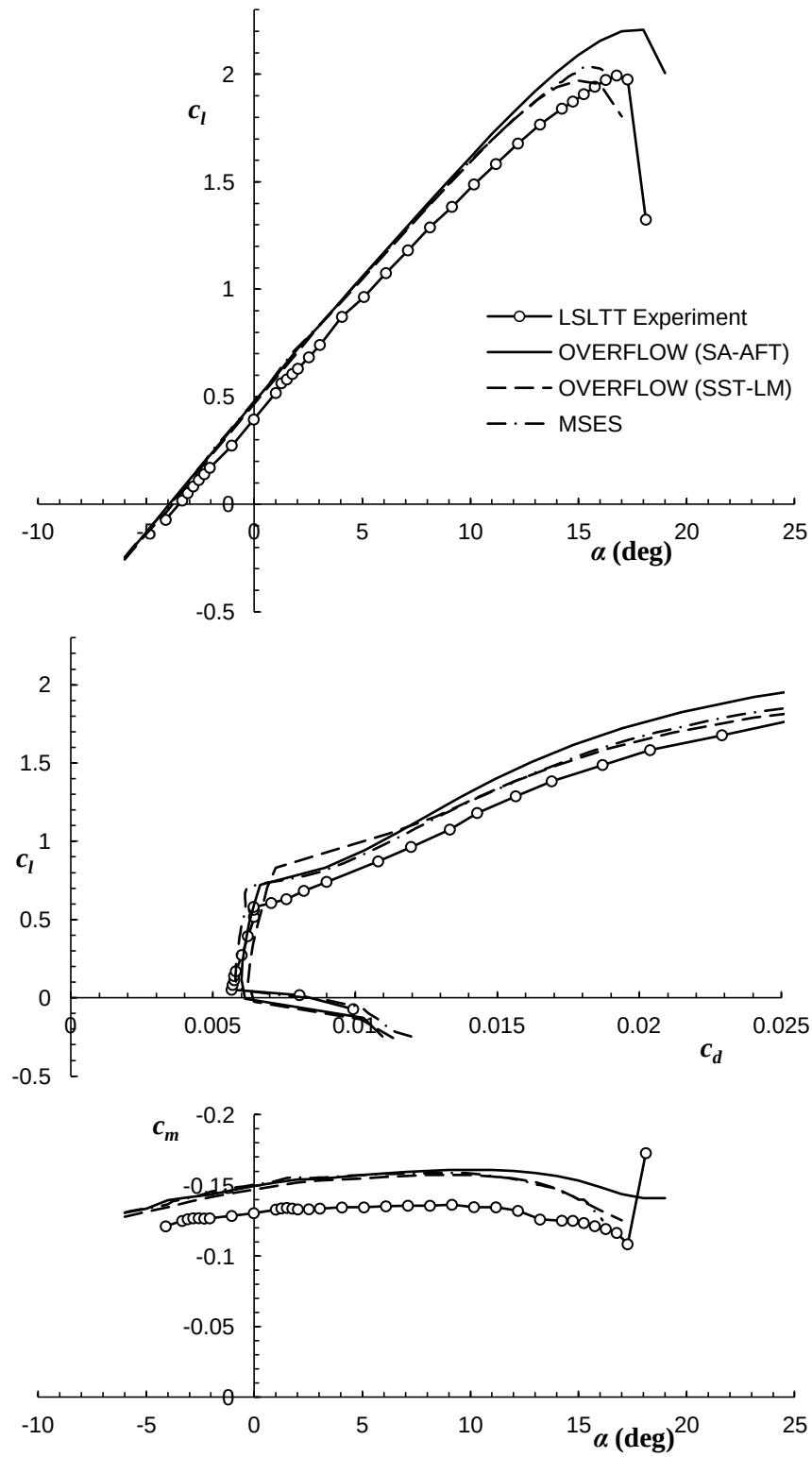


Figure 5-20. Predicted and measured [99] section aerodynamic characteristics of the S414 airfoil, $R = 1.5 \times 10^6$.

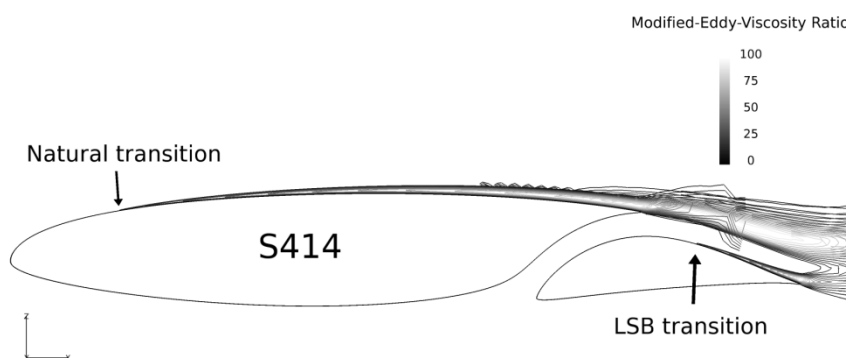


Figure 5-21. Contours of modified eddy-viscosity ratio around the S414 airfoil, $c_l = 1.043$ and $R = 1.5 \times 10^6$.

5.4.4 NLF(1)-0416 Natural-Laminar-Flow, General Aviation Airfoil

The NLF(1)-0416 was designed for operating conditions representative of an advanced, light, single-engine general aviation airplane using two primary objectives [100]. The first was that the airfoil needed to produce a roughness-independent maximum lift coefficient of at least 1.76. The second objective was that the airfoil needed to have low profile-drag coefficients, similar to what was possible with the NACA 6-series airfoils, at the cruise lift coefficient of 0.4 for a Reynolds number of 4.0×10^6 . The low drag coefficients were achieved by designing the airfoil to have long runs of laminar flow that result from favorable gradients in the surface pressure distributions rather than active boundary-layer control methods (“natural laminar flow”). The aerodynamic design of the NLF(1)-0416 was experimentally verified in the NASA Langley Low-Turbulence Pressure Tunnel (LTPT) [100] for Reynolds numbers ranging from 1.0×10^6 to 9.0×10^6 . Experimental measurements from the NASA Langley LTPT have compared favorably with measurements taken in the Penn State LSLTT and Delft LSWT for other airfoils [101], thus it is reasonable to assume that the same transition criteria should apply.

The section aerodynamic characteristics of the NLF(1)-0416 airfoil have been predicted using OVERFLOW with the new transition model (SA-AFT) and using XFOIL for $R = 4.0 \times 10^6$

(the cruise Reynolds number) and are compared in Fig. 5-22. Also compared are the experimental profile-drag data (digitized from Ref. [100]) for the same Reynolds number. The OVERFLOW and XFOIL predictions agree very well with each other. The lift curves are indistinguishable up to a lift coefficient of 1.6, though OVERFLOW predicts a higher maximum lift coefficient than XFOIL. The theoretical predictions for the profile-drag coefficient under-predict the experimental measurements, though they agree very well with each other throughout the operating range of the airfoil. Both methods predict the lowest drag to occur around the design lift coefficient of 0.4. The pitching-moment curve shows very strong agreement between the methods at lower lift coefficients, but the curves diverge at higher lift coefficients.

Also measured in the NASA Langley experiment were laminar-turbulent transition locations for Reynolds numbers up to 4.0×10^6 . The locations were determined by connecting a microphone to the pressure orifices on the upper and lower surfaces and listening for the distinct noise increase associated with turbulent flow. Although this method does not provide exact locations, it does put bounds on the chordwise range in which transition occurs. In the data reported in Ref. [100] for this experiment, curves were faired through the data to represent smooth movement of the transition location within the bounding boxes of the data. The faired curves for the $R = 4.0 \times 10^6$ transition measured were digitized and plotted in Fig. 5-23, along with the transition locations predicted with OVERFLOW and XFOIL. The two theoretical methods agree very well with each for all lift coefficients, although the experimentally measured transition location is more upstream than was predicted. For the upper-surface measurements, the reported data stop below the predicted upstream movement of transition. The lower-surface predictions capture the bend in the transition curves, although at a lower lift coefficient than was observed experimentally.

These data raise a concern as to whether the critical amplification factor of 10.07 is actually representative of this wind tunnel for a Reynolds number of 4.0×10^6 , as transition was

measured consistently upstream of where it is predicted to occur. A second angle-of-attack sweep was run with OVERFLOW and the new transition model using $N_{crit} = 7.18$ (equivalent to $Tu_{\infty} = 0.15\%$ using Eq. 4.50), which was estimated based on the transition location discrepancies. The predicted transition locations from this set are plotted in Fig. 5-24 with experiment and, for comparison, the original $N_{crit} = 10.07$ predictions. The new predictions agree much more closely with experiment than did the original for all lift coefficients. The improved agreement also carries over to the profile-drag predictions, which are compared to experiment and the original OVERFLOW predictions in Fig. 5-25. While it is impossible to conclude from this exercise whether the actual turbulence levels in the wind tunnel were around 0.15% for this experiment, it nevertheless underscores the importance of choosing the appropriate value of N_{crit} .

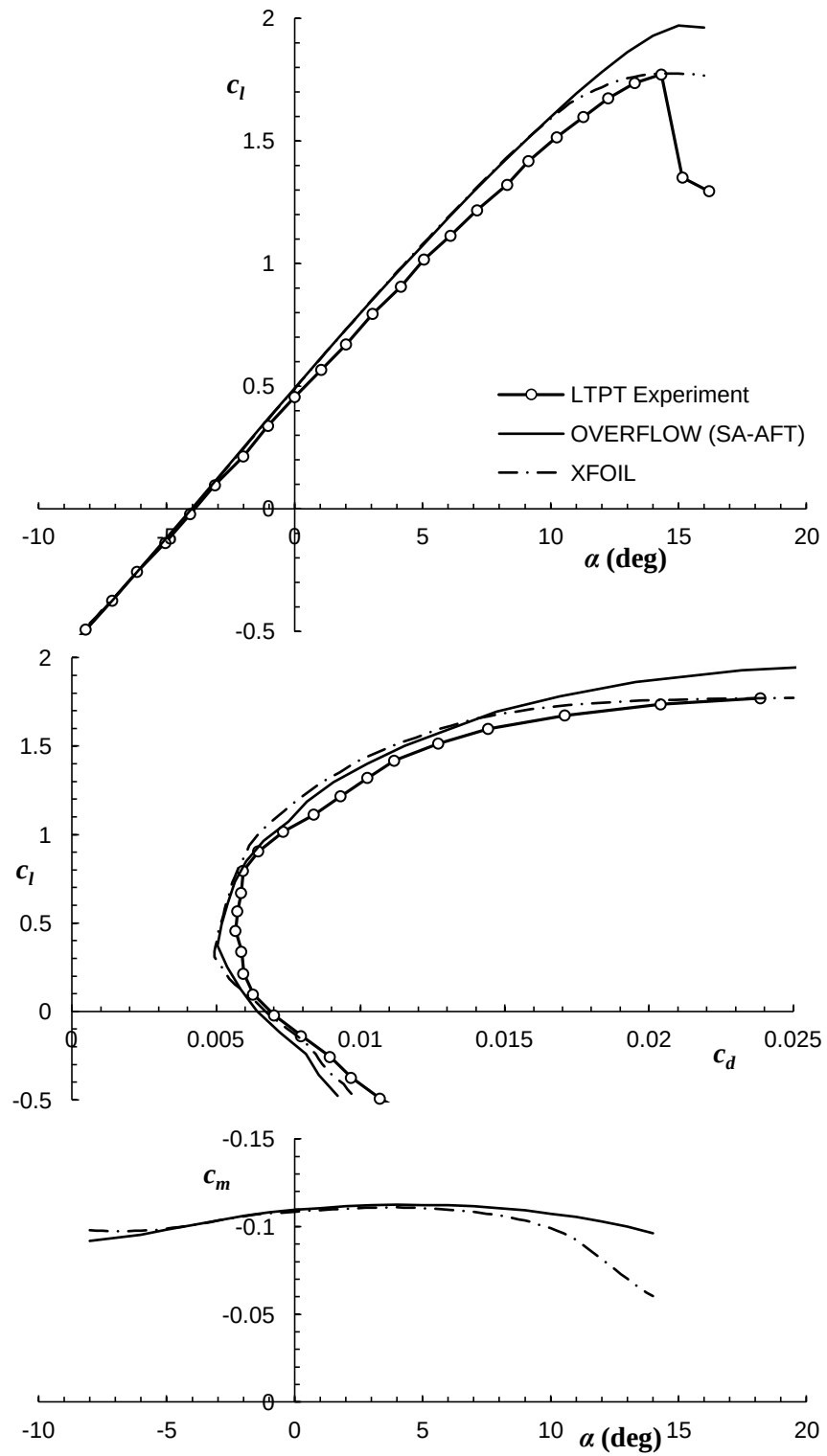


Figure 5-22. Predicted and measured [100] section aerodynamic characteristics of the NLF(1)-0416 airfoil, $R = 4.0 \times 10^6$.

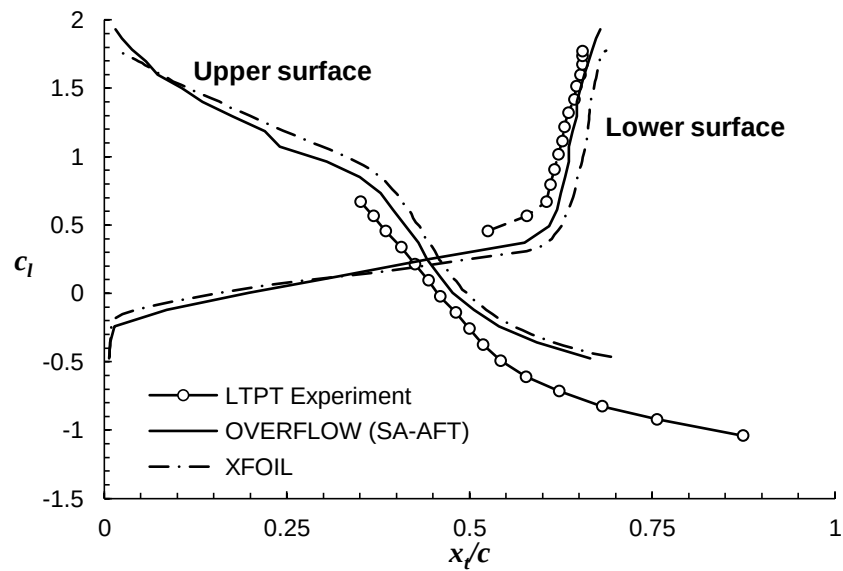


Figure 5-23. Predicted and measured [100] transition locations of the NLF(1)-0416 airfoil, $R = 4.0 \times 10^6$.

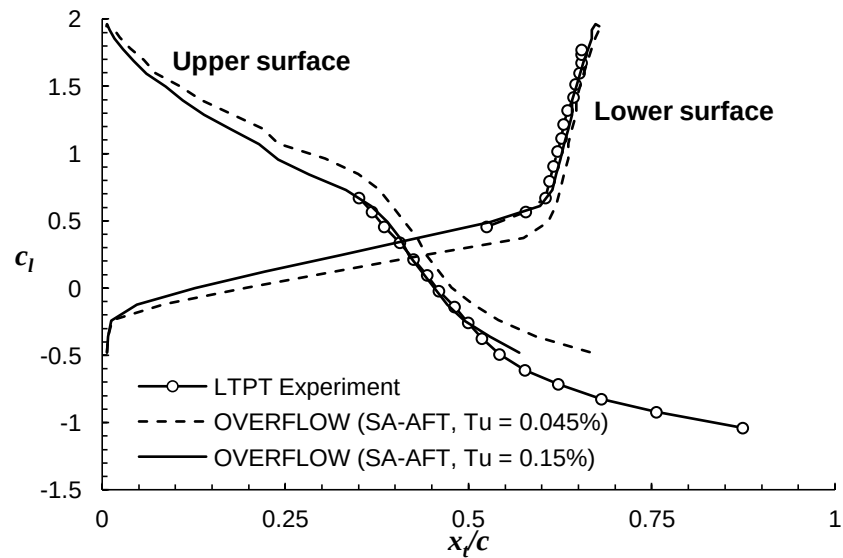


Figure 5-24. Predicted and measured [100] transition locations of the NLF(1)-0416 airfoil, $R = 4.0 \times 10^6$.

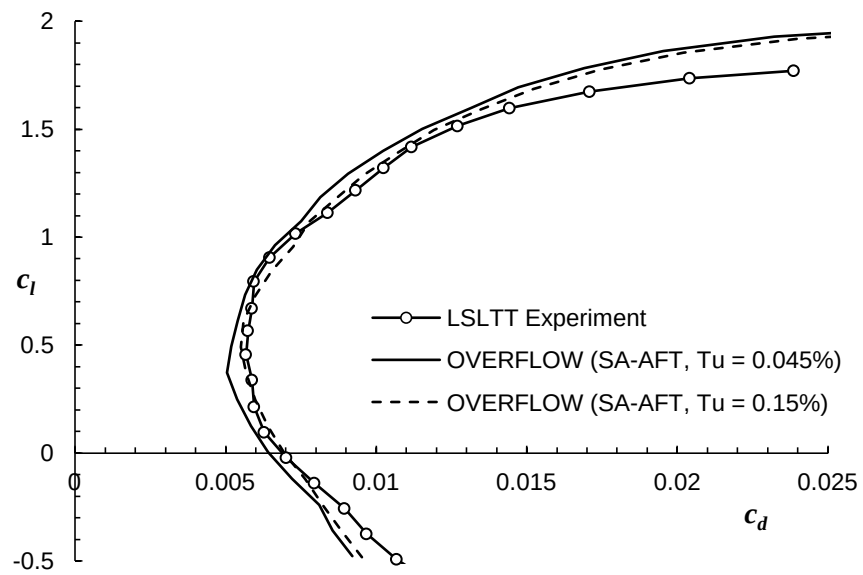


Figure 5-25. Predicted and measured [100] section aerodynamic characteristics of the NLF(1)-0416 airfoil, $R = 4.0 \times 10^6$.

5.5 Three-Dimensional Verification

5.5.1 Discus 2cx

The wing/winglet geometry for the 18-m Schempp-Hirth Discus 2cx sailplane was used as a relatively “simple” three-dimensional test case for the transition model. A picture of the Discus 2cx is included as Fig. 5-26 (from Ref. [102]). The wing is high aspect ratio and exhibits no appreciable spanwise pressure gradients inboard of the winglet through most of the flight regime. There are two goals of this study: provide a qualitative assessment of the transition model behavior in a three-dimensional environment, and explore the ability of transitional CFD to predict aircraft performance across its flight regime compared to fully turbulent methods.



Figure 5-26. 18-m Discus 2cx (from Ref. [102]).

For this study, a single-domain O-O surface mesh was generated from the production IGES model of the geometry using Gridgen [103]. The original IGES definition of the wing begins at the wing root, which does not lie on the aircraft symmetry plane. The surface mesh was subsequently extrapolated to the aircraft centerline. This allows a true symmetry plane to be applied while preserving the true span of the wing. The fuselage and empennage geometries are not modeled. The final volume mesh was generated using HYPGEN [90]. The resulting mesh has 425 chordwise and 37 spanwise surface points with 101 surface-normal points with the far-field boundaries located approximately 20 spans from the surface.

The chordwise resolution of the grid is finer than would be typically used for a fully turbulent, swept-wing performance analysis [104]. Conversely, the spanwise resolution is coarser than would typically be used, but a large portion of the wing is assumed to have very small spanwise flow gradients. Nevertheless, grid clustering was used around the winglet juncture and towards the tip. The off-wall spacing was set such that the y^+ value of the first point is less than one at the highest Reynolds numbers (following best practices for fully turbulent analyses [91]) while the stretching ratio was kept below 1.2.

Lift-coefficient sweeps for the Discus 2cx were generated in OVERFLOW using the new transition model (SA-AFT) and, for comparison, the fully turbulent Spalart-Allmaras model (SA-noft2). The sweep represents a fully ballasted, maximum-weight flight condition (565 kg) for the sailplane and assuming that all lift is generated by the wing. To account for the wing generating constant lift for all flight speeds, the solutions were run such that the product $RC_L^{1/2}$ is held constant at 1.4142×10^6 for all lift coefficients. The specific lift coefficients were attained using the TARGCL feature of OVERFLOW with a desired lift-coefficient tolerance of ± 0.0001 .

With regard to the goal of a qualitative assessment of three-dimensional transition behavior, a good example is included as Fig. 5-27, which shows skin friction contours and surface

streamlines in the winglet juncture for a wing lift coefficient of 0.5. Inboard of this, the flow is largely two dimensional and well behaved. In this juncture, however, there is more three-dimensionality to the flow due to the rapid dihedral change and the leading-edge sweep of the winglet. Nevertheless, the new transition model is well behaved and produces results that are qualitatively consistent with expectation and unpublished flight-test results.

To assess the goal of using the new transition prediction methodology to predict performance, resulting drag polars are plotted in Fig. 5-28 with the “ideal” induced drag subtracted out. This drag is represented mathematically by

$$C_D - C_{D,ideal} = C_D - \frac{C_L^2}{\pi AR} \quad (5.1)$$

This plotting technique allows for a better understanding of profile drag effects on a three-dimensional geometry. It must be emphasized that the use of a span efficiency of one in Eq. 5.1 is not intended to represent the actual span efficiency of the Discus 2cx (the span efficiency for this wing is actually greater than one). Rather, this plotting method takes advantage of the fact that typical span efficiencies are of order unity to collapse the drag scale across the flight regime.

The three-dimensional $(L/D)_{\max}$ for the wing alone is predicted to be 60.6 using the new transition model, while the fully turbulent approach predicts 40.6. This is a significant difference, particularly when one considers that the fully turbulent $(L/D)_{\max}$ for the wing alone is lower than that of the entire actual airplane. The difference in the performance predictions can be further quantified by considering the predicted sink rates for the wing. These are plotted in Fig. 5-29 with forward velocity, which was calculated assuming the wing is the only source of lift for the sailplane. The minimum sink rate predicted using the fully turbulent approach is 40% greater than that predicted with the new transition model.

Another strong impact of including transition modeling compared to fully turbulent analyses is the influence of transition on the surface pressure distributions. For the Reynolds

numbers typical of a sailplane wing, transition commonly occurs via a laminar separation bubble. In such cases, this is visible in the surface pressure distribution as a streamwise plateau relative to the inviscid distribution that would occur for the same flight conditions. Fully turbulent pressure distributions for the same conditions exhibit no such pressure plateaus, allowing the laminar separation bubbles to be identified by comparison. This particular behavior is observed in the OVERFLOW solutions plotted in Fig. 5-30 for the 75%-semispan station at $C_L = 0.5$. Note that the pressures are nearly identical between the two solutions upstream of the upper- and lower-surface bubbles.

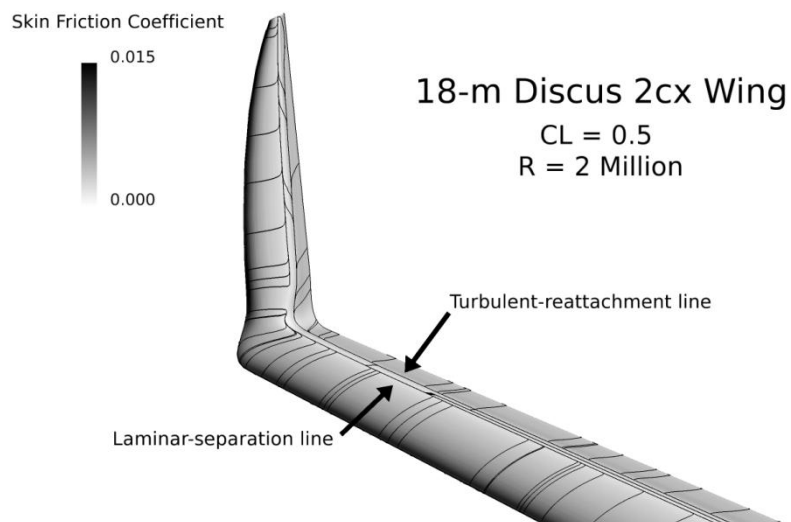


Figure 5-27. Predicted upper-surface streamlines and skin friction contours around the Discus 2cx winglet juncture.

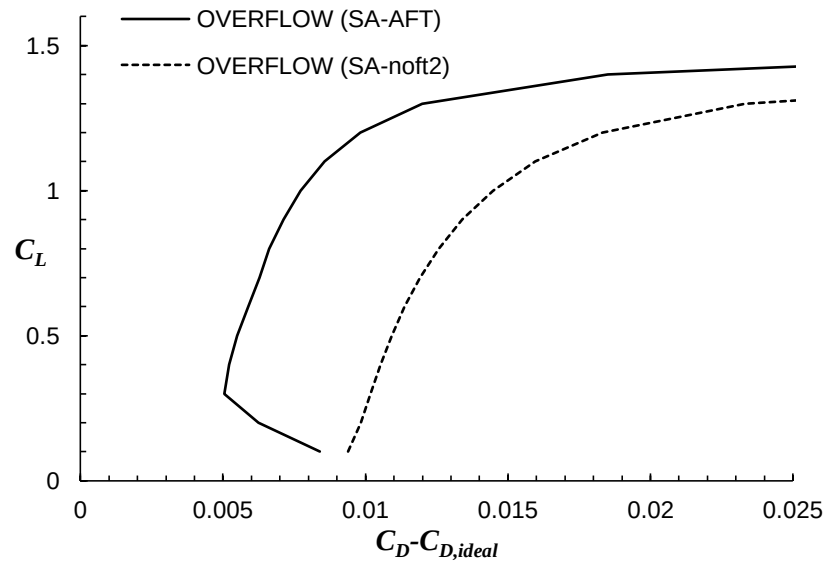


Figure 5-28. Predicted Discus 2cx wing-only drag polars for $RC_L^{1/2} = 1.4142 \times 10^6$.

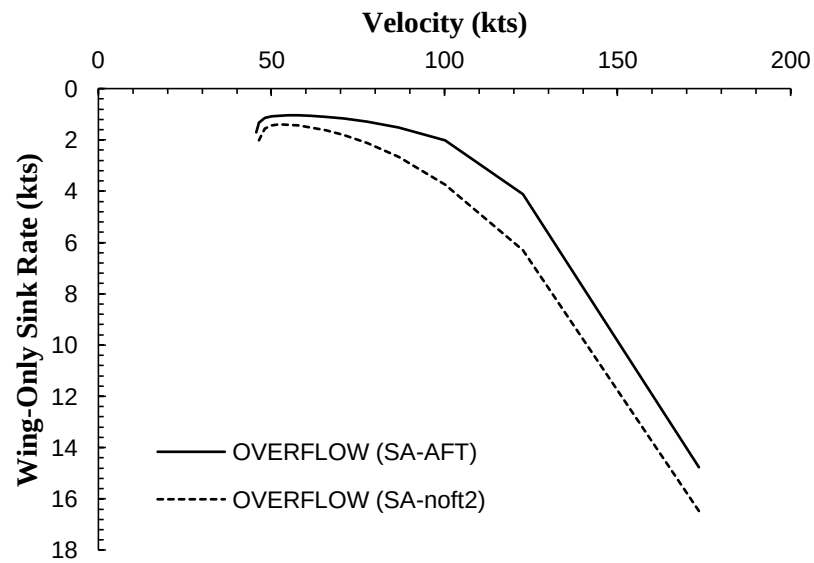


Figure 5-29. Predicted Discus 2cx wing-only sink-rate polars for $RC_L^{1/2} = 1.4142 \times 10^6$ and $W = 565$ kg.

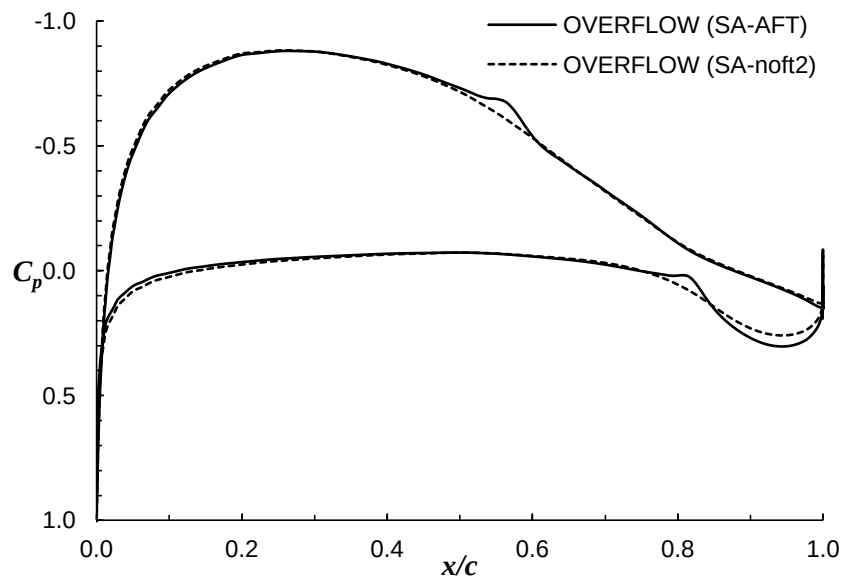


Figure 5-30. Predicted surface pressure distributions at the 75%-semispan station for $C_L = 0.5$.

5.5.2 NASA Common Research Model

A more complex three-dimensional case used to test the qualitative behavior of the transition model is the NASA common research model (CRM) wing-body used in the fifth AIAA drag prediction workshop [105]. The grid used for this example is the medium overset grid from the common grid sequence. More details on these grids may be found in Refs. [104,106]. The case was run in OVERFLOW using both the new transition model (SA-AFT) and the Langtry-Menter model for the flight conditions specified for the workshop, which are a lift coefficient of 0.5, a free-stream Mach number of 0.85, and a Reynolds number based on a mean aerodynamic chord of 5×10^6 . For both transition models, the turbulence intensity was set to 0.07%, which corresponds to $N_{crit} = 9$ according to Eq. 4.50. It must be noted that trip dots were used for the wind-tunnel tests of the CRM with the specific intent of making the boundary layer fully turbulent for CFD validation purposes [107], and experimental data with unforced transition are, at present, unavailable.

The predicted surface skin-friction coefficients are shown in Fig. 5-31 as filled contours on the wing-body surface. Transition on the fuselage is predicted by the new transition model to occur at the “cockpit window” features of the geometry, whereas the Langtry-Menter model predicts it to be further aft. At the wing root, the flow is predicted by both models to be fully turbulent due to leading-edge contamination from the fuselage. Immediately outboard of this region, transition is predicted by both models to occur naturally in an adverse pressure gradient that is present upstream of the shock wave. This behavior continues outboard until the laminar run reaches the shock wave. Through the shock wave, there is a sharp adverse gradient and a small amount of separation, initiating the transition process. At the wing tip, the new transition model predicts the flow to be fully turbulent due to the tip vortex, whereas the Langtry-Menter model predicts a laminarizing effect. It must be emphasized that both transition models used for

this study account only for Tollmien-Schlichting instabilities and a more complete analysis of transition on this geometry would require treatment of crossflow instabilities.

Both models, each based on different underlying theories, predict essentially the same transition patterns on this geometry with only slight discrepancies. There is a noticeable difference in the predicted skin friction downstream of transition, which is attributable to a combination of the transition models being coupled with different eddy-viscosity models and the near-wall behavior of the Langtry-Menter intermittency equation. Following the same computational benchmarking procedure used with the PSU 94-097 airfoil, it was found that the per-iteration solution time for the amplification factor transport equation is 46% that of the two Langtry-Menter transition equations.

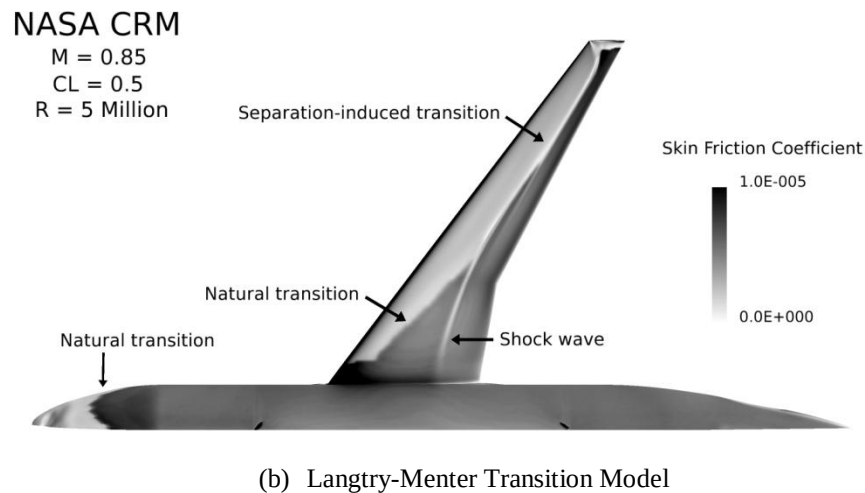
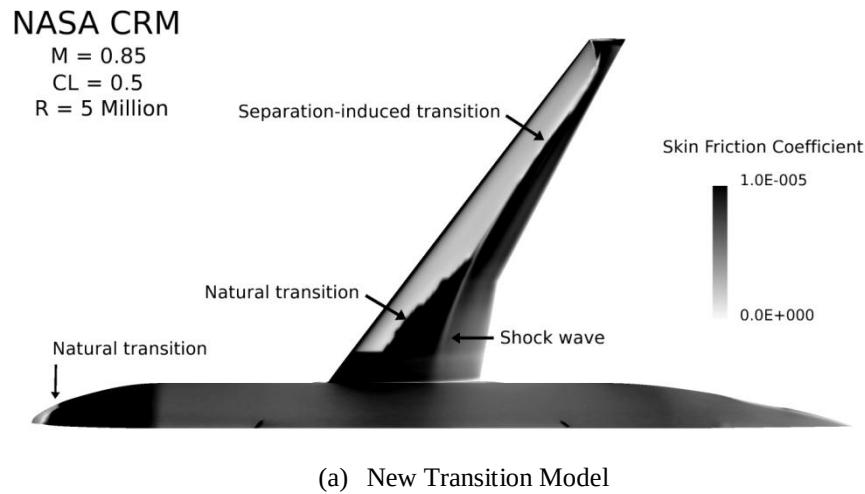


Figure 5-31. Contours of predicted skin-friction coefficient on the NASA CRM wing-body.

5.5.3 DLR-F11 High-Lift Geometry

The DLR-F11 geometry is a high-lift wing-body configuration designed by Airbus-Deutschland for the EUROLIFT program [108,109] and was subsequently used for the 2nd AIAA High-Lift Prediction Workshop (HiLiftPW-2) [86,109]. This model is a departure from the NASA Trap Wing high-lift geometry used for the first workshop [110], as the DLR-F11 is representative of a commercial transport and has an aspect ratio of 9.353 and a quarter-chord

sweep of 30° . The high-lift system for the DLR-F11 features a three-element wing, which includes a leading-edge slat and a trailing-edge Fowler flap. Two sets of wind-tunnel tests were performed for the geometry to measure the aerodynamic forces and moments. The first was an atmospheric test in the low-speed wind tunnel at Airbus-Deutschland (B-LSWT) with a Reynolds number of 1.35×10^6 and transition free. In addition to the aerodynamic characteristics, surface oil flow patterns were photographed for qualitative comparisons. The second was under cryogenic conditions at $R = 15.1 \times 10^6$ in the European Transonic Wind Tunnel (ETW). Both experiments were run with a Mach number of 0.175.

The cases for HiLiftPW-2 focused on slat and flap deflections of 26.5° and 32° , respectively, for the two Reynolds numbers. In addition to the effect of Reynolds number, the influence of geometric fidelity in the computational analysis was considered. The different levels of fidelity were wing/body/high-lift system with side-of-body seal (Config. 2), the addition of slat and flap support brackets (Config. 4), and the further addition of slat pressure-tube bundles (Config. 5). The brackets and pressure-tube bundles were part of the wind-tunnel model, making Config. 5 the geometry that is most representative of what was tested. Although the workshop focused on fully turbulent CFD analyses, an optional case for the workshop was the inclusion of transition, either modeled or prescribed, for Config 4 at a Reynolds number of 1.35×10^6 .

This case was analyzed using OVERFLOW with the SA-AFT variant of the new transition model as well as with the fully turbulent Spalart-Allmaras model (SA-nof2). The aerodynamic predictions are plotted with the B-LSWT measurements in Fig. 5-32. For this comparison, it is more useful to plot the drag with the ideal induced drag removed to provide insights into the influence of transition on the profile drag. Neither the fully turbulent nor the transitional solutions perform well in predicting the aerodynamic characteristics of this configuration, though the quality of these predictions is consistent with those of the other participants in HiLiftPW-2 [86]. Including transition made a significant difference in HiLiftPW-1

and improved the predictions for the Trap Wing [111,112], but the same level of improvement is not realized in these data. This is due to the eddy-viscosity model seemingly not capturing all relevant flow physics about the DLR-F11, as evidenced by the lack of qualitative agreement between the fully turbulent predictions and experiment. Good qualitative agreement between fully turbulent predictions and experiment was observed for the Trap Wing, however [111,112].

The surface flow patterns from the OVERFLOW SA-AFT predictions for $\alpha = 18.5^\circ$ are shown in Figs. 5-33 and 5-34 along with photographs from the B-LSWT experiment. There is good qualitative agreement between experiment and prediction, particularly in predicting the location and length of the laminar separation bubbles on the upper surfaces of the slat and main wing element. Cross-flow transition does not seem to be dominant at these conditions as the experimental surface flows show transition to occur primarily due to laminar separation in a manner consistent with the CFD predictions.

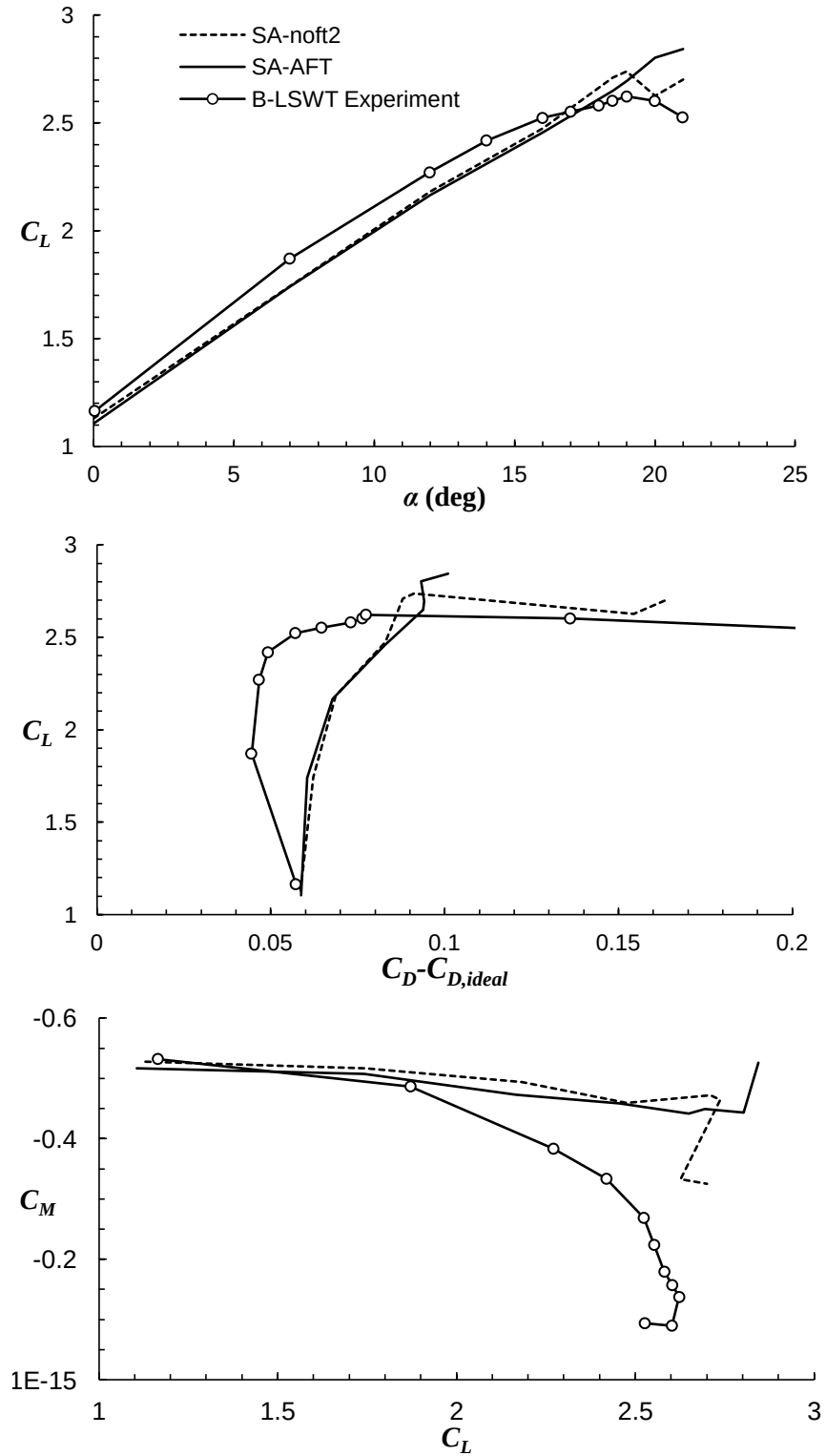


Figure 5-32. Predicted and measured [109] aerodynamic coefficients of the DLR-F11 geometry for $R = 1.35 \times 10^6$.

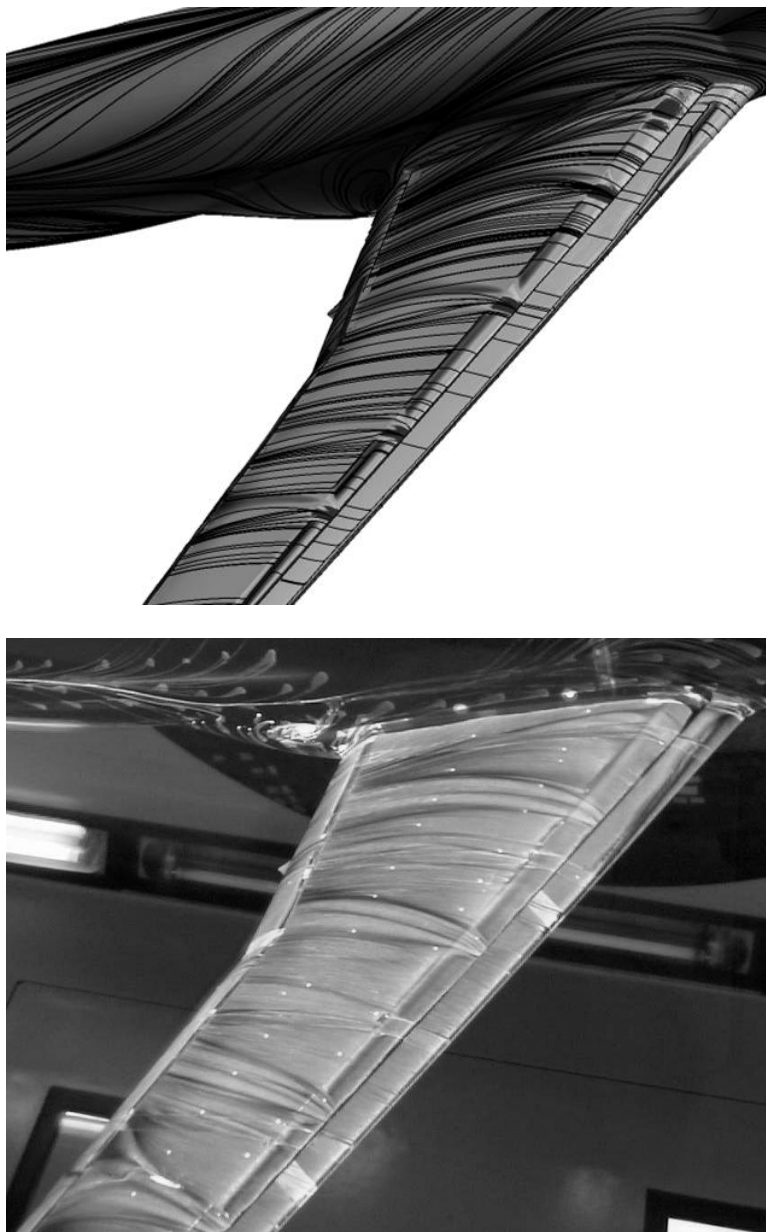


Figure 5-33. Predicted (top) and photographed (bottom) [108] surface streamlines on the DLR-F11 wing for $R = 1.35 \times 10^6$ and $\alpha = 18.5^\circ$.

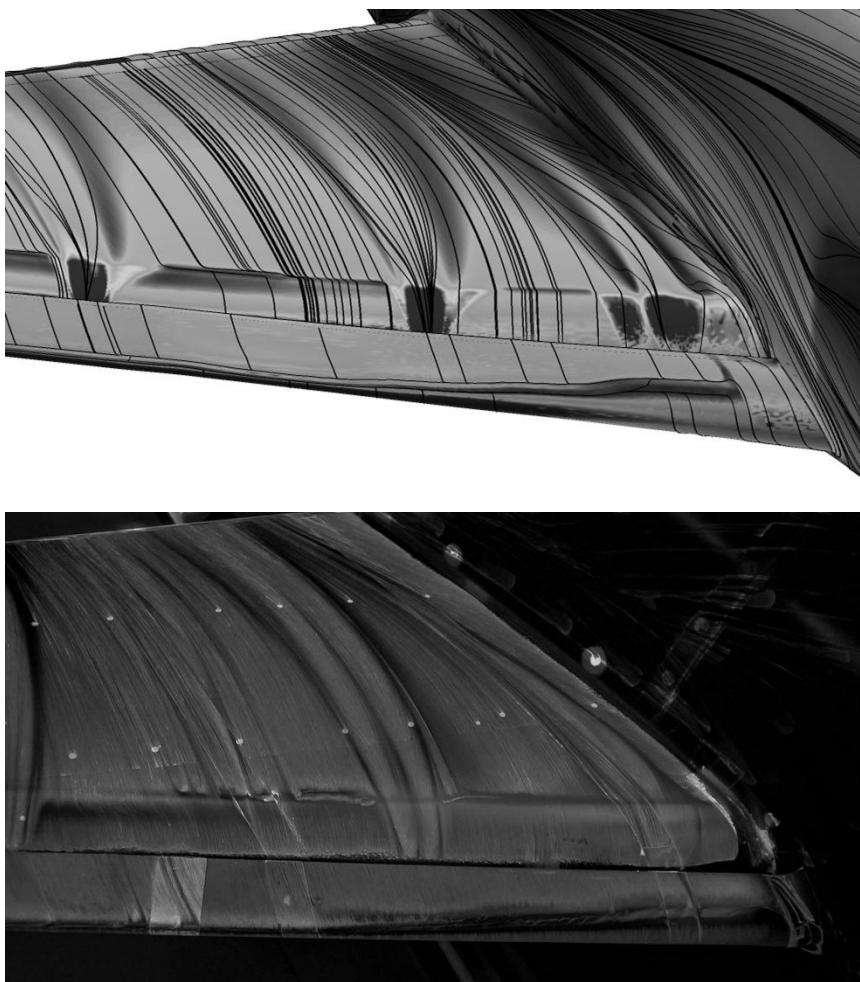


Figure 5-34. Predicted (top) and photographed (bottom) [108] surface streamlines on the DLR-F11 wing for $R = 1.35 \times 10^6$ and $\alpha = 18.5^\circ$.

Chapter 6

Conclusions

A new boundary-layer transition model based on the approximate-envelope method of linear-stability theory has been developed that is solvable within the framework of modern, massively parallelizable Reynolds-averaged Navier-Stokes solvers. It involves the solution of a single amplification-factor transport equation that is locally defined, requiring no knowledge of the global flow solution. The only geometric data necessary is the distance to the nearest solid wall; however, this quantity is available in flow solvers as it is also used in the solution of commonly used eddy-viscosity models. The free-stream flow conditions are also required to provide closure to the model, but these data are constant for the entire computational domain and do not depend on any details of the actual flow solution. The source term of the transport equation depends only on the local velocity field and not the amplification factor itself, which also allows it to capture upstream flow history in a manner consistent with linear-stability theory. Viscous diffusion is numerically necessary in the equation, even though it is non-physical.

The amplification-factor transport equation does not explicitly depend on how the turbulence is calculated in the solution. For this work, the one-equation Spalart-Allmaras and the two-equation Menter SST eddy-viscosity models were of specific interest as they are very commonly used for aeronautical applications. The methodology of incorporating transition into these models stems from the observation that the algebraic f_{t2} turbulence suppression term developed by Spalart and Allmaras behaves very similarly to the converged intermittency equation of the Langtry-Menter transitional SST model. Thus, a modification to the f_{t2} function was introduced that suppresses the function near transition and allows it to behave in its original

manner elsewhere. For the Spalart-Allmaras model, no changes are necessary in how f_t^2 modifies the transport equation, whereas with Menter SST the function is used in lieu of the intermittency.

Aerodynamic predictions for various test cases were made with the new transition model as applied to both eddy-viscosity models (SA-AFT and SST-AFT) and implemented into the OVERFLOW solver. For the airfoils that were analyzed, the results obtained using the new transition model agreed very well with experimental data. The new model outperformed the four-equation Langtry-Menter model in predicting the low-drag lift-coefficient range of the airfoils. The new transition model has similar influence on the predicted maximum lift coefficient as the Langtry-Menter model, and the predictions are heavily dependent on the underlying eddy-viscosity model. Good qualitative behavior of the model was demonstrated for three-dimensional configurations. In particular, the model was able to predict the expected presence of laminar separation bubbles and the existence of a low-drag lift-coefficient range in the profile drag for a Discus 2cx sailplane wing. A prediction made for the NASA Common Research Model shows that the SA-AFT model can converge at transonic Mach numbers and give predictions consistent with those of the Langtry-Menter model, but with less than half of the computational cost per iteration in evaluating the transition/turbulence closure. Finally, solutions about the DLR-F11 high-lift configuration agreed well qualitatively with experimental oil flow photographs in the existence of laminar separation bubbles on the upper surface of the three-element wing.

In its current form, the new transition model does not directly account for the effects of bypass and cross-flow transition mechanisms. While these are not deficiencies per se, they are capabilities that should be further developed in future efforts with the amplification-factor transport equation. Bypass transition is characterized by non-linear growth and secondary instabilities dominating the transition process, and is prominent when the ambient turbulence levels are much higher than typically expected in external aerodynamics. For some complex configurations, however, wakes from upstream elements can interact with downstream elements,

creating a mixture of natural and bypass transition in the same flow field. Although bypass transition falls outside of the applicability of linear-stability theory, the framework of the amplification-factor transport equation is based on the local estimation of the boundary layer shape factor and any physically relevant “progress” toward transition can be used. It may also prove desirable to develop an additional source term that maintains a physically meaningful amplification factor in wakes.

The transition model can also be expanded to include cross-flow transition, which is dominant on swept-wing aircraft. While cross-flow transition is predictable using linear stability theory, it is more complex than Tollmien-Schlichting transition since the flow is not aligned with the local pressure gradient. Locally characterizing the boundary-layer profile would therefore require an additional estimated shape factor and/or an estimate of the effective “sweep” angle, which is the difference between the flow direction at the edge of the boundary layer and the direction of the pressure gradient. The growth rate of the cross-flow instability envelope is also required, and should be calculable using the locally estimated boundary-layer properties. An additional amplification factor transport equation is likely necessary due to the inherent physical differences between Tollmien-Schlichting and cross-flow instabilities.

The amplification factor transport equation has not been validated for unsteady aerodynamic applications. The underlying theory of the approximate envelope method assumes that the flow field is steady in the Galilean-invariant sense, and the correlations used and developed in this dissertation can be written as such using local grid motion without loss of generality or applicability. It is unknown at this time how much impact unsteadiness in the flow field will have on the physical relevance of the steady transition correlations or on the ability to estimate the pertinent boundary-layer properties, and comparisons of such quasi-steady predictions with unsteady experimental transition measurements should be performed.

A detailed investigation into the predicted behavior of laminar separation bubbles is also highly suggested. This should include, but not be limited to, validating the predicted change in boundary-layer properties through the bubble. Improvements to this would, in turn, improve the profile-drag predictions in the low-drag lift-coefficient range of airfoils and wings and, possibly, the predicted limits of this lift-coefficient range.

Nevertheless, the benefits of the amplification factor transport equation developed in this research make it an improvement over the current state-of-the-art for transport-equation-based transition modeling in the RANS CFD environment. The new model is rooted in linear-stability theory rather than using local, algebraic transition correlations. This allows it to capture upstream flow history and more accurately and reliably predict the transition location and its movement with changing lift than the widely implemented Langtry-Menter model. A further benefit of the new model is that it requires the solution of only a single additional transport equation while not being dependent on a specific eddy-viscosity closure. This allows it to be used with the most appropriate eddy-viscosity model for a given aerodynamic situation. When coupled with the Spalart-Allmaras model, the computational overhead of evaluating the transition/turbulence closure is less than half that of the four-equation Langtry-Menter model per iteration with a very similar solution convergence rate.

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VITA

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Biography

James George Coder was born in State College, Pennsylvania and raised in nearby Bellefonte. After high school, he attended the Pennsylvania State University as a member of the Schreyer Honors College, majoring in Aerospace Engineering graduating with highest distinction in 2008. He continued his studies in Aerospace Engineering at Penn State under Professor Mark Maughmer, earning a Master's degree in 2010. Jim has interned with NASA Dryden Flight Research Center (2007) and twice with Boeing Research and Technology (2008, 2013). When Jim is not solving partial differential equations, he enjoys spending time with his wife, Julie, and their son, William, as well as playing golf.

Selected Publications

Coder, J. G. and Maughmer, M. D., "Numerical Validation of the Squire-Young Formula for Profile Drag Prediction," *Journal of Aircraft*.

Coder, J. G., Maughmer, M. D., and Somers, D. M., "Theoretical and Experimental Results for the S414, Slotted, Natural-Laminar-Flow Airfoil," *Journal of Aircraft*.

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Sclafani, A. J, Vassberg, J. C., Winkler, C., Dorgan, A. J., Mani, M., Olsen, M. E., and Coder, J. G., "DPW-5 Analysis of the CRM in a Wing-Body Configuration Using Structured and Unstructured Meshes," *Journal of Aircraft*.

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